Fourier Analysis and Applications

Lecture notes, Chapters 1–8 and Appendix

Based on E. M. Stein and R. Shakarchi, *Fourier Analysis: An Introduction*

Contents

Chapter 1. The Genesis of Fourier Analysis 4

1.1 The vibrating string 4
1.1.1 Derivation of the wave equation . . . 4
1.1.2 Solution to the wave equation . . . 4
1.1.3 Example: the plucked string . . . 5

1.2 The heat equation 5
1.2.1 Derivation of the heat equation . . . 5
1.2.2 Steady-state heat equation in the disc . . . 6

1.3 Selected exercises 6

Chapter 2. Basic Properties of Fourier Series 9

2.1 Examples and formulation of the problem 9
2.1.1 Main definitions and some examples . . . 9

2.2 Uniqueness of Fourier series 10

2.3 Convolutions 11

2.4 Good kernels 12

2.5 Cesàro and Abel summability: applications to Fourier series 13
2.5.1 Cesàro means and summation . . . 13
2.5.2 Fejér's theorem . . . 13
2.5.3 Abel means and summation . . . 14
2.5.4 The Poisson kernel and Dirichlet's problem in the unit disc . . . 14

2.6 Selected exercises 15

Chapter 3. Convergence of Fourier Series 18

3.1 Mean-square convergence of Fourier series 18
3.1.1 Vector spaces and inner products . . . 18
3.1.2 Proof of mean-square convergence . . . 18

3.2 Return to pointwise convergence 20
3.2.1 A local result . . . 20
3.2.2 A continuous function with diverging Fourier series . . . 20

3.3 Selected exercises 21

Chapter 4. Some Applications of Fourier Series **24**

4.1 The isoperimetric inequality **24**

4.2 Weyl's equidistribution theorem **24**

4.3 A continuous but nowhere differentiable function **26**

4.4 The heat equation on the circle **27**

4.5 Selected exercises **27**

Chapter 5. The Fourier Transform on $\mathbb{R}$ **29**

5.1 Elementary theory of the Fourier transform **29**
5.1.1 Integration of functions on the real line . . . 29
5.1.2 Definition of the Fourier transform . . . 29
5.1.3 The Schwartz space . . . 29
5.1.4 The Fourier transform on $\mathcal{S}$ . . . 29
5.1.5 The Fourier inversion . . . 31
5.1.6 The Plancherel formula . . . 31
5.1.7 Extension to functions of moderate decrease . . . 32
5.1.8 The Weierstrass approximation theorem . . . 32

5.2 Applications to some partial differential equations **32**
5.2.1 The time-dependent heat equation on the real line . . . 32
5.2.2 The steady-state heat equation in the upper half-plane . . . 33

5.3 The Poisson summation formula **34**
5.3.1 Theta and zeta functions . . . 35
5.3.2 Heat kernels . . . 35
5.3.3 Poisson kernels . . . 36

5.4 The Heisenberg uncertainty principle **36**

Chapter 6. The Fourier Transform on $\mathbb{R}^d$ **37**

6.1 Preliminaries **37**
6.1.1 Symmetries . . . 37
6.1.2 Integration on $\mathbb{R}^d$ . . . 37

6.2 Elementary theory of the Fourier transform **37**

6.3 The wave equation in $\mathbb{R}^d \times \mathbb{R}$ **38**
6.3.1 Solution in terms of Fourier transforms . . . 38
6.3.2 The wave equation in $\mathbb{R}^3 \times \mathbb{R}$ . . . 39
6.3.3 The wave equation in $\mathbb{R}^2 \times \mathbb{R}$: descent . . . 40

6.4 Radial symmetry and Bessel functions **40**

6.5 The Radon transform and some of its applications **41**
6.5.1 The X-ray transform in $\mathbb{R}^2$ . . . 41
6.5.2 The Radon transform in $\mathbb{R}^3$ . . . 41

6.6 Selected exercises **42**

Chapter 7. Finite Fourier Analysis **44**

7.1 Fourier analysis on $\mathbb{Z}(N)$ **44**
7.1.1 The group $\mathbb{Z}(N)$ . . . 44
7.1.2 Fourier inversion theorem and Plancherel identity on $\mathbb{Z}(N)$ . . . 44
7.1.3 The fast Fourier transform . . . 44

7.2 Fourier analysis on finite abelian groups **45**
7.2.1 Abelian groups . . . 45
7.2.2 Characters . . . 45
7.2.3 The orthogonality relations . . . 46
7.2.4 Characters as a total family . . . 46
7.2.5 Fourier inversion and Plancherel formula . . . 47

7.3 Selected exercises **47**

Chapter 8. Dirichlet's Theorem **50**

8.1 A little elementary number theory **50**
8.1.1 The fundamental theorem of arithmetic . . . 50
8.1.2 The infinitude of primes . . . 51

8.2 Dirichlet's theorem **52**
8.2.1 Fourier analysis, Dirichlet characters, and reduction of the theorem . . . 52
8.2.2 Dirichlet L-functions . . . 53

8.3 Proof of the theorem **54**
8.3.1 Logarithms . . . 54
8.3.2 L-functions . . . 54
8.3.3 Non-vanishing of the L-function . . . 56

Appendix. Integration **61**

A.1 Definition of the Riemann integral **61**
A.1.1 Basic properties . . . 61
A.1.2 Sets of measure zero and discontinuities of integrable functions . . . 63

A.2 Multiple integrals **64**
A.2.1 The Riemann integral in $\mathbb{R}^d$ . . . 64
A.2.2 Repeated integrals . . . 64
A.2.3 The change of variables formula . . . 65
A.2.4 Spherical coordinates . . . 66

A.3 Improper integrals. Integration over $\mathbb{R}^d$ **66**
A.3.1 Integration of functions of moderate decrease . . . 66
A.3.2 Repeated integrals . . . 67
A.3.3 Spherical coordinates . . . 67

Chapter 1. The Genesis of Fourier Analysis

1.1 The vibrating string

Motivation. The vibrating string and heat flow lead to linear PDEs whose separated solutions are sines and cosines, so the general solution is a Fourier series; which functions admit one?

Simple harmonic motion. A mass m on a spring with constant k obeys $-ky = my''$. With $c = \sqrt{k/m}$, $y(t) = a\cos ct + b\sin ct = A\cos(ct - \varphi)$: **amplitude** $A = \sqrt{a^2+b^2}$, **period** $2\pi/c$, **frequency** $c/2\pi$, $a = y(0)$, $b = y'(0)/c$. Euler: $e^{ix} = \cos x + i\sin x$, $\cos x = \frac{e^{ix}+e^{-ix}}{2}$, $\sin x = \frac{e^{ix}-e^{-ix}}{2i}$.

Standing and travelling waves. A **travelling wave** is $u(x,t) = F(x-ct)$ (a profile moving at speed c); a **standing wave** is $u(x,t) = \varphi(x)\psi(t)$ (fixed shape, varying amplitude).

1.1.1 Derivation of the wave equation

Model a string of length L, density ρ, tension τ, fixed at its ends, by N particles of mass ρh ($h = L/N$) at $x_n = nh$ joined by springs, with vertical displacements $y_n(t)$. Hooke's law for the vertical tension components gives

$$\rho h\, y_n''(t) = \frac{\tau}{h}\big(y_{n+1}(t) + y_{n-1}(t) - 2y_n(t)\big).$$

As $h \to 0$, $y_n(t) \to u(x,t)$, the right side over h tends to $\tau\,\partial_x^2 u$: the **wave equation**

$$\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \qquad c = \sqrt{\tau/\rho}. \tag{1.1}$$

(Rescaling $x \mapsto x/L$, $t \mapsto ct/L$ gives $c = 1$, $L = \pi$.) It is linear and invariant under translations in x and t.

1.1.2 Solution to the wave equation

d'Alembert's solution. With $\xi = x + t$, $\eta = x - t$, $v(\xi,\eta) = u(x,t)$, the equation becomes $\frac{\partial^2 v}{\partial\xi\partial\eta} = 0$, so $v = F(\xi) + G(\eta)$:

$$u(x,t) = F(x+t) + G(x-t),$$

two travelling waves. On $\mathbb{R}$ with $u(x,0) = f(x)$, $\partial_t u(x,0) = g(x)$:

$$u(x,t) = \frac{f(x+t) + f(x-t)}{2} + \frac{1}{2}\int_{x-t}^{x+t} g(y)\,dy. \tag{1.2}$$

Check [Supplement]: $F + G = f$, $F' - G' = g$ give $F = \frac{f}{2} + \frac{1}{2}\int_0^x g + C$, $G = \frac{f}{2} - \frac{1}{2}\int_0^x g - C$. On $[0,\pi]$ with $u(0,t) = u(\pi,t) = 0$, extend f, g oddly to $[-\pi,\pi]$ and 2π-periodically; then (1.2) satisfies the boundary conditions.

Superposition of standing waves. $u = \varphi(x)\psi(t)$ gives $\frac{\psi''}{\psi} = \frac{\varphi''}{\varphi} = \lambda$. Boundedness forces $\lambda = -m^2 \le 0$, so $\psi = A\cos mt + B\sin mt$, $\varphi = \tilde{A}\cos mx + \tilde{B}\sin mx$, and $\varphi(0) = \varphi(\pi) = 0$ forces $\varphi = \sin mx$, $m = 1, 2, \ldots$:

$$u_m(x,t) = (A_m\cos mt + B_m\sin mt)\sin mx,$$

the m-th **harmonic** ($m = 1$ **fundamental tone**, $m \ge 2$ **overtones** of frequency $m/2\pi$). By linearity

$$u(x,t) = \sum_{m=1}^{\infty}(A_m\cos mt + B_m\sin mt)\sin mx \tag{1.3}$$

is a candidate, and $u(x,0)=f(x)$ requires

$$f(x)=\sum_{m=1}^{\infty} A_m \sin mx. \tag{1.4}$$

Fourier's question: can every reasonable f on $[0,\pi]$ with $f(0)=f(\pi)=0$ be written as (1.4)? If so, $\int_0^\pi \sin mx \sin nx\,dx = \frac{\pi}{2}\delta_{mn}$ forces

$$A_m=\frac{2}{\pi}\int_0^\pi f(x)\sin mx\,dx,$$

and B_m is fixed by $\partial_t u(x,0)=g(x)=\sum mB_m \sin mx$. Extending f oddly, (1.4) is a special case of

$$F(x)=\sum_{m=-\infty}^{\infty} a_m e^{imx}, \qquad a_m=\frac{1}{2\pi}\int_{-\pi}^{\pi} F(x)e^{-imx}\,dx. \tag{1.5}$$

1.1.3 Example: the plucked string

Pluck at $p\in(0,\pi)$ to height h, zero initial velocity: $f(x)=\frac{xh}{p}$ on $[0,p]$, $f(x)=\frac{h(\pi-x)}{\pi-p}$ on $[p,\pi]$, $g=0$. Then $B_m=0$ and, by parts,

$$A_m=\frac{2h\sin mp}{m^2 p(\pi-p)}, \qquad u(x,t)=\sum_{m\geq 1} A_m \cos mt \sin mx.$$

As $\sum |A_m|<\infty$, the series converges uniformly and u is continuous. In general (1.3) need not be twice differentiable, so it solves the wave equation only in a generalised sense (compare d'Alembert).

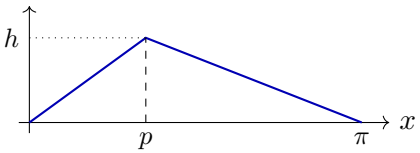


Figure. *The plucked string at $t=0$.*

1.2 The heat equation

Motivation. Steady heat flow in a disc, separated in polar coordinates, leads to Fourier series in the angle θ.

1.2.1 Derivation of the heat equation

For a plate $\mathbb{R}^2$ with temperature $u(x,y,t)$, specific heat σ and conductivity κ, the heat in a small square S is $\sigma\iint_S u$ and the flux across ∂S is $\kappa\int_{\partial S}\partial u/\partial n$ (Fourier's law). Equating the rate of change with the flux, dividing by the area and shrinking S:

$$\frac{\partial u}{\partial t}=\frac{\kappa}{\sigma}\Big(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\Big)=\frac{\kappa}{\sigma}\,\triangle u. \tag{1.6}$$

A **steady state** ($\partial u/\partial t=0$) satisfies the **Laplace equation** $\triangle u=0$; its solutions are **harmonic functions**.

1.2.2 Steady-state heat equation in the disc

Dirichlet problem. Find u with $\triangle u = 0$ in $D = \{x^2 + y^2 < 1\}$ and $u = f$ on the circle C. In polar coordinates

$$\triangle u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2}.$$

$u = F(r)G(\theta)$ gives $\frac{r^2F'' + rF'}{F} = -\frac{G''}{G} = \lambda$. G is 2π-periodic, so $\lambda = m^2$, $G = e^{\pm im\theta}$; $r^2F'' + rF' - m^2F = 0$ gives $F = r^{\pm m}$ $(m \geq 1)$ or $1, \log r$ $(m = 0)$, and boundedness at 0 leaves $F = r^{|m|}$:

$$u(r,\theta) = \sum_{m=-\infty}^{\infty} a_m r^{|m|} e^{im\theta}, \qquad f(\theta) = \sum a_m e^{im\theta}. \tag{1.7}$$

Inserting $a_m = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\varphi)e^{-im\varphi}d\varphi$ and summing formally,

$$u(r,\theta) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(\theta - \varphi) f(\varphi)\, d\varphi, \qquad P_r(\gamma) = \sum_{m\in\mathbb{Z}} r^{|m|}e^{im\gamma} = \frac{1 - r^2}{1 - 2r\cos\gamma + r^2},$$

the **Poisson kernel** (made rigorous in Chapter 2).

1.3 Selected exercises

Motivation. Book exercises justifying the separated solutions, the discrete-to-continuous limit, the polar Laplacian and the plucked-string coefficients.

Exercise 1.6 (Solutions of $f'' + c^2 f = 0$)

Let $c > 0$ and let f be a C^2 function on $\mathbb{R}$ with $f''(t) + c^2 f(t) = 0$. Prove that there are constants a, b with $f(t) = a\cos ct + b\sin ct$ for all t.

Solution.

Step 1. Let $g(t) = f(t)\cos ct - c^{-1}f'(t)\sin ct$ and $h(t) = f(t)\sin ct + c^{-1}f'(t)\cos ct$, both C^1. Using $f'' = -c^2 f$,

$$g'(t) = f'\cos ct - cf\sin ct - c^{-1}f''\sin ct - f'\cos ct = -cf\sin ct + cf\sin ct = 0,$$
$$h'(t) = f'\sin ct + cf\cos ct + c^{-1}f''\cos ct - f'\sin ct = cf\cos ct - cf\cos ct = 0.$$

Step 2. So $g \equiv a = f(0)$ and $h \equiv b = f'(0)/c$ are constant.

Step 3. Then, for every t,

$$a\cos ct + b\sin ct = f(\cos^2 ct + \sin^2 ct) + c^{-1}f'(-\sin ct\cos ct + \sin ct\cos ct) = f(t).$$

This justifies ψ and φ in §1.1.2. ∎

Exercise 1.8 (Second difference quotient)

Let F be C^2 on (a,b) and $x \in (a,b)$. Show that for $x + h \in (a,b)$

$$F(x+h) = F(x) + hF'(x) + \tfrac{h^2}{2}F''(x) + h^2\varphi(h), \qquad \varphi(h) \to 0 \ (h \to 0),$$

and deduce that $\big(F(x+h) + F(x-h) - 2F(x)\big)/h^2 \to F''(x)$.

Solution.

Step 1. $F(x+h)-F(x)=\int_x^{x+h}F'(y)\,dy$, $F'(y)=F'(x)+\int_x^y F''(s)\,ds$; write $F''(s)=F''(x)+(F''(s)-F''(x))$ and use $\int_x^{x+h}(y-x)\,dy=h^2/2$:

$$F(x+h)=F(x)+hF'(x)+\tfrac{h^2}{2}F''(x)+R(h),\qquad R(h)=\int_x^{x+h}\!\!\int_x^{y}\big(F''(s)-F''(x)\big)\,ds\,dy.$$

Step 2. Let $\omega(\delta)=\sup_{|s-x|\le\delta}|F''(s)-F''(x)|\to 0$ (F'' continuous). The inner integral is at most $|y-x|\,\omega(|h|)$, so $|R(h)|\le\frac{h^2}{2}\omega(|h|)$ for either sign of h.

Step 3. With $\varphi(h)=R(h)/h^2$, $|\varphi(h)|\le\frac12\omega(|h|)\to 0$.

Step 4. Adding the expansions at $\pm h$, the first-order terms cancel:

$$\frac{F(x+h)+F(x-h)-2F(x)}{h^2}=F''(x)+\varphi(h)+\varphi(-h)\to F''(x).$$

This is the limit giving (1.1). ∎

Exercise 1.9 (Missing harmonics of the plucked string)

For the plucked string of §1.1.3, derive $A_m=\frac{2h\sin mp}{m^2p(\pi-p)}$. For which p are the 2nd, 4th, 6th, … harmonics missing? For which p are the 3rd, 6th, 9th, … harmonics missing?

Solution.

Step 1. f is continuous, piecewise linear, $f(0)=f(\pi)=0$, so by parts

$$\int_0^\pi f(x)\sin mx\,dx=\frac1m\int_0^\pi f'(x)\cos mx\,dx=\frac1m\Big(\frac{h}{p}\int_0^p\cos mx\,dx-\frac{h}{\pi-p}\int_p^\pi\cos mx\,dx\Big).$$

Step 2. The integrals are $\pm\frac{\sin mp}{m}$, so the bracket is $\frac{h\sin mp}{m}(\frac1p+\frac{1}{\pi-p})=\frac{\pi h\sin mp}{m\,p(\pi-p)}$; multiply by $\frac{2}{\pi}$.

Step 3. $A_m=0$ iff $\sin mp=0$. All even harmonics vanish iff $\sin 2kp=0$ for all k; $k=1$ forces $p=\pi/2$, which conversely works. (Odd ones remain: $\sin(m\pi/2)=\pm1$.)

Step 4. Multiples of 3 vanish iff $\sin 3p=0$, i.e. $p=\pi/3$ or $2\pi/3$. ∎

Exercise 1.10 (Laplacian in polar coordinates)

Let u be C^2 on an open set of $\mathbb{R}^2\setminus\{0\}$ and $v(r,\theta)=u(r\cos\theta,r\sin\theta)$. Prove

$$u_{xx}+u_{yy}=v_{rr}+\frac1r v_r+\frac{1}{r^2}v_{\theta\theta},\qquad |u_x|^2+|u_y|^2=|v_r|^2+\frac{1}{r^2}|v_\theta|^2.$$

Solution. Write $c=\cos\theta$, $s=\sin\theta$; derivatives of u are taken at (rc,rs).

Step 1 (first derivatives). By the chain rule $v_r=cu_x+su_y$ and $v_\theta=-rsu_x+rcu_y$.

Step 2 (gradient identity). Hence $\frac1r v_\theta=-su_x+cu_y$ and

$$|v_r|^2+\tfrac{1}{r^2}|v_\theta|^2=|cu_x+su_y|^2+|-su_x+cu_y|^2=(c^2+s^2)(|u_x|^2+|u_y|^2),$$

the cross terms $\pm2cs\,\mathrm{Re}(u_x\bar u_y)$ cancelling.

Step 3 (second derivatives). Differentiating Step 1, with $u_{xy}=u_{yx}$,

$$v_{rr}=c^2u_{xx}+2csu_{xy}+s^2u_{yy},$$
$$v_{\theta\theta}=-rcu_x-rsu_y+r^2(s^2u_{xx}-2csu_{xy}+c^2u_{yy})=-rv_r+r^2(s^2u_{xx}-2csu_{xy}+c^2u_{yy}).$$

Step 4. So $v_{rr}+\frac1r v_r+\frac{1}{r^2}v_{\theta\theta}=(c^2+s^2)(u_{xx}+u_{yy})$. ∎

Exercise 1.11 (The Euler equation $r^2F'' + rF' - n^2F = 0$)

Let $n \in \mathbb{Z}$. Show that the twice differentiable solutions on $r > 0$ of $r^2F''(r)+rF'(r)-n^2F(r) = 0$ are exactly the linear combinations of r^n, r^{-n} if $n \neq 0$, and of $1, \log r$ if $n = 0$.

Solution. Only n^2 enters, so let $n \geq 0$.

Step 1 (these are solutions). $F = r^{\pm n}$: $r^2(\pm n)(\pm n - 1)r^{\pm n-2} + r(\pm n)r^{\pm n-1} - n^2r^{\pm n} = (n^2 \mp n \pm n - n^2)r^{\pm n} = 0$. $F = \log r$: $r^2(-r^{-2}) + r \cdot r^{-1} = 0$; constants trivially. Combinations by linearity.

Step 2 (reduction). For a solution F let $g = r^{-n}F$. From $F = r^n g$,

$$r^2F'' + rF' - n^2F = r^n\big[r^2g'' + 2nrg' + n(n-1)g + rg' + ng - n^2g\big] = r^{n+1}\big[rg'' + (2n+1)g'\big],$$

so $rg'' + (2n+1)g' = 0$ on $(0, \infty)$.

Step 3. Then $\big(r^{2n+1}g'\big)' = r^{2n}\big(rg'' + (2n+1)g'\big) = 0$, so $r^{2n+1}g' = C$.

Step 4. $n \geq 1$: $g = -\frac{C}{2n}r^{-2n} + D$, $F = Dr^n - \frac{C}{2n}r^{-n}$. $n = 0$: $F = g = C\log r + D$. This is the classification used in §1.2.2 (and in Theorem 5.7). ■

Chapter 2. Basic Properties of Fourier Series

2.1 Examples and formulation of the problem

Motivation. For Riemann integrable f on the circle: in what sense does its Fourier series converge back to f?

2.1.1 Main definitions and some examples

A bounded f on $[a,b]$ is (Riemann) **integrable** if $\sup_P L(P,f) = \inf_P U(P,f)$; piecewise continuous functions are integrable (Appendix), and complex $f = u + iv$ is integrable if u, v are. A function on the circle is a 2π-periodic function on $\mathbb{R}$, equivalently $F(e^{i\theta}) = f(\theta)$ on $\{|z| = 1\}$; it is integrable on the circle if integrable on one (hence every) period.

Definition (Fourier coefficients and series)

For f integrable on $[a,b]$, $L = b - a$, the n-th **Fourier coefficient** is

$$\hat{f}(n) = \frac{1}{L}\int_a^b f(x)e^{-2\pi inx/L}\,dx, \qquad n \in \mathbb{Z},$$

and the **Fourier series** is $f(x) \sim \sum_{n=-\infty}^{\infty} \hat{f}(n)e^{2\pi inx/L}$. On the circle ($L = 2\pi$): $\hat{f}(n) = a_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\theta)e^{-in\theta}\,d\theta$. The **$N$-th partial sum** is $S_N(f)(\theta) = \sum_{|n|\le N} \hat{f}(n)e^{in\theta}$. A **trigonometric polynomial** is $\sum_{|n|\le N} c_n e^{in\theta}$, of degree N if $c_{\pm N} \neq 0$.

Basic question. In what sense does $S_N(f) \to f$ as $N \to \infty$?

Example (Sawtooth)

For $f(\theta) = \frac{\pi - \theta}{2}$ on $0 < \theta < 2\pi$ (extended periodically, $f(0) = 0$): $\hat{f}(0) = 0$ and, integrating by parts, $\hat{f}(n) = \frac{1}{2in}$ for $n \neq 0$. So $f(\theta) \sim \sum_{n\neq 0} \frac{e^{in\theta}}{2in} = \sum_{n\ge 1} \frac{\sin n\theta}{n}$.

Example

For $f(\theta) = |\theta|$ on $[-\pi,\pi]$: $\hat{f}(0) = \frac{\pi}{2}$ and $\hat{f}(n) = \frac{-1+(-1)^n}{\pi n^2}$ for $n \neq 0$, so $f(\theta) \sim \frac{\pi}{2} - \frac{4}{\pi}\sum_{n \text{ odd}\ge 1} \frac{\cos n\theta}{n^2}$.

Example

For $f(\theta) = \theta$ on $(-\pi,\pi)$: $\hat{f}(0) = 0$, $\hat{f}(n) = \frac{(-1)^{n+1}}{in}$, so $f \sim 2\sum_{n\ge 1}(-1)^{n+1}\frac{\sin n\theta}{n}$.

Example (Dirichlet kernel)

$D_N(\theta) = \sum_{|n|\le N} e^{in\theta} = \frac{\sin((N+\frac{1}{2})\theta)}{\sin(\theta/2)}$, so $\hat{D}_N(n) = 1$ for $|n| \le N$, 0 otherwise. *Proof:* with $\omega = e^{i\theta}$, $\sum_0^N \omega^n = \frac{1-\omega^{N+1}}{1-\omega}$, $\sum_{-N}^{-1} \omega^n = \frac{\omega^{-N}-1}{1-\omega}$, so $D_N = \frac{\omega^{-N}-\omega^{N+1}}{1-\omega}$; multiply through by $\omega^{-1/2}$.

Example (Poisson kernel)

For $0 \le r < 1$, $P_r(\theta) = \sum_{n\in\mathbb{Z}} r^{|n|} e^{in\theta} = \frac{1-r^2}{1-2r\cos\theta+r^2}$ (sum two geometric series).

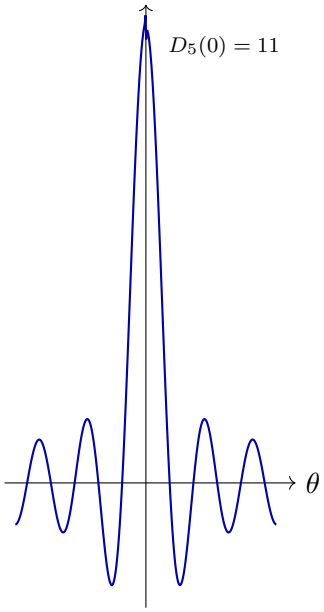


Figure 1: The Dirichlet kernel D_5; $\int |D_N|$ grows like $\log N$.

2.2 Uniqueness of Fourier series

Motivation. Vanishing Fourier coefficients force $f = 0$ at points of continuity, giving uniform convergence when $\sum |\hat{f}(n)| < \infty$.

Theorem 2.1

Let f be integrable on the circle with $\hat{f}(n) = 0$ for all n. Then $f(\theta_0) = 0$ whenever f is continuous at θ_0.

Proof.

Step 1 (setup). Let f be real. Translating (which multiplies $\hat{f}(n)$ by a unimodular factor), take f on $[-\pi, \pi]$, $\theta_0 = 0$, and suppose $f(0) > 0$. Then $\int_{-\pi}^{\pi} fp = 0$ for every trigonometric polynomial p; we construct p_k with $\int fp_k \to \infty$.

Step 2 (the polynomials). Take $0 < \delta \le \pi/2$ with $f > f(0)/2$ on $|\theta| < \delta$; $p(\theta) = \epsilon + \cos\theta$ with $\epsilon > 0$ so small that $|p| < 1 - \epsilon/2$ on $\delta \le |\theta| \le \pi$; $0 < \eta < \delta$ with $p \ge 1 + \epsilon/2$ on $|\theta| < \eta$; $p_k = p^k$; and $|f| \le B$.

Step 3 (estimates). On $\delta \le |\theta| \le \pi$: $\left|\int fp_k\right| \le 2\pi B(1-\epsilon/2)^k \to 0$. On $\eta \le |\theta| < \delta$: $p \ge 0$ and $f > 0$, so $\int fp_k \ge 0$. On $|\theta| < \eta$: $\int fp_k \ge 2\eta \frac{f(0)}{2}(1+\epsilon/2)^k \to \infty$. So $\int fp_k \to \infty$, a contradiction; thus $f(0) \le 0$, and by $-f$, $f(0) = 0$.

Step 4 (complex f). $u = \frac{f+\bar{f}}{2}$, $v = \frac{f-\bar{f}}{2i}$ have vanishing coefficients ($\hat{\bar{f}}(n) = \overline{\hat{f}(-n)}$) and are continuous at θ_0. ∎

Corollary 2.2

If f is continuous on the circle and $\hat{f}(n) = 0$ for all n, then $f = 0$.

Proof. Theorem 2.1 at every point. ■

Corollary 2.3

If f is continuous on the circle and $\sum_n |\hat{f}(n)| < \infty$, then $S_N(f) \to f$ uniformly.

Proof. By the M-test $S_N(f) \to g = \sum \hat{f}(n)e^{in\theta}$ uniformly, g continuous, and termwise integration gives $\hat{g}(n) = \hat{f}(n)$; Corollary 2.2 gives $f = g$. ■

Corollary 2.4

If f is twice continuously differentiable on the circle, then $\hat{f}(n) = O(1/|n|^2)$, so the Fourier series of f converges absolutely and uniformly to f.

Proof. Integrating by parts twice, $2\pi\hat{f}(n) = \frac{1}{in}\int f' e^{-in\theta}d\theta = \frac{-1}{n^2}\int f'' e^{-in\theta}d\theta$, so $|n|^2|\hat{f}(n)| \leq \frac{1}{2\pi}\int |f''|$; apply Corollary 2.3. ■

Remark (on the hypotheses)

$f \in C^1$ suffices (book exercise), and $f \in C^k$ gives $\hat{f}(n) = o(1/|n|^k)$: smoother f, faster decay.

2.3 Convolutions

Motivation. Partial sums and means are convolutions $f * K$, which multiply Fourier coefficients and smooth.

Definition (Convolution)

For 2π-periodic integrable f, g, $(f * g)(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)g(x-y)\,dy$. E.g. $S_N(f)(x) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(y)\sum_{|n|\leq N} e^{in(x-y)}dy = (f * D_N)(x)$.

Proposition 3.1

Let f, g, h be 2π-periodic integrable functions. Then:
(i) $f * (g + h) = (f * g) + (f * h)$.
(ii) $(cf) * g = c(f * g) = f * (cg)$ for $c \in \mathbb{C}$.
(iii) $f * g = g * f$.
(iv) $(f * g) * h = f * (g * h)$.
(v) $f * g$ is continuous.
(vi) $\widehat{f * g}(n) = \hat{f}(n)\hat{g}(n)$.

Lemma 3.2

Let f be integrable on the circle with $|f| \leq B$. There are continuous functions f_k on the circle with $\sup |f_k| \leq B$ and $\int_{-\pi}^{\pi} |f(x) - f_k(x)|\,dx \to 0$.

Proof. Take f real. Given $\epsilon > 0$ take a partition $-\pi = x_0 < \cdots < x_N = \pi$ with $U(P,f) - L(P,f) < \epsilon$, and $f^* = \sup_{[x_{j-1},x_j]} f$ on $[x_{j-1}, x_j)$; then $f^* \geq f$ and $\int (f^* - f) \leq U - L < \epsilon$. On each $[x_j - \delta, x_j + \delta]$ replace f^* by the segment joining $f^*(x_j - \delta)$ to $f^*(x_j + \delta)$ (periodically at $\pm\pi$). The result $\tilde{f}$ is continuous, $|\tilde{f}| \leq B$, and differs from f^* on length $\leq 2N\delta$ by at most $2B$; for small δ, $\int |f - \tilde{f}| < 2\epsilon$. Take $\epsilon = 1/k$. ∎

Proof of Proposition 3.1.

Step 1 (continuous f, g). (i), (ii): linearity. (iii): substitute $y \mapsto x - y$, use periodicity. (iv): interchange integrals (continuous integrand) and substitute. (v): by uniform continuity, $|f * g(x_1) - f * g(x_2)| \leq \sup_y |f(x_1 - y) - f(x_2 - y)| \sup |g| \to 0$ (using (iii)). (vi):

$$\begin{aligned}\widehat{f * g}(n) &= \frac{1}{2\pi} \int \Big(\frac{1}{2\pi} \int f(y) g(x - y)\, dy \Big) e^{-inx}\, dx \\ &= \frac{1}{2\pi} \int f(y) e^{-iny} \Big(\frac{1}{2\pi} \int g(x - y) e^{-in(x-y)}\, dx \Big) dy = \hat{f}(n)\hat{g}(n).\end{aligned}$$

Step 2 (approximation). For integrable f, g take continuous $f_k \to f$, $g_k \to g$ (Lemma 3.2, bound B). Then $f * g - f_k * g_k = (f - f_k) * g + f_k * (g - g_k)$, $\sup_x |(f - f_k) * g| \leq \frac{B}{2\pi} \int |f - f_k| \to 0$, and likewise the second term. So $f_k * g_k \to f * g$ uniformly, giving (v), and (iii), (iv) in the limit. For (vi), $|\hat{f}(n) - \hat{f}_k(n)| \leq \frac{1}{2\pi} \int |f - f_k| \to 0$ and $\widehat{f_k * g_k}(n) \to \widehat{f * g}(n)$. ∎

2.4 Good kernels

Motivation. Kernels of integral 1, bounded L^1 norm and concentrating mass reproduce f; D_N fails the middle condition, Fejér and Poisson pass.

Definition (Good kernels)

A family $\{K_n\}$ of integrable 2π-periodic functions is a family of **good kernels** if (i) $\frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(x)\, dx = 1$ for all n;
(ii) $\int_{-\pi}^{\pi} |K_n(x)|\, dx \leq M$ for all n;
(iii) for every $\delta > 0$, $\int_{\delta \leq |x| \leq \pi} |K_n(x)|\, dx \to 0$ as $n \to \infty$. (If $K_n \geq 0$, (ii) follows from (i).)
Likewise for continuous parameters.

Theorem 4.1

If $\{K_n\}$ is a family of good kernels and f is integrable on the circle, then $(f * K_n)(x) \to f(x)$ at every point of continuity x of f. If f is continuous, the convergence is uniform.

Proof. Let f be continuous at x, $\epsilon > 0$, and $|f(x - y) - f(x)| < \epsilon$ for $|y| < \delta$. By (i),

$$(f * K_n)(x) - f(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(y)[f(x - y) - f(x)]\, dy,$$

so with $|f| \leq B$,

$$\begin{aligned}|(f * K_n)(x) - f(x)| &\leq \frac{1}{2\pi} \int_{|y|<\delta} |K_n||f(x - y) - f(x)| + \frac{1}{2\pi} \int_{\delta \leq |y| \leq \pi} |K_n||f(x - y) - f(x)| \\ &\leq \frac{\epsilon M}{2\pi} + \frac{2B}{2\pi} \int_{\delta \leq |y| \leq \pi} |K_n(y)| dy,\end{aligned}$$

by (ii), and the last term is $< \epsilon$ for large n by (iii). For continuous f, δ is uniform in x. ∎

$\{D_N\}$ satisfies (i) but not (ii): [Supplement] $|D_N(\theta)| \geq \frac{2}{|\theta|}|\sin((N+\frac{1}{2})\theta)|$, and summing over the $2N$ subintervals where $|\sin((N+\frac{1}{2})\theta)| \geq \frac{1}{2}$ gives $\int |D_N| \geq c \log N$. This is the root of the pointwise difficulties.

2.5 Cesàro and Abel summability: applications to Fourier series

Motivation. Cesàro and Abel means replace D_N by the good Fejér and Poisson kernels: trigonometric polynomials are dense, and the Poisson integral solves Dirichlet's problem.

2.5.1 Cesàro means and summation

For $s_n = \sum_{k=0}^{n} c_k$, the **Cesàro mean** is $\sigma_N = \frac{s_0+\cdots+s_{N-1}}{N}$, and $\sum c_k$ is **Cesàro summable** to σ if $\sigma_N \to \sigma$. Convergent series are Cesàro summable to their sum; $1-1+1-\ldots$ is Cesàro summable to $\frac{1}{2}$.

2.5.2 Fejér's theorem

$\sigma_N(f)(x) = \frac{S_0(f)(x)+\cdots+S_{N-1}(f)(x)}{N} = (f * F_N)(x)$, where $F_N = \frac{D_0+\cdots+D_{N-1}}{N}$ is the **Fejér kernel**.

Lemma 5.1

$F_N(x) = \frac{1}{N}\frac{\sin^2(Nx/2)}{\sin^2(x/2)}$, and $\{F_N\}$ is a family of good kernels.

Proof. With $\omega = e^{ix}$, $D_n = \frac{\omega^{-n}-\omega^{n+1}}{1-\omega}$, so

$$NF_N = \frac{1}{1-\omega}\Big(\sum_{n=0}^{N-1}\omega^{-n} - \omega\sum_{n=0}^{N-1}\omega^{n}\Big) = \frac{1}{1-\omega}\Big(\frac{1-\omega^{-N}}{1-\omega^{-1}} - \frac{\omega(1-\omega^N)}{1-\omega}\Big) = \frac{\omega^{-N}-2+\omega^N}{(1-\omega)(1-\omega^{-1})},$$

with $\omega^{-N}-2+\omega^N = -(2\sin(Nx/2))^2$ and $(1-\omega)(1-\omega^{-1}) = (2\sin(x/2))^2$ (signs match, $NF_N \geq 0$). Good: $F_N \geq 0$; $\frac{1}{2\pi}\int F_N = 1$ (average of D_n's); on $\delta \leq |x| \leq \pi$, $\sin^2(x/2) \geq c_\delta > 0$, so $F_N \leq \frac{1}{Nc_\delta}$. ■

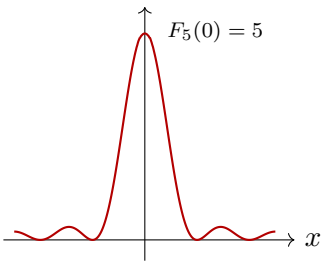


Figure 2: The Fejér kernel F_5: non-negative, integral 1, concentrating at 0.

Theorem 5.2 (Fejér)

If f is integrable on the circle, its Fourier series is Cesàro summable to f at every point of continuity. If f is continuous, the Fourier series is uniformly Cesàro summable to f.

Proof. $\sigma_N(f) = f * F_N$ with $\{F_N\}$ good (Lemma 5.1); apply Theorem 4.1. ■

Corollary 5.3

If f is integrable on the circle and $\hat{f}(n) = 0$ for all n, then $f = 0$ at all points of continuity of f.

Proof. All partial sums, hence Cesàro means, vanish; apply Theorem 5.2. ■

Corollary 5.4

Continuous functions on the circle can be uniformly approximated by trigonometric polynomials.

Proof. $\sigma_N(f)$ is a trigonometric polynomial, and $\sigma_N(f) \to f$ uniformly (Theorem 5.2). ■

2.5.3 Abel means and summation

$\sum c_k$ is **Abel summable** to s if $A(r) = \sum c_k r^k$ converges for $0 \le r < 1$ and $A(r) \to s$ as $r \to 1^-$. Cesàro summable implies Abel summable to the same limit (not conversely: $\sum (-1)^k (k+1)$ is Abel summable to $\frac{1}{4}$ only). For $f \sim \sum a_n e^{in\theta}$ the **Abel means** are $A_r(f)(\theta) = \sum r^{|n|} a_n e^{in\theta} = (f * P_r)(\theta)$ (interchange, as $|a_n| \le \sup |f|$).

2.5.4 The Poisson kernel and Dirichlet's problem in the unit disc

Lemma 5.5

If $0 \le r < 1$, then $P_r(\theta) = \frac{1-r^2}{1-2r\cos\theta+r^2}$, and $\{P_r\}_{0\le r<1}$ is a family of good kernels as $r \to 1^-$.

Proof. With $\omega = re^{i\theta}$, $P_r = \frac{1}{1-\omega} + \frac{\bar{\omega}}{1-\bar{\omega}} = \frac{1-|\omega|^2}{|1-\omega|^2}$. Good: $P_r > 0$; $\frac{1}{2\pi}\int P_r = 1$ termwise; for $\frac{1}{2} \le r < 1$, $\delta \le |\theta| \le \pi$, $1 - 2r\cos\theta + r^2 = (1-r)^2 + 2r(1-\cos\theta) \ge c_\delta > 0$, so $P_r \le \frac{1-r^2}{c_\delta} \to 0$. ■

Theorem 5.6

The Fourier series of an integrable function on the circle is Abel summable to f at every point of continuity. If f is continuous, it is uniformly Abel summable to f.

Proof. $A_r(f) = f * P_r$ with $\{P_r\}$ good; apply Theorem 4.1. ■

Theorem 5.7

Let f be integrable on the unit circle and $u(r,\theta) = (f * P_r)(\theta)$ for $0 \le r < 1$. Then:
(i) u is C^2 in the unit disc and $\triangle u = 0$.
(ii) If θ is a point of continuity of f, then $u(r,\theta) \to f(\theta)$ as $r \to 1$. If f is continuous, $u(r,\theta) \to f(\theta)$ uniformly as $r \to 1$.
(iii) If f is continuous, then u is the unique solution of $\triangle u = 0$ in the disc that satisfies (ii).

Proof.

(i) $u = \sum a_n r^{|n|} e^{in\theta}$, $|a_n| \le B$. For $r \le \rho < 1$ the series and its first and second r, θ derivatives (factors $|n|^2$) converge uniformly, so u is C^2 (indeed C^∞), differentiable termwise, and each $r^{|n|}e^{in\theta}$ is harmonic (§1.2.2).

(ii) Theorem 5.6.

(iii) [Supplement] (Book exercise.) For another solution v, $w = u - v$ is harmonic with $w \to 0$ uniformly as $r \to 1$, so extends by 0 continuously to the closed disc. *Maximum principle:* $w_\epsilon = w + \epsilon(x^2 + y^2)$ has $\triangle w_\epsilon = 4\epsilon > 0$, so no interior maximum (there $\triangle w_\epsilon \le 0$); hence $w_\epsilon \le \epsilon$. Let $\epsilon \to 0$: $w \le 0$; by $-w$, $w = 0$ (complex w: real and imaginary parts). ■

2.6 Selected exercises

Motivation. Cesàro implies Abel, a Tauberian converse, Abel means at a jump, and why Theorem 5.7(iii) needs uniformity.

Exercise 2.13 (Convergent ⇒ Abel, Cesàro ⇒ Abel)

Let $\sum_{k\ge0} c_k$ have partial sums $s_n = \sum_{k=0}^{n} c_k$ and Cesàro means $\sigma_N = \frac{1}{N}(s_0 + \cdots + s_{N-1})$.
(a) If $s_n \to s$, show that $\sum c_k$ is Abel summable to s.
(b) If $\sigma_N \to \sigma$, show that $\sum c_k$ is Abel summable to σ.

Solution.

(a)

Step 1. $|s_n| \le M$, so $|c_k| \le 2M$ and $A(r)$ converges absolutely. With $s_{-1} = 0$,

$$\sum_{k=0}^{N} c_k r^k = \sum_{k=0}^{N} (s_k - s_{k-1}) r^k = (1-r) \sum_{k=0}^{N-1} s_k r^k + s_N r^N.$$

As $s_N r^N \to 0$, $A(r) = (1-r)\sum_{k\ge0} s_k r^k$.

Step 2. $(1-r)\sum r^k = 1$ gives $A(r) - s = (1-r)\sum_k (s_k - s) r^k$. With $|s_k - s| < \epsilon$ for $k \ge K$,

$$|A(r) - s| \le (1-r)\, 2MK + \epsilon(1-r) \sum_{k\ge K} r^k \le 2MK(1-r) + \epsilon,$$

so $\limsup_{r\to1^-} |A(r) - s| \le \epsilon$.

(b)

Step 1. Let $t_N = N\sigma_N = s_0 + \cdots + s_{N-1}$ ($t_0 = 0$), $|\sigma_N| \le M$. Then $|t_N| \le MN$, $|s_n| = |t_{n+1} - t_n| \le M(2n+1)$, $|c_n| \le 4M(n+1)$, so $A(r)$ converges absolutely, $s_N r^N \to 0$, and $A(r) = (1-r)\sum_{k\ge0} s_k r^k$ as in (a).

Step 2. Likewise, from $s_k = t_{k+1} - t_k$,

$$\sum_{k=0}^{N} s_k r^k = t_{N+1} r^N + (1-r) \sum_{k=1}^{N} t_k r^{k-1},$$

and $t_{N+1} r^N \to 0$, so

$$A(r) = (1-r)^2 \sum_{k\ge1} k\sigma_k r^{k-1}.$$

Step 3. $(1-r)^2 \sum_{k\ge1} k r^{k-1} = 1$, so $A(r) - \sigma = (1-r)^2 \sum_k k(\sigma_k - \sigma) r^{k-1}$; with $|\sigma_k - \sigma| < \epsilon$ for $k \ge K$,

$$|A(r) - \sigma| \le (1-r)^2\, 2MK^2 + \epsilon(1-r)^2 \sum_{k\ge K} k r^{k-1} \le 2MK^2(1-r)^2 + \epsilon,$$

and conclude as in (a). This gives the chain convergent ⇒ Cesàro ⇒ Abel of §2.5.3. ■

Exercise 2.14 (A Tauberian theorem)

If $\sum_{k\ge 0} c_k$ is Cesàro summable to σ and $kc_k \to 0$, prove that $\sum c_k$ converges to σ.

Solution.

Step 1. With the notation of Exercise 2.13,

$$s_{N-1} - \sigma_N = \frac{1}{N}\sum_{j=0}^{N-1}(s_{N-1} - s_j) = \frac{1}{N}\sum_{j=0}^{N-1}\sum_{k=j+1}^{N-1} c_k = \frac{1}{N}\sum_{k=1}^{N-1} k c_k,$$

as c_k occurs once for each $j < k$.

Step 2. $b_k = kc_k \to 0$, $|b_k| \le B$; with $|b_k| < \epsilon$ for $k \ge K$, $|\frac{1}{N}\sum_{k=1}^{N-1} b_k| \le \frac{KB}{N} + \epsilon < 2\epsilon$ for large N.

Step 3. So $s_{N-1} = \sigma_N + (s_{N-1} - \sigma_N) \to \sigma$. ∎

Exercise 2.17 (Abel means at a jump)

Let f be integrable on the circle with a jump discontinuity at θ: the limits $f(\theta^{\pm}) = \lim_{h\to 0^+} f(\theta \pm h)$ exist. Prove that $A_r(f)(\theta) \to \frac{1}{2}(f(\theta^+) + f(\theta^-))$ as $r \to 1^-$.

Solution.

Step 1. P_r is even, so $\frac{1}{2\pi}\int_{-\pi}^{0} P_r = \frac{1}{2\pi}\int_0^{\pi} P_r = \frac{1}{2}$ by Lemma 5.5.

Step 2. $A_r(f)(\theta) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_r(y) f(\theta - y)\, dy$, with $\theta - y < \theta$ for $y > 0$. By Step 1,

$$A_r(f)(\theta) - \frac{f(\theta^+) + f(\theta^-)}{2} = \frac{1}{2\pi}\int_0^{\pi} P_r(y)\big[f(\theta - y) - f(\theta^-)\big]dy + \frac{1}{2\pi}\int_{-\pi}^{0} P_r(y)\big[f(\theta - y) - f(\theta^+)\big]dy.$$

Step 3. With $|f| \le B$, choose δ with $|f(\theta - y) - f(\theta^-)| < \epsilon$ for $0 < y < \delta$ and $|f(\theta - y) - f(\theta^+)| < \epsilon$ for $-\delta < y < 0$. As $P_r > 0$,

$$\Big|A_r(f)(\theta) - \frac{f(\theta^+) + f(\theta^-)}{2}\Big| \le \frac{\epsilon}{2\pi}\int_{-\pi}^{\pi} P_r + \frac{2B}{2\pi}\int_{\delta \le |y| \le \pi} P_r(y)\, dy = \epsilon + o(1) \quad (r \to 1^-),$$

by property (iii) of $\{P_r\}$. ∎

Exercise 2.18 (Pointwise boundary values do not determine u)

Let $u(r,\theta) = \partial_\theta P_r(\theta)$ for $0 \le r < 1$. Show that $\triangle u = 0$ in the disc and $u(r,\theta) \to 0$ as $r \to 1$ for each θ, but $u \not\equiv 0$.

Solution.

Step 1 (P is harmonic). With $z = x + iy = re^{i\theta}$,

$$\frac{1+z}{1-z} = \frac{(1+z)(1-\bar z)}{|1-z|^2} = \frac{1 - |z|^2 + (z - \bar z)}{|1-z|^2}, \qquad \text{so} \qquad P_r(\theta) = \operatorname{Re}\frac{1+z}{1-z}.$$

The real part U of a holomorphic $G = U + iV$ is C^∞ and harmonic: $U_x = V_y$, $U_y = -V_x$ give $U_{xx} + U_{yy} = V_{yx} - V_{xy} = 0$.

Step 2 (u is harmonic). $\partial_\theta = x\partial_y - y\partial_x$, so $u = xP_y - yP_x$, and $\triangle(xg) = x\triangle g + 2g_x$ gives

$$\triangle u = x\,\triangle P_y + 2P_{xy} - y\,\triangle P_x - 2P_{xy} = x\,\partial_y(\triangle P) - y\,\partial_x(\triangle P) = 0.$$

Step 3 (boundary limits). Differentiating the closed form,

$$u(r,\theta) = -\frac{2r(1-r^2)\sin\theta}{(1-2r\cos\theta+r^2)^2}.$$

For $\theta \in 2\pi\mathbb{Z}$, $u(r,\theta) = 0$; otherwise the denominator tends to $(2-2\cos\theta)^2 > 0$ and the numerator to 0.

Step 4. But $u(\tfrac{1}{2}, \tfrac{\pi}{2}) = -\frac{3/4}{(5/4)^2} = -\frac{12}{25} \neq 0$. So the convergence is not uniform, else Theorem 5.7(iii) with $f = 0$ would force $u \equiv 0$. ∎

Chapter 3. Convergence of Fourier Series

3.1 Mean-square convergence of Fourier series

Motivation. $S_N(f)$ is the orthogonal projection onto trigonometric polynomials of degree $\le N$, which are dense, so every integrable f is the mean-square limit of its Fourier series.

Theorem 1.1 (Mean-square convergence)

If f is integrable on the circle, then

$$\frac{1}{2\pi}\int_0^{2\pi} |f(\theta) - S_N(f)(\theta)|^2\, d\theta \to 0 \quad \text{as } N \to \infty.$$

3.1.1 Vector spaces and inner products

An **inner product** on a complex vector space V is linear in the first variable, $(Y, X) = \overline{(X, Y)}$, $(X, X) \ge 0$; **strictly positive definite** if $(X, X) = 0$ only for $X = 0$. $\|X\| = (X, X)^{1/2}$. For any inner product:

- **Pythagoras:** if $X \perp Y$ ($(X, Y) = 0$), then $\|X + Y\|^2 = \|X\|^2 + \|Y\|^2$.
- **Cauchy–Schwarz:** $|(X, Y)| \le \|X\|\|Y\|$.
- **Triangle inequality:** $\|X + Y\| \le \|X\| + \|Y\|$.

Proofs [Supplement]. Pythagoras: expand. Cauchy–Schwarz: if $\|Y\| = 0$, $0 \le \|X - tY\|^2 = \|X\|^2 - 2t\,\mathrm{Re}(X, Y)$ for all real t forces $\mathrm{Re}(X, Y) = 0$, likewise $\mathrm{Im}(X, Y) = 0$. Otherwise $c = (X, Y)/\|Y\|^2$ gives $X - cY \perp Y$ and $\|X\|^2 = \|X - cY\|^2 + |c|^2\|Y\|^2 \ge |(X, Y)|^2/\|Y\|^2$. Triangle: $\|X + Y\|^2 = \|X\|^2 + 2\,\mathrm{Re}(X, Y) + \|Y\|^2 \le (\|X\| + \|Y\|)^2$.

Examples. (1) $\mathbb{R}^d$, $\mathbb{C}^d$ with the usual inner products.

(2) $\ell^2(\mathbb{Z})$: sequences $(a_n)_{n\in\mathbb{Z}}$ with $\sum |a_n|^2 < \infty$, $(A, B) = \sum a_n \bar{b}_n$ (finite by Cauchy–Schwarz). A vector space ($|a + b|^2 \le 2|a|^2 + 2|b|^2$), and **complete**.

(3) $\mathcal{R}$: integrable functions on $[0, 2\pi]$, $(f, g) = \frac{1}{2\pi}\int_0^{2\pi} f\bar{g}$. Not strictly positive definite ($f = 0$ off one point has $\|f\| = 0$) and not complete; its completion is Lebesgue's L^2.

3.1.2 Proof of mean-square convergence

$e_n(\theta) = e^{in\theta}$ is orthonormal in $\mathcal{R}$: $(e_n, e_m) = \frac{1}{2\pi}\int e^{i(n-m)\theta} d\theta = \delta_{nm}$. With $a_n = \hat{f}(n) = (f, e_n)$, $S_N(f) = \sum_{|n|\le N} a_n e_n$.

Key orthogonality. $(f - S_N(f), e_n) = a_n - a_n = 0$ for $|n| \le N$, so $f - S_N(f) \perp \sum_{|n|\le N} b_n e_n$, and by Pythagoras

$$\|f\|^2 = \|f - S_N(f)\|^2 + \sum_{|n|\le N} |a_n|^2. \tag{3.1}$$

Lemma 1.2 (Best approximation)

If f is integrable on the circle with Fourier coefficients a_n, then $\|f - S_N(f)\| \le \|f - \sum_{|n|\le N} c_n e_n\|$ for all complex c_n, with equality iff $c_n = a_n$ for all $|n| \le N$.

Proof. $f-\sum c_n e_n = (f-S_N(f))+\sum b_n e_n$, $b_n = a_n - c_n$, orthogonal summands, so $\|f-\sum c_n e_n\|^2 = \|f-S_N(f)\|^2 + \sum |b_n|^2$. ∎

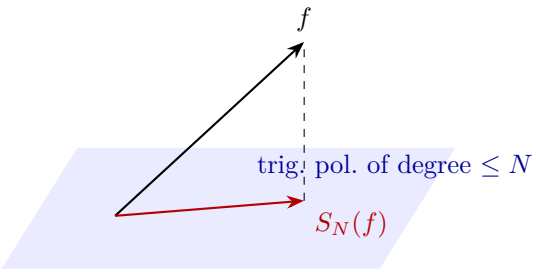


Figure. *$S_N(f)$ is the orthogonal projection of f, the best approximation (Lemma 1.2).*

Proof of Theorem 1.1.

Step 1 (continuous f). Corollary 5.4 gives a trigonometric polynomial P of degree M with $|f-P| < \epsilon$, so $\|f-P\| < \epsilon$, and by Lemma 1.2 $\|f-S_N(f)\| < \epsilon$ for $N \geq M$.

Step 2 (integrable f). Let $|f| \leq B$. Lemma 3.2 gives continuous g, $|g| \leq B$, $\int |f-g| < \epsilon^2$, so $\|f-g\|^2 \leq \frac{2B}{2\pi}\int |f-g| < C\epsilon^2$. With $\|g-P\| < \epsilon$ (Step 1), $\|f-P\| < C'\epsilon$, and Lemma 1.2 gives $\|f-S_N(f)\| < C'\epsilon$ for large N. ∎

Theorem 1.3 (Parseval)

Let $f \sim \sum a_n e^{in\theta}$ be integrable on the circle. Then:
(i) Mean-square convergence: $\frac{1}{2\pi}\int_0^{2\pi} |f(\theta) - S_N(f)(\theta)|^2 d\theta \to 0$.
(ii) Parseval's identity: $\sum_{n=-\infty}^{\infty} |a_n|^2 = \frac{1}{2\pi}\int_0^{2\pi} |f(\theta)|^2 d\theta$.
(iii) The map $f \mapsto (a_n)_{n\in\mathbb{Z}}$ is "unitary": $(f,g) = \sum a_n \bar{b}_n$ for $g \sim \sum b_n e^{in\theta}$ (Lemma 1.5).

Proof. (i) is Theorem 1.1.

(ii) Let $N \to \infty$ in (3.1).

(iii) is Lemma 1.5 below. ∎

Bessel's inequality $\sum |a_n|^2 \leq \|f\|^2$ follows from (3.1).

Theorem 1.4 (Riemann–Lebesgue lemma)

If f is integrable on the circle, then $\hat{f}(n) \to 0$ as $|n| \to \infty$. Equivalently, $\int_0^{2\pi} f(\theta)\sin N\theta\, d\theta \to 0$ and $\int_0^{2\pi} f(\theta)\cos N\theta\, d\theta \to 0$ as $N \to \infty$.

Proof. Parseval gives $\sum |\hat{f}(n)|^2 < \infty$; for the reformulation write $\cos N\theta$, $\sin N\theta$ via $e^{\pm iN\theta}$. ∎

Lemma 1.5

If $F \sim \sum a_n e^{in\theta}$ and $G \sim \sum b_n e^{in\theta}$ are integrable on the circle, then $\frac{1}{2\pi}\int_0^{2\pi} F(\theta)\overline{G(\theta)}\, d\theta = \sum_{n=-\infty}^{\infty} a_n \bar{b}_n$, the series converging absolutely.

Proof. [Supplement] Absolute convergence: $|a_n\bar{b}_n| \leq \frac{1}{2}(|a_n|^2 + |b_n|^2)$ and Parseval. Polarization: $(F,G) = \frac{1}{4}\sum_{k=0}^{3} i^k \|F + i^k G\|^2$ and likewise $\sum a_n\bar{b}_n = \frac{1}{4}\sum_k i^k \sum_n |a_n + i^k b_n|^2$; $F + i^k G$ has coefficients $a_n + i^k b_n$, so Parseval applies termwise. ∎

3.2 Return to pointwise convergence

Motivation. Differentiability at a point gives convergence there, and convergence is local; continuity alone does not suffice.

3.2.1 A local result

Theorem 2.1

If f is integrable on the circle and differentiable at θ_0, then $S_N(f)(\theta_0) \to f(\theta_0)$.

Proof. Define, for $|t| \le \pi$,

$$F(t) = \begin{cases} \dfrac{f(\theta_0 - t) - f(\theta_0)}{t} & t \neq 0, \\ -f'(\theta_0) & t = 0. \end{cases}$$

F is bounded near 0 and integrable on $[-\pi, \pi]$ (Proposition 1.4). As $\frac{1}{2\pi}\int D_N = 1$,

$$S_N(f)(\theta_0) - f(\theta_0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} (f(\theta_0 - t) - f(\theta_0)) D_N(t)\, dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} F(t)\, t\, D_N(t)\, dt.$$

With $\sin((N + \frac{1}{2})t) = \sin Nt \cos\frac{t}{2} + \cos Nt \sin\frac{t}{2}$,

$$S_N(f)(\theta_0) - f(\theta_0) = \frac{1}{2\pi}\int F(t)\frac{t}{\sin(t/2)} \cos\tfrac{t}{2}\, \sin Nt\, dt + \frac{1}{2\pi}\int F(t)\, t \cos Nt\, dt.$$

$F(t)\frac{t\cos(t/2)}{\sin(t/2)}$ and $tF(t)$ are integrable ($t/\sin(t/2)$ is bounded), so both integrals tend to 0 by Theorem 1.4. ∎

(The proof only needs $|f(\theta) - f(\theta_0)| \le M|\theta - \theta_0|$.)

Theorem 2.2 (Riemann localization principle)

If f, g are integrable on the circle and $f = g$ on an open interval I containing θ_0, then $S_N(f)(\theta_0) - S_N(g)(\theta_0) \to 0$.

Proof. $f - g$ vanishes on I, so it is differentiable at θ_0 with value 0; apply Theorem 2.1. ∎

3.2.2 A continuous function with diverging Fourier series

Symmetry breaking. The odd sawtooth $f(\theta) = i(\pi - \theta)$ on $0 < \theta < \pi$ has $f \sim \sum_{n\neq 0} \frac{e^{in\theta}}{n}$. The half $\sum_{n\le -1} \frac{e^{in\theta}}{n}$ is the Fourier series of no integrable $\tilde f$: else $|A_r(\tilde f)(0)| = \sum_{n\ge 1} \frac{r^n}{n} \to \infty$, while $|A_r(\tilde f)(0)| \le \sup|\tilde f|$. For $N \ge 1$ let

$$f_N(\theta) = \sum_{1\le|n|\le N} \frac{e^{in\theta}}{n}, \qquad \tilde f_N(\theta) = \sum_{-N\le n\le -1} \frac{e^{in\theta}}{n}.$$

Then (i) $|\tilde f_N(0)| = \sum_{n=1}^{N} \frac{1}{n} \ge \log N$, and (ii) $f_N(\theta)$ is bounded uniformly in N, θ by the lemma below.

Lemma 2.3

Suppose the Abel means $A_r = \sum_{n\geq 1} r^n c_n$ of $\sum c_n$ are bounded as $r \to 1^-$, and $c_n = O(1/n)$. Then the partial sums $S_N = \sum_{n=1}^{N} c_n$ are bounded.

Proof. Let $r = 1 - 1/N$, $n|c_n| \leq M$. Then

$$|S_N - A_r| \leq \sum_{n=1}^{N} |c_n|(1-r^n) + \sum_{n>N} r^n |c_n| \leq M \sum_{n=1}^{N} (1-r) + \frac{M}{N} \sum_{n>N} r^n \leq MN(1-r) + \frac{M}{N} \frac{1}{1-r} = 2M,$$

as $1 - r^n \leq n(1-r)$. So $|A_r| \leq M$ gives $|S_N| \leq 3M$. ∎

Apply it with $c_n = \frac{e^{in\theta}}{n} - \frac{e^{-in\theta}}{n}$ (partial sums $f_N(\theta)$, $|c_n| \leq 2/n$): the Abel means $(f * P_r)(\theta)$ are bounded by $\sup |f|$. So $|f_N(\theta)| \leq C$.

Shift frequencies by $2N$: $P_N = e^{i2N\theta} f_N$, $\tilde{P}_N = e^{i2N\theta} \tilde{f}_N$, with frequencies in $N \leq n \leq 3N$, $n \neq 2N$ (resp. $N \leq n \leq 2N - 1$).

Lemma 2.4

$S_M(P_N) = P_N$ if $M \geq 3N$; $S_M(P_N) = \tilde{P}_N$ if $M = 2N$; $S_M(P_N) = 0$ if $M < N$.

Proof. S_M keeps the frequencies $|n| \leq M$; those of P_N lie in $[N, 3N]$, and those $\leq 2N$ are the frequencies of $\tilde{P}_N$. ∎

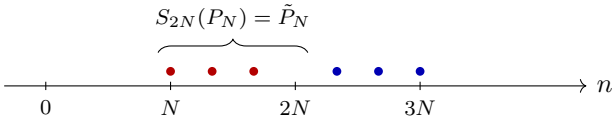


Figure. *Frequencies of P_N; S_{2N} keeps the left half.*

The construction. Take $\alpha_k > 0$, $\sum \alpha_k < \infty$, and $N_{k+1} > 3N_k$ with $\alpha_k \log N_k \to \infty$; e.g. $\alpha_k = 1/k^2$, $N_k = 3^{2^k}$ ($\alpha_k \log N_k = \frac{2^k \log 3}{k^2} \to \infty$). Let

$$f(\theta) = \sum_{k=1}^{\infty} \alpha_k P_{N_k}(\theta).$$

As $|P_N| = |f_N| \leq C$, the series converges uniformly: f is continuous with termwise coefficients. By Lemma 2.4 ($2N_m < N_k$ for $k > m$, $2N_m \geq 3N_k$ for $k < m$),

$$S_{2N_m}(f) = \sum_{k<m} \alpha_k P_{N_k} + \alpha_m \tilde{P}_{N_m}.$$

At 0: $|\sum_{k<m} \alpha_k P_{N_k}(0)| \leq C \sum \alpha_k$ and $|\tilde{P}_{N_m}(0)| \geq \log N_m$, so $|S_{2N_m}(f)(0)| \geq \alpha_m \log N_m - C' \to \infty$: *the Fourier series of the continuous f diverges at 0.*

3.3 Selected exercises

Motivation. Consequences of Parseval and Riemann–Lebesgue: Wirtinger–Poincaré inequalities, the Dirichlet integral, and absolute convergence for C^1 and Lipschitz f.

Exercise 3.11 (Wirtinger and Poincaré inequalities)

(a) Let f be T-periodic, continuous and piecewise C^1 with $\int_0^T f = 0$. Show that $\int_0^T |f|^2 \le \frac{T^2}{4\pi^2}\int_0^T |f'|^2$, with equality iff $f(t) = A\sin(2\pi t/T) + B\cos(2\pi t/T)$.
(b) Let f be C^1 on $[a,b]$ with $f(a) = f(b) = 0$. Show that $\int_a^b |f|^2 \le \frac{(b-a)^2}{\pi^2}\int_a^b |f'|^2$, with equality iff $f(t) = A\sin\left(\pi\frac{t-a}{b-a}\right)$; in particular the constant is optimal.

Solution.

(a)

Step 1 (reduction to $T = 2\pi$). $g(s) = f(Ts/2\pi)$ has $\int_0^{2\pi}|g|^2 = \frac{2\pi}{T}\int_0^T |f|^2$ and $\int_0^{2\pi}|g'|^2 = \frac{T}{2\pi}\int_0^T |f'|^2$, so the claim becomes $\int_0^{2\pi}|g|^2 \le \int_0^{2\pi}|g'|^2$, with corresponding extremals.

Step 2 (coefficients of g'). For $T = 2\pi$, integrating by parts on each C^1 piece (boundary terms telescope and cancel by periodicity), $\hat{f}'(n) = in\hat{f}(n)$; and $\hat{f}(0) = 0$.

Step 3. Parseval for f and f':

$$\frac{1}{2\pi}\int_0^{2\pi}|f|^2 = \sum_{n\neq 0}|\hat{f}(n)|^2 \le \sum_{n\neq 0} n^2|\hat{f}(n)|^2 = \frac{1}{2\pi}\int_0^{2\pi}|f'|^2.$$

Step 4 (equality). Iff $(n^2-1)|\hat{f}(n)|^2 = 0$ for all n, i.e. $\hat{f}(n) = 0$ for $|n| \ge 2$; then $f = \hat{f}(-1)e^{-it} + \hat{f}(1)e^{it} = A\sin t + B\cos t$ by Corollary 2.2. Conversely these give equality.

(b)

Step 1. Take $a = 0$, $L = b$. Extend f oddly to $[-L, L]$ and $2L$-periodically; $f(0) = f(L) = 0$ makes $\tilde{f}$ continuous and piecewise C^1, with $\int_{-L}^{L}\tilde{f} = 0$.

Step 2. By (a) with $T = 2L$, $\int_{-L}^{L}|\tilde{f}|^2 \le \frac{L^2}{\pi^2}\int_{-L}^{L}|\tilde{f}'|^2$, and each side is twice its integral over $[0, L]$.

Step 3. Equality forces $\tilde{f} = A\sin(\pi t/L) + B\cos(\pi t/L)$, and oddness $B = 0$; $\sin(\pi t/L)$ attains it, so L^2/π^2 is optimal. ■

Exercise 3.12 (The Dirichlet integral)

Prove that $\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$ (as an improper Riemann integral).

Solution.

Step 1 (existence). For $R > 1$, by parts $\int_1^R \frac{\sin x}{x}\,dx = \cos 1 - \frac{\cos R}{R} - \int_1^R \frac{\cos x}{x^2}\,dx$, convergent as $R \to \infty$; the integrand is continuous on $(0, 1]$ with limit 1 at 0. So $I = \lim_R \int_0^R \frac{\sin x}{x}\,dx$ exists; take $R_N = (N + \frac{1}{2})\pi$.

Step 2 (splitting D_N). $\frac{1}{2\pi}\int_{-\pi}^{\pi} D_N = 1$. Let $h(\theta) = \frac{1}{\sin(\theta/2)} - \frac{2}{\theta}$ for $0 < |\theta| \le \pi$, $h(0) = 0$. As

$$h(\theta) = \frac{\theta - 2\sin(\theta/2)}{\theta\sin(\theta/2)} = \frac{\theta^3/24 + O(\theta^5)}{\theta^2/2 + O(\theta^4)} \to 0,$$

h is continuous on $[-\pi, \pi]$.

Step 3 (Riemann–Lebesgue). Expanding $\sin((N + \frac{1}{2})\theta)$ as in Theorem 2.1, Theorem 1.4 gives $\int_{-\pi}^{\pi} h(\theta)\sin((N + \frac{1}{2})\theta)\,d\theta \to 0$, so

$$\int_{-\pi}^{\pi}\frac{2\sin((N+\frac{1}{2})\theta)}{\theta}\,d\theta = 2\pi - \int_{-\pi}^{\pi} h(\theta)\sin\left((N+\tfrac{1}{2})\theta\right)d\theta \to 2\pi.$$

Step 4. By evenness $\int_0^{\pi}\frac{\sin((N+\frac{1}{2})\theta)}{\theta}d\theta \to \frac{\pi}{2}$, which is $\int_0^{R_N}\frac{\sin x}{x}\,dx$ after $x = (N + \frac{1}{2})\theta$. So $I = \frac{\pi}{2}$. ■

Exercise 3.14 (C^1 functions have absolutely convergent Fourier series)

Prove that the Fourier series of a C^1 function f on the circle converges absolutely, hence uniformly to f.

Solution.

Step 1. As in Exercise 3.11, $\hat{f'}(n) = in\hat{f}(n)$, so $\sum_n n^2|\hat{f}(n)|^2 = \frac{1}{2\pi}\int |f'|^2 < \infty$.

Step 2. By Cauchy–Schwarz,

$$\sum_{0<|n|\le N} |\hat{f}(n)| = \sum_{0<|n|\le N} |n\hat{f}(n)| \cdot \frac{1}{|n|} \le \Big(\sum_n n^2|\hat{f}(n)|^2\Big)^{1/2}\Big(\sum_{n\neq 0}\frac{1}{n^2}\Big)^{1/2}.$$

uniformly in N, so $\sum|\hat{f}(n)| < \infty$.

Step 3. Corollary 2.3 gives uniform convergence, improving Corollary 2.4 (C^2). ∎

Exercise 3.16 (Lipschitz functions: Bernstein's argument)

Let f be 2π-periodic with $|f(x)-f(y)| \le K|x-y|$. Prove that $\sum_n |\hat{f}(n)| < \infty$, so the Fourier series of f converges absolutely and uniformly to f.

Solution.

Step 1. For $h > 0$, $g_h(x) = f(x+h)-f(x-h)$ has $\hat{g}_h(n) = 2i\sin(nh)\hat{f}(n)$. Parseval and $|g_h| \le 2Kh$ give

$$\sum_n 4\sin^2(nh)|\hat{f}(n)|^2 = \frac{1}{2\pi}\int_0^{2\pi}|g_h|^2 \le 4K^2h^2, \qquad \text{so} \qquad \sum_n \sin^2(nh)|\hat{f}(n)|^2 \le K^2h^2.$$

Step 2. For $p \ge 0$, $h = \pi/2^{p+1}$ and $2^{p-1} < |n| \le 2^p$: $\frac{\pi}{4} < |n|h \le \frac{\pi}{2}$, so $\sin^2(nh) \ge \frac{1}{2}$ and

$$\sum_{2^{p-1}<|n|\le 2^p} |\hat{f}(n)|^2 \le 2K^2h^2 = \frac{K^2\pi^2}{2^{2p+1}}.$$

Step 3. The block has at most 2^{p+1} integers, so by Cauchy–Schwarz

$$\sum_{2^{p-1}<|n|\le 2^p} |\hat{f}(n)| \le (2^{p+1})^{1/2}\Big(\frac{K^2\pi^2}{2^{2p+1}}\Big)^{1/2} = \frac{K\pi}{2^{p/2}}.$$

Step 4. The blocks cover $n \neq 0$ and $\sum_p 2^{-p/2} < \infty$, so $\sum_n |\hat{f}(n)| < \infty$; Corollary 2.3 gives uniform convergence. (Likewise for Hölder exponent $\alpha > \frac{1}{2}$.) ∎

Chapter 4. Some Applications of Fourier Series

4.1 The isoperimetric inequality

Motivation. In Fourier coefficients, length and area become sums of squares, and Parseval reduces the isoperimetric inequality to $|n| \le n^2$.

Curves. A **parametrized curve** $\gamma : [a,b] \to \mathbb{R}^2$, $\gamma(s) = (x(s), y(s))$, is **simple** if injective on $[a,b)$, **closed** if $\gamma(a) = \gamma(b)$, and C^1 (closed curves $(b-a)$-periodic); the **curve** Γ is its image. Its **length** $\ell = \int_a^b |\gamma'(s)|\, ds$ is invariant under C^1 reparametrization, and if $\gamma' \neq 0$ it can be parametrized by **arc-length**, $|\gamma'| = 1$, $\ell = b - a$. For a simple closed C^1 curve, Green's theorem motivates the definition of the enclosed **area**

$$\mathcal{A} = \frac{1}{2}\Big|\int_\Gamma (x\,dy - y\,dx)\Big| = \frac{1}{2}\Big|\int_a^b (x(s)y'(s) - y(s)x'(s))ds\Big|,$$

Theorem 1.1 (Isoperimetric inequality)

If Γ is a simple closed C^1 curve in $\mathbb{R}^2$ of length ℓ enclosing area $\mathcal{A}$, then $\mathcal{A} \le \frac{\ell^2}{4\pi}$, with equality iff Γ is a circle.

Proof.

Step 1 (normalization). Scaling multiplies ℓ by δ and $\mathcal{A}$ by δ^2, so take $\ell = 2\pi$ and show $\mathcal{A} \le \pi$. With arc-length on $[0, 2\pi]$, $x'^2 + y'^2 = 1$ and

$$\frac{1}{2\pi}\int_0^{2\pi} (|x'(s)|^2 + |y'(s)|^2)ds = 1.$$

Step 2 (Fourier coefficients). With $x \sim \sum a_n e^{ins}$, $y \sim \sum b_n e^{ins}$, we have $x' \sim \sum a_n in e^{ins}$, $y' \sim \sum b_n in e^{ins}$, and Parseval gives

$$\sum_n |n|^2(|a_n|^2 + |b_n|^2) = 1.$$

As x, y are real, $\overline{a_n} = a_{-n}$, $\overline{b_n} = b_{-n}$, and by Lemma 1.5

$$\mathcal{A} = \frac{1}{2}\Big|\int_0^{2\pi} (xy' - yx')\, ds\Big| = \pi\Big|\sum_n n(a_n\overline{b_n} - b_n\overline{a_n})\Big|.$$

Step 3 (the inequality). $|a_n\overline{b_n} - b_n\overline{a_n}| \le 2|a_n||b_n| \le |a_n|^2 + |b_n|^2$ and $|n| \le |n|^2$, so

$$\mathcal{A} \le \pi\sum_n |n|(|a_n|^2 + |b_n|^2) \le \pi\sum_n |n|^2(|a_n|^2 + |b_n|^2) = \pi.$$

Step 4 (equality). $\mathcal{A} = \pi$ forces $a_n = b_n = 0$ for $|n| \ge 2$, so $x = a_{-1}e^{-is} + a_0 + a_1 e^{is}$, $y = b_{-1}e^{-is} + b_0 + b_1 e^{is}$, with $2(|a_1|^2 + |b_1|^2) = 1$. Equality in Step 3 requires $2|a_1||b_1| = |a_1|^2 + |b_1|^2$, so $|a_1| = |b_1| = \frac{1}{2}$, and $|a_1\overline{b_1} - \overline{a_1}b_1| = 2|a_1||b_1|$. With $a_1 = \frac{1}{2}e^{i\alpha}$, $b_1 = \frac{1}{2}e^{i\beta}$: $|\sin(\alpha - \beta)| = 1$, $\alpha - \beta = k\pi/2$, k odd. So $x = a_0 + \cos(\alpha + s)$, $y = b_0 \pm \sin(\alpha + s)$, a unit circle; conversely circles give equality. ∎

4.2 Weyl's equidistribution theorem

Motivation. For irrational γ, averages of $e^{2\pi ikn\gamma}$ are geometric sums tending to 0, so $\langle n\gamma \rangle$ fills $[0,1)$ uniformly.

$[x]$ is the integer part and $\langle x\rangle = x - [x] \in [0,1)$ the **fractional part**; $x = y \bmod \mathbb{Z}$ if $x - y \in \mathbb{Z}$. For rational γ, $\langle n\gamma\rangle$ takes finitely many values; for irrational γ they are distinct ($(n_1 - n_2)\gamma \notin \mathbb{Z}$).

Definition (Equidistribution)

A sequence $\xi_1, \xi_2, \ldots$ in $[0,1)$ is **equidistributed** if for every interval $(a,b) \subseteq [0,1)$,

$$\lim_{N\to\infty} \frac{\#\{1 \le n \le N : \xi_n \in (a,b)\}}{N} = b - a.$$

Theorem 2.1 (Weyl)

If γ is irrational, then $\langle\gamma\rangle, \langle 2\gamma\rangle, \langle 3\gamma\rangle, \ldots$ is equidistributed in $[0,1)$. In particular $\{\langle n\gamma\rangle\}$ is dense in $[0,1)$ (**Kronecker's theorem**).

With $\chi_{(a,b)}$ the 1-periodic extension of the indicator of $(a,b) \subseteq [0,1)$, $\#\{n \le N : \langle n\gamma\rangle \in (a,b)\} = \sum_{n=1}^{N} \chi_{(a,b)}(n\gamma)$, and the theorem says

$$\frac{1}{N}\sum_{n=1}^{N} \chi_{(a,b)}(n\gamma) \to \int_0^1 \chi_{(a,b)}(x)\,dx. \tag{4.1}$$

Lemma 2.2

If f is continuous, 1-periodic, and γ is irrational, then $\frac{1}{N}\sum_{n=1}^{N} f(n\gamma) \to \int_0^1 f(x)\,dx$.

Proof.

Step 1 (exponentials). $f = 1$ is trivial. For $f = e^{2\pi ikx}$, $k \neq 0$: $\int_0^1 f = 0$ and $e^{2\pi ik\gamma} \neq 1$, so

$$\frac{1}{N}\sum_{n=1}^{N} e^{2\pi ikn\gamma} = \frac{e^{2\pi ik\gamma}}{N}\frac{1 - e^{2\pi ikN\gamma}}{1 - e^{2\pi ik\gamma}}, \qquad \left|\frac{1}{N}\sum_{n=1}^{N} e^{2\pi ikn\gamma}\right| \le \frac{2}{N|1 - e^{2\pi ik\gamma}|} \to 0.$$

By linearity the lemma holds for trigonometric polynomials.

Step 2 (approximation). Corollary 5.4 gives P with $|f - P| < \epsilon/3$; then the averages, the integrals, and (for large N) $\frac{1}{N}\sum P(n\gamma)$ versus $\int P$ each differ by $< \epsilon/3$. ∎

Corollary 2.3

The conclusion of Lemma 2.2 holds for every 1-periodic f that is Riemann integrable on $[0,1]$.

Proof. Let f be real. Take step functions $f_U \ge f \ge f_L$ with $\int(f_U - f_L) < \epsilon$ and modify near the partition points (as in Lemma 3.2) to continuous 1-periodic $f_+ \ge f \ge f_-$ with $\int(f_+ - f_-) < 2\epsilon$. By Lemma 2.2,

$$\limsup \frac{1}{N}\sum f(n\gamma) \le \lim \frac{1}{N}\sum f_+(n\gamma) = \int f_+ \le \int f + 2\epsilon,$$

and similarly $\liminf \ge \int f - 2\epsilon$. ∎

Proof of Theorem 2.1. Apply Corollary 2.3 to $f = \chi_{(a,b)}$, which gives (4.1). ∎

Weyl's criterion (one direction above, the other a book exercise): $\{\xi_n\} \subseteq [0,1)$ is equidistributed iff $\frac{1}{N}\sum_{n=1}^{N} e^{2\pi ik\xi_n} \to 0$ for all $k \neq 0$; e.g. $\langle n^2\gamma\rangle$ for irrational γ. *Geometrically:* a light ray in a square mirror box is periodic for rational slope and equidistributed for irrational slope.

4.3 A continuous but nowhere differentiable function

Motivation. Delayed means isolate one lacunary term, whose derivative is far larger than the $O(\log N)$ that differentiability would allow.

Theorem 3.1 (Weierstrass)

For $0 < \alpha < 1$, $f_\alpha(x) = f(x) = \sum_{n=0}^{\infty} 2^{-n\alpha} e^{i2^n x}$ is continuous but nowhere differentiable.

The series converges absolutely and uniformly ($\sum 2^{-n\alpha} < \infty$), so f is continuous. For nowhere differentiability use the **delayed means**

$$\triangle_N(f) = 2\sigma_{2N}(f) - \sigma_N(f).$$

Facts. (a) As $\sigma_M(f) = \sum_{|n|<M} (1 - \frac{|n|}{M}) a_n e^{inx}$, the coefficient of $a_n e^{inx}$ in $\triangle_N(f)$ is 1 for $|n| \le N$ and $2 - \frac{|n|}{N}$ for $N < |n| < 2N$:

$$\triangle_N(f) = S_N(f) + \sum_{N<|n|<2N} \Big(2 - \frac{|n|}{N}\Big) a_n e^{inx}.$$

(b) For lacunary f (frequencies 2^k) and $N = 2^{n-1}$, no frequency lies strictly between N and $2N$, so $\triangle_N(f) = S_N(f)$.

Lemma 3.2

If g is continuous and differentiable at x_0, then $\sigma_N(g)'(x_0) = O(\log N)$, hence $\triangle_N(g)'(x_0) = O(\log N)$.

Proof.

Step 1. As $\int F_N' = 0$, $\sigma_N(g)'(x_0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} F_N'(t)(g(x_0 - t) - g(x_0))dt$, and differentiability plus boundedness give $|g(x_0 - t) - g(x_0)| \le C|t|$.

Step 2 (bounds for F_N'). $F_N = \sum_{|n|<N}(1 - \frac{|n|}{N})e^{int}$ gives $|F_N'| \le N^2$; and differentiating $F_N = \frac{\sin^2(Nt/2)}{N \sin^2(t/2)}$ [Supplement],

$$|F_N'(t)| \le \frac{|\sin(Nt/2)\cos(Nt/2)|}{\sin^2(t/2)} + \frac{\sin^2(Nt/2)|\cos(t/2)|}{N|\sin(t/2)|^3} \le \frac{A}{t^2} + \frac{A}{N|t|^3} \le \frac{2A}{t^2} \quad (|t| \ge 1/N).$$

Step 3. So $|\sigma_N(g)'(x_0)| \le C \int_{|t|\le 1/N} N^2|t|\,dt + C \int_{1/N\le|t|\le\pi} \frac{2A}{|t|}\,dt = O(\log N)$; and $\triangle_N(g)' = 2\sigma_{2N}(g)' - \sigma_N(g)'$. ■

Lemma 3.3

If $2N = 2^n$, then $\triangle_{2N}(f) - \triangle_N(f) = 2^{-n\alpha} e^{i2^n x}$.

Proof. By Fact (b), $\triangle_{2N}(f) = S_{2N}(f)$, $\triangle_N(f) = S_N(f)$, and the only frequency in $(N, 2N]$ is 2^n. ■

Proof of Theorem 3.1. If f were differentiable at x_0, Lemma 3.3 gives $|\triangle_{2N}(f)'(x_0) - \triangle_N(f)'(x_0)| = |i2^{n(1-\alpha)} e^{i2^n x_0}| = (2N)^{1-\alpha}$, while Lemma 3.2 gives $O(\log N)$; but $N^{1-\alpha}/\log N \to \infty$. ■

4.4 The heat equation on the circle

Motivation. On a ring $u = \sum a_n e^{-4\pi^2 n^2 t} e^{2\pi i n x} = f * H_t$: high frequencies die fast, and H_t is a good kernel.

With $\theta = 2\pi x$, $x \in [0,1)$, and rescaled time, $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $u(x,0) = f(x)$. $u = A(x)B(t)$ gives $\frac{B'}{B} = \frac{A''}{A} = \lambda$; periodicity forces $\lambda = -4\pi^2 n^2$, $A = e^{\pm 2\pi i n x}$, $B = e^{-4\pi^2 n^2 t}$. Superposing,

$$u(x,t) = \sum_{n=-\infty}^{\infty} a_n e^{-4\pi^2 n^2 t} e^{2\pi i n x}, \qquad a_n = \hat{f}(n).$$

For integrable f the a_n are bounded, the series converges with all derivatives for $t > 0$ and solves the equation, and $u(\cdot,t) \to f$ in mean square as $t \to 0$ (Parseval). In convolution form $u = f * H_t$ with the **heat kernel**

$$H_t(x) = \sum_{n=-\infty}^{\infty} e^{-4\pi^2 n^2 t} e^{2\pi i n x}.$$

$\{H_t\}$ is a family of good kernels as $t \to 0$ (Corollary 3.4), so $u \to f$ at points of continuity, uniformly for continuous f. H_t has no elementary closed form; its positivity and concentration follow from Poisson summation (Chapter 5).

4.5 Selected exercises

Motivation. The converse of Weyl's criterion and a uniform version, a non-equidistributed sequence, and mean-square initial values for the heat equation.

Exercise 4.5 (Powers of the golden ratio)

Let $\varphi = \frac{1+\sqrt{5}}{2}$ and $\gamma_n = \langle \varphi^n \rangle$. Prove that $\{\gamma_n\}$ is not equidistributed in $[0,1)$.

Solution.

Step 1. $\psi = \frac{1-\sqrt{5}}{2}$ and φ solve $x^2 = x + 1$, so $U_n = \varphi^n + \psi^n$ satisfies $U_{n+1} = U_n + U_{n-1}$, $U_0 = 2$, $U_1 = 1$: all U_n are integers.

Step 2. $0 < |\psi| < 1$ and $\varphi^n = U_n - \psi^n$: for odd n, $\gamma_n = |\psi|^n$; for even n, $\gamma_n = 1 - |\psi|^n$.

Step 3. With $|\psi|^{n_0} < \frac{1}{4}$, $\gamma_n \in [0, \frac{1}{4}) \cup (\frac{3}{4}, 1)$ for $n \geq n_0$, so

$$\frac{\#\{1 \leq n \leq N : \gamma_n \in (\frac{1}{4}, \frac{3}{4})\}}{N} \leq \frac{n_0}{N} \to 0 \neq \frac{1}{2}.$$

■

Exercise 4.7 (Weyl's criterion, converse direction)

Let $\{\xi_n\} \subseteq [0,1)$ be equidistributed. Prove that $\frac{1}{N}\sum_{n=1}^{N} e^{2\pi i k \xi_n} \to 0$ for every integer $k \neq 0$.

Solution. Write $\#_N(E) = \#\{1 \leq n \leq N : \xi_n \in E\}$.

Step 1 (single points are negligible). For $c = 0$: $\#_N(\{0\}) \leq N - \#_N((0,1))$, $\#_N((0,1))/N \to 1$. For $c > 0$: $\#_N(\{c\}) \leq \#_N((c-\delta, c+\delta))$, ratio $\to 2\delta$. So $\#_N(\{c\})/N \to 0$.

Step 2 (continuous f). For real continuous f, $|f| \le B$, choose $0 = x_0 < \cdots < x_M = 1$ with $|f(x) - f(x_j)| < \epsilon$ on $[x_{j-1}, x_j]$. Each ξ_n lies in some (x_{j-1}, x_j) or equals some x_j, $j < M$, so

$$\Big|\frac{1}{N}\sum_{n=1}^{N} f(\xi_n) - \sum_{j=1}^{M} f(x_j)\frac{\#_N((x_{j-1}, x_j))}{N}\Big| \le \epsilon + B\sum_{j=0}^{M-1}\frac{\#_N(\{x_j\})}{N}.$$

Step 3. As $N \to \infty$ the sum tends to the Riemann sum $\sum_j f(x_j)(x_j - x_{j-1})$, within ϵ of $\int_0^1 f$, and the error to ϵ (Step 1). So $\frac{1}{N}\sum_{n=1}^{N} f(\xi_n) \to \int_0^1 f$.

Step 4. Apply Step 3 to $\cos(2\pi kx)$, $\sin(2\pi kx)$, $k \neq 0$. ■

Exercise 4.10 (Uniform averages along an equidistributed sequence)

Let $\{\xi_n\}$ be equidistributed in $[0, 1)$ and f continuous and 1-periodic with $\int_0^1 f = 0$. Prove that $\frac{1}{N}\sum_{n=1}^{N} f(x + \xi_n) \to 0$ uniformly in x.

Solution.

Step 1 (exponentials). For $k \neq 0$,

$$\Big|\frac{1}{N}\sum_{n=1}^{N} e^{2\pi ik(x+\xi_n)}\Big| = \Big|\frac{1}{N}\sum_{n=1}^{N} e^{2\pi ik\xi_n}\Big| \to 0$$

by Exercise 4.7, independently of x; by linearity the claim holds uniformly for trigonometric polynomials Q with zero constant term.

Step 2 (approximation). Corollary 5.4 (period 1) gives P with $\sup|f - P| < \epsilon$; its constant term has $|c_0| = |\int_0^1 (P - f)| < \epsilon$, so $Q = P - c_0$ has $\sup|f - Q| < 2\epsilon$.

Step 3. For all x,

$$\Big|\frac{1}{N}\sum_{n=1}^{N} f(x + \xi_n)\Big| \le 2\epsilon + \Big|\frac{1}{N}\sum_{n=1}^{N} Q(x + \xi_n)\Big| < 3\epsilon$$

for $N \ge N_0$, N_0 independent of x by Step 1. ■

Exercise 4.11 (Mean-square initial values for the heat equation)

Let f be integrable on $[0, 1]$ (1-periodic) and $u(x, t) = (f * H_t)(x)$. Prove that $\int_0^1 |u(x, t) - f(x)|^2 dx \to 0$ as $t \to 0^+$.

Solution.

Step 1. For $t > 0$, $u(\cdot, t) = \sum_n a_n e^{-4\pi^2 n^2 t} e^{2\pi inx}$ converges uniformly, so $u(\cdot, t) - f$ has coefficients $a_n(e^{-4\pi^2 n^2 t} - 1)$.

Step 2. By Parseval (Theorem 1.3, period 1),

$$\int_0^1 |u(x, t) - f(x)|^2 dx = \sum_n |a_n|^2 (1 - e^{-4\pi^2 n^2 t})^2.$$

Step 3. Each factor lies in $[0, 1]$ and tends to 0. With $\sum_{|n|>M} |a_n|^2 < \epsilon$ ($\sum |a_n|^2 = \int_0^1 |f|^2$), the sum is at most $\sum_{|n|\le M} |a_n|^2 (1 - e^{-4\pi^2 M^2 t})^2 + \epsilon < 2\epsilon$ for small t. ■

Chapter 5. The Fourier Transform on $\mathbb{R}$

5.1 Elementary theory of the Fourier transform

Motivation. On $\mathcal{S}(\mathbb{R})$ the transform $\hat{f}(\xi) = \int f(x)e^{-2\pi ix\xi}dx$ swaps differentiation and multiplication by x, fixes the Gaussian, and is an L^2-isometric bijection.

5.1.1 Integration of functions on the real line

f is of **moderate decrease** if continuous with $|f(x)| \le \frac{A}{1+x^2}$; then $\int_{-N}^{N} f$ is Cauchy and $\int_{-\infty}^{\infty} f = \lim_N \int_{-N}^{N} f$. These form a vector space $\mathcal{M}(\mathbb{R})$.

Proposition 1.1

For f, g of moderate decrease:
(i) Linearity: $\int(af + bg) = a\int f + b\int g$.
(ii) Translation invariance: $\int f(x-h)\,dx = \int f(x)\,dx$ for $h \in \mathbb{R}$.
(iii) Scaling: $\delta \int f(\delta x)\,dx = \int f(x)\,dx$ for $\delta > 0$.
(iv) Continuity: $\int |f(x-h) - f(x)|\,dx \to 0$ as $h \to 0$.

Proof. (i) Clear from finite intervals.

(ii) $\int_{-N}^{N} f(x-h)dx = \int_{-N-h}^{N-h} f$, and the end pieces tend to 0 by the decay.

(iii) $\delta \int_{-N}^{N} f(\delta x)dx = \int_{-\delta N}^{\delta N} f$.

(iv) Choose N with $\int_{|x|\ge N} |f(x)|, \int_{|x|\ge N} |f(x-h)| < \epsilon$ for $|h| \le 1$; by uniform continuity on $[-N, N]$, $\int_{-N}^{N} |f(x-h) - f(x)| < \epsilon$ for small h. ∎

5.1.2 Definition of the Fourier transform

For $f \in \mathcal{M}(\mathbb{R})$, $\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi ix\xi}\,dx$ is bounded and continuous. (If $\hat{f} \in \mathcal{M}(\mathbb{R})$ too, inversion and Plancherel hold, §5.1.7.)

5.1.3 The Schwartz space

$\mathcal{S}(\mathbb{R})$ consists of the C^∞ f with $\sup_x |x|^k |f^{(\ell)}(x)| < \infty$ for all $k, \ell \ge 0$ (**rapidly decreasing**); it is closed under differentiation and multiplication by polynomials. It contains the **Gaussian** e^{-ax^2}, $a > 0$ (derivatives $P(x)e^{-ax^2}$), and smooth compactly supported functions such as the **bump function** $e^{-1/(1-x^2)}$ on $|x| < 1$, 0 elsewhere.

5.1.4 The Fourier transform on $\mathcal{S}$

Proposition 1.2

If $f \in \mathcal{S}(\mathbb{R})$, writing $f(x) \to \hat{f}(\xi)$:
(i) $f(x+h) \to \hat{f}(\xi)e^{2\pi ih\xi}$ for $h \in \mathbb{R}$.
(ii) $f(x)e^{-2\pi ixh} \to \hat{f}(\xi+h)$ for $h \in \mathbb{R}$.
(iii) $f(\delta x) \to \delta^{-1}\hat{f}(\delta^{-1}\xi)$ for $\delta > 0$.
(iv) $f'(x) \to 2\pi i\xi\hat{f}(\xi)$.

(v) $-2\pi i x f(x) \to \frac{d}{d\xi}\hat{f}(\xi)$.

Proof. (i)–(iii) by Proposition 1.1(ii), (iii).

(iv) By parts on $[-N,N]$: $\int_{-N}^{N} f'(x)e^{-2\pi i x\xi}dx = [f(x)e^{-2\pi i x\xi}]_{-N}^{N} + 2\pi i\xi\int_{-N}^{N} f(x)e^{-2\pi i x\xi}dx$, with boundary term $\to 0$.

(v) Let $g(x) = -2\pi i x f(x) \in \mathcal{S}$. For $\epsilon > 0$,

$$\frac{\hat{f}(\xi+h)-\hat{f}(\xi)}{h} - \hat{g}(\xi) = \int f(x)e^{-2\pi i x\xi}\Big[\frac{e^{-2\pi i x h}-1}{h} + 2\pi i x\Big]dx.$$

Choose N with $\int_{|x|\geq N}|f|, \int_{|x|\geq N}|x||f| < \epsilon$. For $|x| \leq N$ the bracket is $\leq C|h|$ (Taylor), and always $|\frac{e^{-2\pi i x h}-1}{h}| \leq 2\pi|x|$; so the difference is $\leq C'\epsilon + C''N|h| < C\epsilon$ for small $|h|$. ∎

Theorem 1.3

If $f \in \mathcal{S}(\mathbb{R})$, then $\hat{f} \in \mathcal{S}(\mathbb{R})$.

Proof. $\hat{f}$ is bounded. By Proposition 1.2(iv), (v), the function $\xi^k(\frac{d}{d\xi})^\ell\hat{f}$ is the transform of $\frac{1}{(2\pi i)^k}(\frac{d}{dx})^k[(-2\pi i x)^\ell f] \in \mathcal{S}$, hence bounded; $\hat{f} \in C^\infty$ by repeated (v). ∎

Theorem 1.4

If $f(x) = e^{-\pi x^2}$, then $\hat{f}(\xi) = f(\xi)$.

Proof. $F = \hat{f}$ has $F(0) = \int e^{-\pi x^2}dx = 1$. By (v), $f' = -2\pi x f$ and (iv),

$$F'(\xi) = \int f(x)(-2\pi i x)e^{-2\pi i x\xi}dx = i\int f'(x)e^{-2\pi i x\xi}dx = i(2\pi i\xi)F(\xi) = -2\pi\xi F(\xi).$$

So $G = Fe^{\pi\xi^2}$ has $G' = 0$, $G \equiv 1$. ∎

($\int e^{-\pi x^2}dx = 1$: square it and use polar coordinates.)

Corollary 1.5

If $\delta > 0$ and $K_\delta(x) = \delta^{-1/2}e^{-\pi x^2/\delta}$, then $\widehat{K_\delta}(\xi) = e^{-\pi\delta\xi^2}$.

Proof. Proposition 1.2(iii) with δ replaced by $\delta^{-1/2}$ and Theorem 1.4. ∎

As $\delta \to 0$, K_δ concentrates while $\widehat{K_\delta}$ flattens: the uncertainty principle in miniature (§5.4).

Definition (Good kernels on $\mathbb{R}$)

$\{K_\delta\}_{\delta>0}$ is a family of **good kernels** as $\delta \to 0$ if (i) $\int K_\delta = 1$;
(ii) $\int |K_\delta| \leq M$;
(iii) for every $\eta > 0$, $\int_{|x|>\eta}|K_\delta| \to 0$.

Theorem 1.6

$\{K_\delta\}_{\delta>0}$, $K_\delta(x) = \delta^{-1/2}e^{-\pi x^2/\delta}$, is a family of good kernels as $\delta \to 0$.

Proof. $K_\delta > 0$, $\int K_\delta = \widehat{K_\delta}(0) = 1$, and with $x = \delta^{1/2}y$, $\int_{|x|>\eta} K_\delta = \int_{|y|>\eta/\sqrt{\delta}} e^{-\pi y^2} dy \to 0$. ■

For $f, g \in \mathcal{S}(\mathbb{R})$, $(f * g)(x) = \int f(x-t)g(t)\,dt$.

Corollary 1.7

If $f \in \mathcal{S}(\mathbb{R})$, then $(f * K_\delta)(x) \to f(x)$ uniformly in x as $\delta \to 0$.

Proof. f is uniformly continuous ($|f| < \epsilon/4$ for $|x| \geq R$, uniform continuity on $[-R-1, R+1]$); take η with $|f(x-t) - f(x)| < \epsilon$ for $|t| < \eta$. Then

$$|(f * K_\delta)(x) - f(x)| \leq \int_{|t|<\eta} K_\delta(t)|f(x-t) - f(x)|dt + \int_{|t|\geq\eta} K_\delta(t)\, 2\sup|f|\, dt \leq \epsilon + C\int_{|t|\geq\eta} K_\delta < 2\epsilon$$

for δ small, uniformly in x. ■

5.1.5 The Fourier inversion

Proposition 1.8 (Multiplication formula)

If $f, g \in \mathcal{S}(\mathbb{R})$, then $\int f(x)\hat{g}(x)\,dx = \int \hat{f}(y)g(y)\,dy$.

Proof. $F(x,y) = f(x)g(y)e^{-2\pi ixy}$ has $|F| \leq \frac{C}{(1+x^2)(1+y^2)}$, so by Theorem 3.1 the iterated integrals, $\int f\hat{g}$ and $\int \hat{f}g$, agree. ■

Theorem 1.9 (Fourier inversion)

If $f \in \mathcal{S}(\mathbb{R})$, then $f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi ix\xi}\,d\xi$.

Proof.

Step 1 ($x = 0$). $G_\delta(x) = e^{-\pi\delta x^2}$ has $\widehat{G_\delta} = K_\delta$, so $\int fK_\delta = \int \hat{f}G_\delta$ (Proposition 1.8). As $\delta \to 0$ the left side tends to $f(0)$ (Corollary 1.7, K_δ even) and the right to $\int \hat{f}$ ($G_\delta \to 1$ uniformly on compacts, dominated by $|\hat{f}|$).

Step 2 (general x). Apply Step 1 to $f(\cdot + x)$, with transform $\hat{f}(\xi)e^{2\pi ix\xi}$. ■

With $\mathcal{F}(f) = \hat{f}$ and $\mathcal{F}^*(g)(x) = \int g(\xi)e^{2\pi ix\xi}d\xi = \mathcal{F}(g)(-x)$, $\mathcal{F}^*\mathcal{F} = I$ on $\mathcal{S}$.

Corollary 1.10

The Fourier transform is a bijection of $\mathcal{S}(\mathbb{R})$ onto itself.

Proof. $\mathcal{F}(\mathcal{S}) \subseteq \mathcal{S}$ (Theorem 1.3), $\mathcal{F}^*\mathcal{F} = I$ (Theorem 1.9), and $\mathcal{F}\mathcal{F}^* = I$ as $\mathcal{F}^*(g) = \mathcal{F}(g)(-\cdot)$. ■

5.1.6 The Plancherel formula

Proposition 1.11

If $f, g \in \mathcal{S}(\mathbb{R})$:
(i) $f * g \in \mathcal{S}(\mathbb{R})$;
(ii) $f * g = g * f$;
(iii) $\widehat{f * g}(\xi) = \hat{f}(\xi)\hat{g}(\xi)$.

Proof.

(ii) Substitute $t \mapsto x - t$.

(i) $|x|^\ell \le 2^\ell(|x-y|^\ell + |y|^\ell)$ gives $\sup_x |x|^\ell |g(x-y)| \le A_\ell(1+|y|)^\ell$, so $\sup_x |x^\ell (f*g)| \le A_\ell \int |f(y)|(1+|y|)^\ell dy < \infty$; and $\frac{d}{dx}(f*g) = f*g'$, repeated.

(iii) $f(y)g(x-y)e^{-2\pi i x\xi}$ is integrable on $\mathbb{R}^2$, so by Fubini (Theorem 3.1), $\int (f*g)(x)e^{-2\pi i x\xi}dx = \int f(y)e^{-2\pi i y\xi}\Big(\int g(x-y)e^{-2\pi i(x-y)\xi}dx\Big)dy = \hat{f}(\xi)\hat{g}(\xi)$. ■

Theorem 1.12 (Plancherel)

If $f \in \mathcal{S}(\mathbb{R})$, then $\|\hat{f}\| = \|f\|$, where $\|f\| = \big(\int |f|^2\big)^{1/2}$.

Proof. $f^\flat(x) = \overline{f(-x)}$ has $\widehat{f^\flat} = \overline{\hat{f}}$, so $h = f * f^\flat$ has $\hat{h} = |\hat{f}|^2$ and $h(0) = \int |f|^2$; inversion at 0 gives $h(0) = \int |\hat{f}|^2$. ■

5.1.7 Extension to functions of moderate decrease

For $f, g \in \mathcal{M}(\mathbb{R})$, $f*g \in \mathcal{M}(\mathbb{R})$, $\widehat{f*g} = \hat{f}\hat{g}$, and the multiplication formula holds; so inversion and Plancherel hold whenever $f, \hat{f} \in \mathcal{M}(\mathbb{R})$.

5.1.8 The Weierstrass approximation theorem

Theorem 1.13 (Weierstrass)

If f is continuous on $[a,b]$ and $\epsilon > 0$, there is a polynomial P with $\sup_{[a,b]} |f - P| < \epsilon$.

Proof. Take $[-M, M] \supset [a,b]$ in its interior and extend f to continuous g on $\mathbb{R}$, linear to 0 on $[b, M]$ and $[-M, a]$, 0 outside; $|g| \le B$. As in Corollary 1.7, fix δ with $|g * K_\delta - g| < \epsilon/2$. Truncating the Taylor series of $e^{-\pi x^2/\delta}$ (uniform on $|x| \le 2M$) gives a polynomial R with $|K_\delta - R| < \frac{\epsilon}{4MB}$ on $|x| \le 2M$. Then $P(x) = \int_{-M}^{M} g(t)R(x-t)\,dt$ is a polynomial, and for $|x| \le M$,

$$|(g*K_\delta)(x) - P(x)| = \Big|\int_{-M}^{M} g(t)(K_\delta(x-t) - R(x-t))dt\Big| \le 2MB\frac{\epsilon}{4MB} = \frac{\epsilon}{2}.$$

■

5.2 Applications to some partial differential equations

Motivation. ∂_x^2 becomes multiplication by $-4\pi^2\xi^2$, so the heat and Laplace equations are solved by Gaussian and Poisson convolutions; uniqueness needs control at infinity.

5.2.1 The time-dependent heat equation on the real line

Seek u with $\partial_t u = \partial_x^2 u$ $(t > 0)$, $u(x,0) = f(x)$. In x: $\partial_t \hat{u} = -4\pi^2\xi^2\hat{u}$, so $\hat{u} = \hat{f}(\xi)e^{-4\pi^2\xi^2 t}$ and $u = f * \mathcal{H}_t$, the **heat kernel** being

$$\mathcal{H}_t(x) = K_\delta(x) \text{ with } \delta = 4\pi t, \qquad \mathcal{H}_t(x) = \frac{1}{(4\pi t)^{1/2}}e^{-x^2/4t}, \qquad \widehat{\mathcal{H}_t}(\xi) = e^{-4\pi^2 t\xi^2}.$$

Theorem 2.1

Given $f \in \mathcal{S}(\mathbb{R})$, let $u(x,t) = (f * \mathcal{H}_t)(x)$ for $t > 0$. Then:
(i) u is C^2 for $x \in \mathbb{R}$, $t > 0$, and solves the heat equation.
(ii) $u(x,t) \to f(x)$ uniformly in x as $t \to 0$; hence, setting $u(x,0) = f(x)$, u is continuous on $\{t \geq 0\}$.
(iii) $\int |u(x,t) - f(x)|^2 dx \to 0$ as $t \to 0$.

Proof.

(i) $u = \int \hat{f}(\xi) e^{-4\pi^2\xi^2 t} e^{2\pi i x\xi} d\xi$ (Proposition 1.11 and inversion); differentiating under the integral (rapid decay of $\hat{f}$) gives $\partial_t u = \partial_x^2 u$.

(ii) Corollary 1.7 with $\delta = 4\pi t$.

(iii) By Plancherel, $\int |u - f|^2 = \int |\hat{f}|^2 |e^{-4\pi^2\xi^2 t} - 1|^2 d\xi$; split at $|\xi| = N$, using $|e^{-4\pi^2\xi^2 t} - 1| \leq 4\pi^2 N^2 t$ inside and ≤ 1 with $\int_{|\xi|>N} |\hat{f}|^2 < \epsilon$ outside. ∎

Corollary 2.2

$u(\cdot, t) \in \mathcal{S}(\mathbb{R})$ uniformly in t: for every $T > 0$ and $k, \ell \geq 0$,

$$\sup_{x \in \mathbb{R},\ 0<t<T} |x|^k |\partial_x^\ell u(x,t)| < \infty. \tag{5.1}$$

Proof. $\partial_x^\ell u = (\partial_x^\ell f) * \mathcal{H}_t$, so take $\ell = 0$, $|x| \geq 1$, and split $\int f(x-y)\mathcal{H}_t(y)dy$ at $|y| = |x|/2$. For $|y| \leq |x|/2$, $|f(x-y)| \leq C_k |x|^{-k}$ and $\int \mathcal{H}_t = 1$. For $|y| \geq |x|/2$, $\mathcal{H}_t(y) \leq (4\pi t)^{-1/2} e^{-x^2/16t}$, and [Supplement] with $s = x^2/t \geq x^2/T$, $t^{-1/2} e^{-x^2/16t} = |x|^{-1} s^{1/2} e^{-s/16} \leq C|x|^{-1} e^{-x^2/32T} \leq C_{k,T} |x|^{-k}$; with $\int |f| < \infty$, $|u(x,t)| \leq C|x|^{-k}$. ∎

Theorem 2.3 (Uniqueness)

Suppose (i) u is continuous on $\{t \geq 0\}$;
(ii) u satisfies the heat equation for $t > 0$;
(iii) $u(x,0) = 0$;
(iv) $u(\cdot, t) \in \mathcal{S}(\mathbb{R})$ uniformly in t as in (5.1). Then $u = 0$.

Proof. $E(t) = \int |u(x,t)|^2 dx \geq 0$, $E(0) = 0$. Differentiating and integrating by parts (decay kills boundary terms),

$$\frac{dE}{dt} = \int (\partial_t u\, \bar{u} + u\, \overline{\partial_t u}) = \int (\partial_x^2 u\, \bar{u} + u\, \overline{\partial_x^2 u}) = -2 \int |\partial_x u|^2 \leq 0.$$

So E decreases, $E \equiv 0$, and $u = 0$. ∎

(Without decay conditions such as (iv), uniqueness fails.)

5.2.2 The steady-state heat equation in the upper half-plane

Find u on $\mathbb{R}^2_+ = \{y > 0\}$ with $\triangle u = 0$, $u(x,0) = f(x)$. In x: $\partial_y^2 \hat{u} = 4\pi^2\xi^2 \hat{u}$, so $\hat{u} = Ae^{2\pi|\xi|y} + Be^{-2\pi|\xi|y}$; boundedness gives $\hat{u} = \hat{f}(\xi) e^{-2\pi|\xi|y}$ and $u = f * \mathcal{P}_y$, the **Poisson kernel** $\mathcal{P}_y(x) = \frac{1}{\pi} \frac{y}{x^2+y^2}$.

Lemma 2.4

$\int_{-\infty}^{\infty} e^{-2\pi|\xi|y} e^{2\pi i\xi x} d\xi = \mathcal{P}_y(x)$ and $\int_{-\infty}^{\infty} \mathcal{P}_y(x) e^{-2\pi i x\xi} dx = e^{-2\pi|\xi|y}$.

Proof. Splitting at 0: $\int_0^\infty e^{-2\pi\xi(y-ix)}d\xi + \int_0^\infty e^{-2\pi\xi(y+ix)}d\xi = \frac{1}{2\pi(y-ix)} + \frac{1}{2\pi(y+ix)} = \frac{1}{\pi}\frac{y}{x^2+y^2}$. Both functions are of moderate decrease, so inversion (§5.1.7) gives the second identity. ■

Lemma 2.5

$\{\mathcal{P}_y\}$ is a family of good kernels on $\mathbb{R}$ as $y \to 0$.

Proof. $\mathcal{P}_y > 0$; $\int \mathcal{P}_y = \widehat{\mathcal{P}_y}(0) = 1$ (or $\frac{1}{\pi}\int \frac{dx}{1+x^2} = 1$ after $x \mapsto yx$); $\int_{|x|\geq\delta} \mathcal{P}_y = \frac{1}{\pi}\int_{|x|\geq\delta/y} \frac{dx}{1+x^2} \to 0$. ■

Theorem 2.6

Given $f \in \mathcal{S}(\mathbb{R})$, let $u(x,y) = (f * \mathcal{P}_y)(x)$. Then:
(i) u is C^2 in $\mathbb{R}^2_+$ and $\triangle u = 0$;
(ii) $u(x,y) \to f(x)$ uniformly as $y \to 0$;
(iii) $\int |u(x,y) - f(x)|^2 dx \to 0$ as $y \to 0$;
(iv) with $u(x,0) = f(x)$, u is continuous on $\overline{\mathbb{R}^2_+}$ and vanishes at infinity: $u(x,y) \to 0$ as $|x| + y \to \infty$.

Proof.

(i)–(iii) As for Theorem 2.1: $u = \int \hat{f}(\xi) e^{-2\pi|\xi|y} e^{2\pi i x\xi} d\xi$ and $(\partial_x^2 + \partial_y^2) e^{-2\pi|\xi|y} e^{2\pi i x\xi} = 0$; Lemma 2.5; Plancherel.

(iv) Splitting at $|t| = |x|/2$, $|u| \leq C(\frac{1}{1+x^2} + \frac{y}{x^2+y^2})$; also $|u| \leq \sup \mathcal{P}_y \int |f| \leq C/y$. Use the first for $|x| \geq y$, the second for $|x| \leq y$. ■

Theorem 2.7

If u is continuous on $\overline{\mathbb{R}^2_+}$, $\triangle u = 0$ in $\mathbb{R}^2_+$, $u(x,0) = 0$, and u vanishes at infinity, then $u = 0$.

Lemma 2.8 (Mean-value property)

Let u be C^2 with $\triangle u = 0$ in an open $\Omega \subseteq \mathbb{R}^2$. If the closed disc of radius R centred at (x,y) lies in Ω, then $u(x,y) = \frac{1}{2\pi}\int_0^{2\pi} u(x + r\cos\theta, y + r\sin\theta)\, d\theta$ for $0 \leq r \leq R$.

Proof. $U(r,\theta) = u(x + r\cos\theta, y + r\sin\theta)$ satisfies $0 = \partial_\theta^2 U + r\partial_r(r\partial_r U)$. For $F(r) = \frac{1}{2\pi}\int_0^{2\pi} U\, d\theta$, $r\partial_r(r\partial_r F) = -\frac{1}{2\pi}\int \partial_\theta^2 U = 0$, so rF' is constant, $= 0$ at $r = 0$, and $F(r) = F(0) = u(x,y)$. ■

Proof of Theorem 2.7. ($u = y$ shows the decay is needed.) Take u real and suppose $u(x_0,y_0) > 0$. Choose R with $u \leq \frac{1}{2}u(x_0,y_0)$ outside $D_R^+ = \{x^2 + y^2 \leq R^2,\ y \geq 0\}$; the maximum M of u on D_R^+, at (x_1,y_1), is its maximum on $\overline{\mathbb{R}^2_+}$, and $y_1 > 0$. By the mean-value property for $r < y_1$, $M = \frac{1}{2\pi}\int u(x_1 + r\cos\theta, y_1 + r\sin\theta)d\theta \leq M$, so $u = M$ on these circles, hence on the disc of radius y_1, whose boundary meets $y = 0$ where $u = 0$: a contradiction. Similarly $u \geq 0$. ■

5.3 The Poisson summation formula

Motivation. Periodizing f gives a function on the circle with Fourier coefficients $\hat{f}(n)$, linking $\mathbb{R}$ and the circle: the theta functional equation and the circle's heat and Poisson kernels.

For $f \in \mathcal{S}(\mathbb{R})$ the **periodization** $F_1(x) = \sum_{n\in\mathbb{Z}} f(x+n)$ is 1-periodic and continuous; compare $F_2(x) = \sum_n \hat{f}(n) e^{2\pi i n x}$.

Theorem 3.1 (Poisson summation formula)

If $f \in \mathcal{S}(\mathbb{R})$, then $\sum_{n\in\mathbb{Z}} f(x+n) = \sum_{n\in\mathbb{Z}} \hat{f}(n)e^{2\pi inx}$. In particular $\sum_n f(n) = \sum_n \hat{f}(n)$.

Proof. Both sides are continuous and 1-periodic, the right absolutely convergent. The m-th coefficient of the left side is

$$\int_0^1 \sum_n f(x+n)e^{-2\pi imx}dx = \sum_n \int_n^{n+1} f(y)e^{-2\pi imy}dy = \int_{-\infty}^{\infty} f(y)e^{-2\pi imy}dy = \hat{f}(m),$$

by uniform convergence, as for the right side; Corollary 2.2 (period 1) gives equality. ■

(Likewise when $f, \hat{f} \in \mathcal{M}(\mathbb{R})$.)

5.3.1 Theta and zeta functions

The **theta function** is $\vartheta(s) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 s}$ for $s > 0$.

Theorem 3.2

$s^{-1/2}\vartheta(1/s) = \vartheta(s)$ for $s > 0$.

Proof. Poisson summation at $x = 0$ for $f = e^{-\pi sx^2}$, $\hat{f}(\xi) = s^{-1/2}e^{-\pi\xi^2/s}$ (Corollary 1.5, $\delta = 1/s$). ■

Via $\pi^{-s/2}\Gamma(s/2)\zeta(s) = \int_0^\infty t^{s/2-1}\frac{\vartheta(t)-1}{2}dt$ this gives the functional equation of $\zeta(s) = \sum n^{-s}$ (Book II).

5.3.2 Heat kernels

Theorem 3.3

The heat kernel of the circle is the periodization of the heat kernel of the line: $H_t(x) = \sum_{n\in\mathbb{Z}} \mathcal{H}_t(x+n)$.

Proof. Poisson summation for $f = \mathcal{H}_t$, $\hat{f}(\xi) = e^{-4\pi^2 t\xi^2}$: $\sum_n \mathcal{H}_t(x+n) = \sum_n e^{-4\pi^2 tn^2}e^{2\pi inx} = H_t(x)$. ■

Corollary 3.4

H_t is a good kernel as $t \to 0$ (on the circle $\mathbb{R}/\mathbb{Z}$, with integrals over $|x| \le \frac{1}{2}$).

Proof. $H_t > 0$ by Theorem 3.3, $\int_{|x|\le 1/2} H_t = \hat{H}_t(0) = 1$. For $|x| \le \frac{1}{2}$, $H_t = \mathcal{H}_t + E_t$ with $E_t(x) = \sum_{n\neq 0} \mathcal{H}_t(x+n) \le Ct^{-1/2}\sum_{n\ge 1} e^{-cn^2/t} \to 0$ uniformly ($|x+n| \ge |n|/2$), and $\mathcal{H}_t(x) \le (4\pi t)^{-1/2}e^{-\delta^2/4t} \to 0$ for $\delta \le |x| \le \frac{1}{2}$. ■

5.3.3 Poisson kernels

Theorem 3.5

$P_r(2\pi x) = \sum_{n\in\mathbb{Z}} \mathcal{P}_y(x+n)$ with $r = e^{-2\pi y}$.

Proof. Poisson summation for $f = \mathcal{P}_y$, $\hat{f} = e^{-2\pi|\xi|y}$ (moderate decrease): $\sum_n \mathcal{P}_y(x+n) = \sum r^{|n|} e^{in(2\pi x)} = P_r(2\pi x)$. ∎

5.4 The Heisenberg uncertainty principle

Motivation. f and $\hat{f}$ cannot both be sharply localized: the product of their spreads is at least $1/16\pi^2$, with equality only for Gaussians.

Theorem 4.1 (Heisenberg)

If $\psi \in \mathcal{S}(\mathbb{R})$ with $\int |\psi|^2 = 1$, then

$$\Big(\int x^2|\psi(x)|^2dx\Big)\Big(\int \xi^2|\hat{\psi}(\xi)|^2d\xi\Big) \geq \frac{1}{16\pi^2},$$

with equality iff $\psi(x) = Ae^{-Bx^2}$ with $B > 0$ and $|A|^2 = \sqrt{2B/\pi}$. More generally $\big(\int (x-x_0)^2|\psi|^2\big)\big(\int(\xi-\xi_0)^2|\hat{\psi}|^2\big) \geq \frac{1}{16\pi^2}$ for all $x_0, \xi_0 \in \mathbb{R}$.

Proof.

Step 1. By parts (ψ, ψ' rapidly decreasing),

$$1 = \int |\psi|^2 = -\int x\frac{d}{dx}|\psi|^2 = -\int (x\psi'\bar{\psi} + x\psi\overline{\psi'}),$$

so by Cauchy–Schwarz $1 \leq 2\int |x||\psi||\psi'| \leq 2\big(\int x^2|\psi|^2\big)^{1/2}\big(\int |\psi'|^2\big)^{1/2}$.

Step 2. By Proposition 1.2(iv) and Plancherel, $\int |\psi'|^2 = 4\pi^2 \int \xi^2|\hat{\psi}|^2$. Squaring gives the inequality.

Step 3 (equality). Equality in Cauchy–Schwarz forces $\psi' = \beta x\psi$, so $\psi = Ae^{\beta x^2/2}$; $\psi \in \mathcal{S}$ forces $\beta = -2B < 0$, and $\int |\psi|^2 = |A|^2\sqrt{\pi/2B} = 1$.

Step 4. For the general form apply the inequality to $e^{-2\pi i x\xi_0}\psi(x+x_0)$. ∎

Quantum mechanics. For a state ψ, $\int |\psi|^2 = 1$, the particle lies in (a,b) with probability $\int_a^b |\psi|^2$, with mean $\bar{x} = \int x|\psi|^2$ and variance $(\Delta x)^2 = \int (x-\bar{x})^2|\psi|^2$. Momentum is governed likewise by $\hat{\psi}$ (frequency ξ corresponds to momentum $h\xi$), and Theorem 4.1 reads $(\Delta x)(\Delta\xi) \geq \frac{1}{4\pi}$, i.e. $(\Delta\,\text{position})(\Delta\,\text{momentum}) \geq \frac{h}{4\pi}$.

Chapter 6. The Fourier Transform on $\mathbb{R}^d$

6.1 Preliminaries

Motivation. On $\mathbb{R}^d$ rotations join translations and dilations, and polar coordinates split $\mathbb{R}^d$ into radii and directions.

6.1.1 Symmetries

Analysis on $\mathbb{R}^d$ is shaped by **translations** $x \mapsto x+h$, **dilations** $x \mapsto \delta x$ ($\delta > 0$) and **rotations**: linear R with $R(x)\cdot R(y) = x \cdot y$, i.e. $|Rx| = |x|$, or $R^t = R^{-1}$; $\det R = \pm 1$ (**proper** if $+1$). On $\mathbb{R}$: $\pm I$; on $\mathbb{R}^2 = \mathbb{C}$: $z \mapsto e^{i\varphi}z$, $e^{i\varphi}\bar{z}$; for $d = 3$, Euler angles parametrize proper rotations.

6.1.2 Integration on $\mathbb{R}^d$

A continuous f is **rapidly decreasing** if $\sup_x |x|^k |f(x)| < \infty$ for all k; then $\int_{\mathbb{R}^d} f = \lim_N \int_{Q_N} f$ over cubes $Q_N = \{|x_i| \le N/2\}$ (or balls). $\sup |x|^{d+\epsilon}|f(x)| < \infty$ suffices; **moderate decrease** means $\epsilon = 1$. Then

$$\int f(x+h)\,dx = \int f(x)\,dx, \qquad \delta^d \int f(\delta x)\,dx = \int f(x)\,dx, \qquad \int f(Rx)\,dx = \int f(x)\,dx.$$

Polar coordinates. With $x = r\gamma$, $r \ge 0$, $\gamma \in S^{d-1}$, and surface measure $d\sigma$,

$$\int_{\mathbb{R}^d} f(x)\,dx = \int_{S^{d-1}} \int_0^\infty f(r\gamma)\, r^{d-1}\, dr\, d\sigma(\gamma).$$

For $d = 2$: $\int f = \int_0^{2\pi}\int_0^\infty f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta$; for $d = 3$: $x = (r\sin\theta\cos\varphi, r\sin\theta\sin\varphi, r\cos\theta)$, $d\sigma = \sin\theta\, d\theta\, d\varphi$. S^{d-1} has area $\omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ (from $\int e^{-\pi|x|^2}dx = 1$).

6.2 Elementary theory of the Fourier transform

Motivation. Chapter 5 extends with $x\xi$ replaced by $x\cdot\xi$; new is that the transform commutes with rotations, so radial functions have radial transforms.

Multi-indices: $x^\alpha = x_1^{\alpha_1}\cdots x_d^{\alpha_d}$, $(\frac{\partial}{\partial x})^\alpha$, $|\alpha| = \alpha_1 + \cdots + \alpha_d$. $\mathcal{S}(\mathbb{R}^d)$: smooth f with $\sup_x |x^\alpha (\frac{\partial}{\partial x})^\beta f| < \infty$ for all α, β; $\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i x\cdot\xi} dx$.

Proposition 2.1

For $f \in \mathcal{S}(\mathbb{R}^d)$:
(i) $f(x+h) \to \hat{f}(\xi)e^{2\pi i\xi\cdot h}$;
(ii) $f(x)e^{-2\pi i x\cdot h} \to \hat{f}(\xi+h)$;
(iii) $f(\delta x) \to \delta^{-d}\hat{f}(\delta^{-1}\xi)$, $\delta > 0$;
(iv) $(\frac{\partial}{\partial x})^\alpha f(x) \to (2\pi i\xi)^\alpha \hat{f}(\xi)$;
(v) $(-2\pi i x)^\alpha f(x) \to (\frac{\partial}{\partial \xi})^\alpha \hat{f}(\xi)$;
(vi) $f(Rx) \to \hat{f}(R\xi)$ for every rotation R.

Proof. (i)–(v) as in Proposition 1.2, one variable at a time.
(vi) With $y = Rx$ and $R^{-1}y\cdot\xi = y\cdot R\xi$: $\int f(Rx)e^{-2\pi i x\cdot\xi}dx = \int f(y)e^{-2\pi i y\cdot R\xi}dy = \hat{f}(R\xi)$. ∎

Corollary 2.2

The Fourier transform maps $\mathcal{S}(\mathbb{R}^d)$ to itself.

Proof. By (iv), (v), $\xi^\alpha(\partial_\xi)^\beta \hat{f}$ is the transform of a Schwartz function, hence bounded. ∎

Corollary 2.3

The Fourier transform of a radial function ($f(Rx) = f(x)$ for all rotations R) is radial.

Proof. By (vi), $\hat{f}(R\xi) = \hat{f}(\xi)$ for all R. ∎

The Gaussian $e^{-\pi|x|^2} = \prod_j e^{-\pi x_j^2}$ is radial and its own transform, so $K_\delta(x) = \delta^{-d/2} e^{-\pi|x|^2/\delta}$ has $\widehat{K_\delta}(\xi) = e^{-\pi\delta|\xi|^2}$ and is a good kernel as $\delta \to 0$.

Theorem 2.4 (Fourier inversion and Plancherel)

If $f \in \mathcal{S}(\mathbb{R}^d)$, then $f(x) = \int_{\mathbb{R}^d} \hat{f}(\xi) e^{2\pi i x\cdot\xi} d\xi$ and $\int |\hat{f}(\xi)|^2 d\xi = \int |f(x)|^2 dx$. In particular $\mathcal{F}$ is a bijection of $\mathcal{S}(\mathbb{R}^d)$.

Proof. As in Chapter 5: the multiplication formula $\int f\hat{g} = \int \hat{f} g$ (Fubini) with $g = e^{-\pi\delta|x|^2}$ and the good kernels K_δ give inversion at 0; translate. For Plancherel use $h = f * f^\flat$, $f^\flat(x) = \overline{f(-x)}$, and $\widehat{f * g} = \hat{f}\hat{g}$. ∎

6.3 The wave equation in $\mathbb{R}^d \times \mathbb{R}$

Motivation. Fourier transform solves the wave equation and conserves energy; in $\mathbb{R}^3$ spherical means give Huygens' principle, and $\mathbb{R}^2$ follows by descent.

Cauchy problem. Find $u(x,t)$ with $\triangle u = \frac{\partial^2 u}{\partial t^2}$ (speed $c = 1$), $u(x,0) = f(x)$, $\partial_t u(x,0) = g(x)$, $f, g \in \mathcal{S}(\mathbb{R}^d)$.

6.3.1 Solution in terms of Fourier transforms

In x: $\partial_t^2 \hat{u} = -4\pi^2|\xi|^2 \hat{u}$, so $\hat{u} = \hat{f}(\xi)\cos(2\pi|\xi|t) + \hat{g}(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}$.

Theorem 3.1

A solution of the Cauchy problem for the wave equation is

$$u(x,t) = \int_{\mathbb{R}^d} \left[\hat{f}(\xi)\cos(2\pi|\xi|t) + \hat{g}(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}\right] e^{2\pi i x\cdot\xi} d\xi. \tag{6.1}$$

Proof. $\cos(2\pi|\xi|t)$ and $\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}$ are smooth in ξ (even power series in $|\xi|$) and, with their t-derivatives, polynomially bounded for bounded t. As $\hat{f}, \hat{g}$ decay rapidly, differentiate twice under the integral: each term satisfies $\triangle_x = -4\pi^2|\xi|^2 = \partial_t^2$. At $t = 0$ inversion gives $u = f$, $\partial_t u = g$. ∎

The **energy** of u is $E(t) = \int_{\mathbb{R}^d} \left(\left|\frac{\partial u}{\partial t}\right|^2 + \sum_j \left|\frac{\partial u}{\partial x_j}\right|^2\right) dx$.

Theorem 3.2

If u is given by (6.1), then $E(t) = E(0)$ for all t.

Lemma 3.3

For $a, b \in \mathbb{C}$ and $\alpha \in \mathbb{R}$: $|a\cos\alpha + b\sin\alpha|^2 + |-a\sin\alpha + b\cos\alpha|^2 = |a|^2 + |b|^2$.

Proof. $e_1 = (\cos\alpha, \sin\alpha)$, $e_2 = (-\sin\alpha, \cos\alpha)$ form an orthonormal basis of $\mathbb{R}^2 \subseteq \mathbb{C}^2$, and the identity is Pythagoras for the coordinates $(a, b) \cdot e_j$. ∎

Proof of Theorem 3.2. By Plancherel and Proposition 2.1(iv), $\int |\partial_t u|^2 = \int |-2\pi|\xi|\hat{f}\sin(2\pi|\xi|t) + \hat{g}\cos(2\pi|\xi|t)|^2 d\xi$ and $\sum_j \int |\partial_{x_j} u|^2 = \int |2\pi|\xi|\hat{f}\cos(2\pi|\xi|t) + \hat{g}\sin(2\pi|\xi|t)|^2 d\xi$. By Lemma 3.3 ($a = 2\pi|\xi|\hat{f}$, $b = \hat{g}$, $\alpha = 2\pi|\xi|t$) the integrands sum to $4\pi^2|\xi|^2|\hat{f}|^2 + |\hat{g}|^2$, free of t. ∎

Uniqueness [Supplement]

For any solution with Schwartz-type decay, integrating by parts as in Theorem 2.3, $E'(t) = 2\,\mathrm{Re}\int \partial_t u\,\overline{(\partial_t^2 u - \triangle u)} = 0$. So zero data give $E \equiv 0$, u constant, hence 0: (6.1) is the only such solution.

6.3.2 The wave equation in $\mathbb{R}^3 \times \mathbb{R}$

The **spherical mean** $M_t(f)(x) = \frac{1}{4\pi}\int_{S^2} f(x - t\gamma)\, d\sigma(\gamma)$ averages f over the sphere of radius t about x.

Lemma 3.4

If $f \in \mathcal{S}(\mathbb{R}^3)$ and t is fixed, then $M_t(f) \in \mathcal{S}(\mathbb{R}^3)$; moreover $M_t(f)$ is C^∞ in t and each t-derivative lies in $\mathcal{S}(\mathbb{R}^3)$.

Proof. For $|x| \geq 2|t|$, $|x - t\gamma| \geq |x|/2$, so $|f(x - t\gamma)| \leq C_N(1 + |x|)^{-N}$. Derivatives in x, t pass under the integral as spherical means of derivatives of f (times polynomials in γ), likewise rapidly decreasing. ∎

Lemma 3.5

$\frac{1}{4\pi}\int_{S^2} e^{-2\pi i\xi\cdot\gamma} d\sigma(\gamma) = \frac{\sin(2\pi|\xi|)}{2\pi|\xi|}$.

Proof. The left side is radial ($d\sigma$ is rotation invariant), so take $\xi = (0, 0, \rho)$:

$$\frac{1}{4\pi}\int_0^{2\pi}\int_0^{\pi} e^{-2\pi i\rho\cos\theta}\sin\theta\, d\theta\, d\varphi = \frac{1}{2}\int_{-1}^{1} e^{-2\pi i\rho u} du = \frac{\sin(2\pi\rho)}{2\pi\rho}.$$

∎

Hence, by Fubini,

$$\widehat{M_t(f)}(\xi) = \hat{f}(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|t}. \tag{6.2}$$

Theorem 3.6 (Kirchhoff)

For $d = 3$, the solution of the Cauchy problem is $u(x,t) = \frac{\partial}{\partial t}(tM_t(f)(x)) + tM_t(g)(x)$.

Proof. By Theorem 3.1, (6.2) and inversion, the solution with data $(0, g)$ is

$$u_1 = \int \hat{g}(\xi)\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|}e^{2\pi i x\cdot\xi}d\xi = t\int \hat{g}\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|t}e^{2\pi i x\cdot\xi}d\xi = tM_t(g)(x),$$

For data $(f, 0)$, $u_2 = \int \hat{f}\cos(2\pi|\xi|t)e^{2\pi i x\cdot\xi}d\xi = \partial_t\Big(t\int \hat{f}\frac{\sin(2\pi|\xi|t)}{2\pi|\xi|t}e^{2\pi i x\cdot\xi}d\xi\Big) = \partial_t(tM_t(f))$. Superpose. ■

Huygens' principle. In $d = 1$, $u(x,t)$ depends on the data on $[x-t, x+t]$ (only the endpoints if $g = 0$). In $d = 3$ it depends only on the data on the *sphere* $|y - x| = t$: a localized disturbance is felt at distance t exactly at time t and then passes (sharp signals).

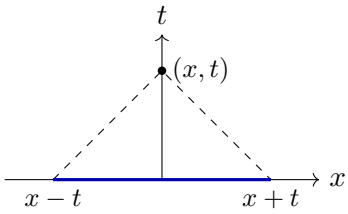


Figure. *Domain of dependence for $d = 1$: the interval $[x-t, x+t]$; for $d = 3$ only the sphere of radius t.*

6.3.3 The wave equation in $\mathbb{R}^2 \times \mathbb{R}$: descent

For $f \in \mathcal{S}(\mathbb{R}^2)$ let $\tilde{M}_t(f)(x) = \frac{1}{2\pi}\int_{|y|\le 1} f(x - ty)(1 - |y|^2)^{-1/2}dy$.

Theorem 3.7

A solution of the Cauchy problem in two dimensions with data $f, g \in \mathcal{S}(\mathbb{R}^2)$ is

$$u(x,t) = \frac{\partial}{\partial t}(t\tilde{M}_t(f)(x)) + t\tilde{M}_t(g)(x). \tag{6.3}$$

Proof.

Step 1 (descent idea). With $\tilde{f}(x_1, x_2, x_3) = f(x_1, x_2)$, a 3-dimensional solution with data independent of x_3 is independent of x_3 and solves the 2-dimensional problem. Projecting the hemispheres $\gamma = (y, \pm\sqrt{1 - |y|^2})$ onto the unit disc, $d\sigma = (1 - |y|^2)^{-1/2}dy$ on each, so $M_t(\tilde{f})(x, x_3) = \frac{2}{4\pi}\int_{|y|\le 1} f(x - ty)(1 - |y|^2)^{-1/2}dy = \tilde{M}_t(f)(x)$.

Step 2 (truncation). As $\tilde{f} \notin \mathcal{S}(\mathbb{R}^3)$, fix $T > 0$, $\eta \in \mathcal{S}(\mathbb{R})$ with $\eta = 1$ on $|x_3| \le 3T$, and $\tilde{f}^\flat = f(x_1, x_2)\eta(x_3)$, $\tilde{g}^\flat$ likewise. By Theorem 3.6, $u^\flat = \partial_t(tM_t(\tilde{f}^\flat)) + tM_t(\tilde{g}^\flat)$ solves the 3-dimensional problem. For $|x_3|, |t| \le T$ the spheres stay in $|x_3| \le 2T$, where $\tilde{f}^\flat = \tilde{f}$, so $u^\flat$ equals (6.3), independent of x_3; thus $\partial_{x_3}^2 u^\flat = 0$ and (6.3) solves the 2-dimensional equation for $|t| < T$, T arbitrary. ■

In two dimensions $u(x,t)$ depends on the whole disc of radius t: Huygens' principle fails (ripples linger).

6.4 Radial symmetry and Bessel functions

Motivation. For radial $f = f_0(|x|)$, $\hat{f} = F_0(|\xi|)$; the relation is elementary for $d = 1, 3$ and involves J_0 for $d = 2$.

- $d = 1$ (radial = even): $F_0(\rho) = \int f(x)e^{-2\pi i x\rho}dx = 2\int_0^\infty \cos(2\pi\rho r) f_0(r)\, dr$.
- $d = 3$: by polar coordinates and Lemma 3.5,

$$F_0(\rho) = \int_0^\infty f_0(r)\Big(\int_{S^2} e^{-2\pi i r\gamma\cdot\xi} d\sigma(\gamma)\Big) r^2 dr$$
$$= \int_0^\infty f_0(r)\frac{2\sin(2\pi\rho r)}{\rho r} r^2 dr = 2\rho^{-1}\int_0^\infty \sin(2\pi\rho r) f_0(r)\, r\, dr.$$

- $d = 2$: the **Bessel function** $J_n(\rho) = \frac{1}{2\pi}\int_0^{2\pi} e^{i\rho\sin\theta}e^{-in\theta}d\theta$ is the n-th Fourier coefficient of $e^{i\rho\sin\theta}$, so $e^{i\rho\sin\theta} = \sum_n J_n(\rho)e^{in\theta}$. With $\xi = (0, -\rho)$,

$$F_0(\rho) = \int_0^{2\pi}\int_0^\infty f_0(r)e^{2\pi i r\rho\sin\theta} r\, dr\, d\theta = 2\pi\int_0^\infty J_0(2\pi r\rho) f_0(r)\, r\, dr.$$

In general F_0 involves a Bessel function of order $d/2 - 1$, elementary only for odd d.

6.5 The Radon transform and some of its applications

Motivation. X-rays measure line or plane integrals of the attenuation; the 1-dimensional transform of the Radon transform is a slice of $\hat{f}$, which allows reconstruction.

6.5.1 The X-ray transform in $\mathbb{R}^2$

A beam of intensity I_0 through an object with attenuation ρ along a line L emerges with $I = I_0 e^{-\int_L \rho}$, so measurements give the **X-ray transform** $X(\rho)(L) = \int_L \rho$. **Reconstruction problem:** recover ρ from $X(\rho)$. In $\mathbb{R}^2$ inversion involves a Hilbert transform; we treat the simpler 3-dimensional analogue.

6.5.2 The Radon transform in $\mathbb{R}^3$

Planes in $\mathbb{R}^3$ are $\mathcal{P}_{t,\gamma} = \{x\cdot\gamma = t\}$, $t\in\mathbb{R}$, $\gamma\in S^2$ ($\mathcal{P}_{t,\gamma} = \mathcal{P}_{-t,-\gamma}$). For $f\in\mathcal{S}(\mathbb{R}^3)$ and $\{e_1, e_2, \gamma\}$ orthonormal,

$$\mathcal{R}(f)(t,\gamma) = \int_{\mathcal{P}_{t,\gamma}} f = \int_{\mathbb{R}^2} f(t\gamma + u_1 e_1 + u_2 e_2)\, du_1\, du_2.$$

Proposition 5.1

For $f\in\mathcal{S}(\mathbb{R}^3)$ and each γ, $\mathcal{R}(f)(t,\gamma)$ does not depend on the choice of e_1, e_2, and $\int_{-\infty}^\infty \mathcal{R}(f)(t,\gamma)\, dt = \int_{\mathbb{R}^3} f(x)\, dx$.

Proof. Another e_1', e_2' differs by a rotation or reflection of $\mathbb{R}^2$, which preserves $\int_{\mathbb{R}^2}$. And $x = t\gamma + u_1 e_1 + u_2 e_2$ is a rotation of $\mathbb{R}^3$, so Fubini gives the identity. ∎

Lemma 5.2

For $f\in\mathcal{S}(\mathbb{R}^3)$ and fixed γ, $\mathcal{R}(f)(\cdot,\gamma)\in\mathcal{S}(\mathbb{R})$, and $\widehat{\mathcal{R}(f)}(s,\gamma) = \hat{f}(s\gamma)$ (one-dimensional transform in t on the left, three-dimensional on the right).

Proof. $|t\gamma + u|^2 = t^2 + |u|^2$ gives $(1+|t|)^N|f(t\gamma+u)| \le C(1+|u|)^{-3}$, so $\mathcal{R}(f)$ decays rapidly in t; t-derivatives are Radon transforms of directional derivatives. Proposition 5.1 for $f(x)e^{-2\pi i s x\cdot\gamma}$ ($= e^{-2\pi i s t} f$ on $\mathcal{P}_{t,\gamma}$) gives $\int \mathcal{R}(f)(t,\gamma)e^{-2\pi i s t}dt = \int_{\mathbb{R}^3} f(x)e^{-2\pi i s x\cdot\gamma}dx = \hat{f}(s\gamma)$. ∎

Corollary 5.3

If $f, g \in \mathcal{S}(\mathbb{R}^3)$ and $\mathcal{R}(f) = \mathcal{R}(g)$, then $f = g$.

Proof. Lemma 5.2 gives $\hat{f}(s\gamma) = \hat{g}(s\gamma)$ for all s, γ, so $\hat{f} = \hat{g}$; invert. ■

The **dual Radon transform** $\mathcal{R}^*(F)(x) = \int_{S^2} F(x \cdot \gamma, \gamma)\, d\sigma(\gamma)$ averages F over the planes through x.

Theorem 5.4 (Radon inversion)

If $f \in \mathcal{S}(\mathbb{R}^3)$, then $\triangle(\mathcal{R}^*\mathcal{R}(f)) = -8\pi^2 f$.

Proof. By Lemma 5.2 and inversion, $\mathcal{R}(f)(t, \gamma) = \int_{\mathbb{R}} \hat{f}(s\gamma) e^{2\pi i t s} ds$, so $\mathcal{R}^*\mathcal{R}(f)(x) = \int_{S^2} \int_{\mathbb{R}} \hat{f}(s\gamma) e^{2\pi i x \cdot \gamma s} ds\, d\sigma(\gamma)$. $\triangle$ multiplies the integrand by $-4\pi^2 s^2$, and $(s, \gamma) \mapsto (-s, -\gamma)$ makes the parts $s < 0$, $s > 0$ equal, so

$$\triangle(\mathcal{R}^*\mathcal{R}(f))(x) = -8\pi^2 \int_{S^2} \int_0^\infty \hat{f}(s\gamma) e^{2\pi i x \cdot s\gamma} s^2 ds\, d\sigma(\gamma) = -8\pi^2 f(x),$$

by inversion in polar coordinates. ■

So $f = -\frac{1}{8\pi^2} \triangle \mathcal{R}^*\mathcal{R}(f)$: plane integrals determine f explicitly, the basis of tomography.

6.6 Selected exercises

Motivation. Rotation invariance, Plancherel with derivatives, and Fubini on $\mathbb{R}^3 \times S^2$ in book exercises.

Exercise 6.5 (Gaussian integrals with a matrix)

Let A be a real symmetric positive definite $d \times d$ matrix. Show that

$$\int_{\mathbb{R}^d} e^{-\pi(x, Ax)}\, dx = (\det A)^{-1/2}.$$

Solution.

Step 1. Take R orthogonal with $R^t A R = D = \operatorname{diag}(\lambda_1, \ldots, \lambda_d)$, $\lambda_i > 0$. With $\lambda = \min_i \lambda_i$, $(x, Ax) \geq \lambda |x|^2$, so $e^{-\pi(x,Ax)}$ is rapidly decreasing.

Step 2. R is a rotation (Section 6.1.1), so $\int f(x)\, dx = \int f(Ry)\, dy$ (Section 6.1.2), and $(Ry, ARy) = (y, Dy) = \sum_i \lambda_i y_i^2$:

$$\int_{\mathbb{R}^d} e^{-\pi(x,Ax)}\, dx = \int_{\mathbb{R}^d} \prod_{i=1}^d e^{-\pi \lambda_i y_i^2}\, dy = \prod_{i=1}^d \int_{\mathbb{R}} e^{-\pi \lambda_i t^2}\, dt,$$

the integral over Q_N factoring before $N \to \infty$.

Step 3. With $s = \sqrt{\lambda_i}\, t$, each factor is $\lambda_i^{-1/2}$, so the integral is $(\det A)^{-1/2}$. ■

Exercise 6.6 (Heisenberg uncertainty principle in $\mathbb{R}^d$)

Let $\psi \in \mathcal{S}(\mathbb{R}^d)$ with $\int |\psi|^2 = 1$. Show that

$$\Big(\int_{\mathbb{R}^d} |x|^2 |\psi(x)|^2 dx\Big)\Big(\int_{\mathbb{R}^d} |\xi|^2 |\hat{\psi}(\xi)|^2 d\xi\Big) \geq \frac{d^2}{16\pi^2}.$$

Solution.

Step 1. Integrating by parts in x_j (other coordinates fixed),

$$\int_{\mathbb{R}} |\psi|^2 \, dx_j = \big[x_j |\psi|^2\big]_{x_j=-\infty}^{\infty} - \int_{\mathbb{R}} x_j \, \partial_j(|\psi|^2) \, dx_j = -\int_{\mathbb{R}} x_j \, \partial_j(|\psi|^2) \, dx_j,$$

the boundary term vanishing by rapid decay; by Fubini, $1 = -\int x_j \partial_j(|\psi|^2)\, dx$.

Step 2. $\partial_j |\psi|^2 = 2\,\mathrm{Re}(\bar{\psi}\, \partial_j \psi)$, so summing over j

$$d = -2\,\mathrm{Re} \int_{\mathbb{R}^d} \bar{\psi}(x) \sum_{j=1}^{d} x_j \, \partial_j \psi(x) \, dx \leq 2 \int_{\mathbb{R}^d} |x| \, |\psi(x)| \, |\nabla \psi(x)| \, dx,$$

as $|\sum_j x_j \partial_j \psi| \leq |x| \, |\nabla \psi|$.

Step 3. By Cauchy–Schwarz for integrals,

$$d^2 \leq 4 \int |x|^2 |\psi|^2 dx \cdot \int |\nabla \psi|^2 dx.$$

Step 4. $\widehat{\partial_j \psi} = 2\pi i \xi_j \hat{\psi}$ and Plancherel (Theorem 2.4) give

$$\int |\nabla \psi|^2 dx = \sum_{j=1}^{d} \int 4\pi^2 \xi_j^2 |\hat{\psi}(\xi)|^2 d\xi = 4\pi^2 \int |\xi|^2 |\hat{\psi}(\xi)|^2 d\xi.$$

so $d^2 \leq 16\pi^2 \int |x|^2 |\psi|^2 \int |\xi|^2 |\hat{\psi}|^2$. ∎

Exercise 6.12 (The dual Radon transform is the adjoint)

For $f \in \mathcal{S}(\mathbb{R}^3)$ and $F \in \mathcal{S}(\mathbb{R} \times S^2)$ prove

$$\int_{\mathbb{R}} \int_{S^2} \mathcal{R}(f)(t,\gamma) \, \overline{F(t,\gamma)} \, d\sigma(\gamma) \, dt = \int_{\mathbb{R}^3} f(x) \, \overline{\mathcal{R}^*(F)(x)} \, dx.$$

Solution.

Step 1. For fixed γ, $h_\gamma(x) = f(x)\overline{F(x \cdot \gamma, \gamma)}$ is continuous, rapidly decreasing, and equals $f(x)\overline{F(t,\gamma)}$ on $\mathcal{P}_{t,\gamma}$, so $\mathcal{R}(h_\gamma)(t,\gamma) = \mathcal{R}(f)(t,\gamma)\overline{F(t,\gamma)}$.

Step 2. The proof of Proposition 5.1 applies to h_γ:

$$\int_{\mathbb{R}} \mathcal{R}(f)(t,\gamma) \overline{F(t,\gamma)} \, dt = \int_{\mathbb{R}^3} f(x) \overline{F(x \cdot \gamma, \gamma)} \, dx.$$

Step 3. Integrate over γ. As $\int_{S^2} \int_{\mathbb{R}^3} |f(x)| \, |F(x \cdot \gamma, \gamma)| \, dx \, d\sigma \leq 4\pi \sup |F| \int |f| < \infty$, Fubini gives

$$\int_{S^2} \int_{\mathbb{R}} \mathcal{R}(f) \overline{F} \, dt \, d\sigma(\gamma) = \int_{\mathbb{R}^3} f(x) \, \overline{\int_{S^2} F(x \cdot \gamma, \gamma) \, d\sigma(\gamma)} \, dx = \int_{\mathbb{R}^3} f \, \overline{\mathcal{R}^*(F)} \, dx.$$

Step 4. On the left, $|\mathcal{R}(f)F| \leq \sup |F| \, \mathcal{R}(|f|)$ and $\int_{\mathbb{R}} \mathcal{R}(|f|)(t,\gamma)\, dt = \int |f|$, so the integrand is integrable on $\mathbb{R} \times S^2$ and the order $\int_{\mathbb{R}} \int_{S^2}$ is allowed: $(\mathcal{R}f, F)_2 = (f, \mathcal{R}^* F)_1$. ∎

Chapter 7. Finite Fourier Analysis

7.1 Fourier analysis on $\mathbb{Z}(N)$

Motivation. On the N-th roots of unity Fourier series are finite sums with an orthogonal basis of exponentials, and even/odd splitting computes them in $O(N \log N)$ (FFT).

7.1.1 The group $\mathbb{Z}(N)$

For $N \geq 1$ let $\zeta = e^{2\pi i/N}$. The roots $\{1, \zeta, \ldots, \zeta^{N-1}\}$ form a multiplicative group, identified via $\zeta^k \mapsto k$ with $\mathbb{Z}(N) = \mathbb{Z}/N\mathbb{Z}$; functions on $\mathbb{Z}(N)$ are N-periodic functions on $\mathbb{Z}$, or functions on N equally spaced points of the circle (*discretization*).

7.1.2 Fourier inversion theorem and Plancherel identity on $\mathbb{Z}(N)$

The functions on $\mathbb{Z}(N)$ form an N-dimensional space V with $(F, G) = \sum_{k=0}^{N-1} F(k)\overline{G(k)}$, $\|F\|^2 = \sum |F(k)|^2$. Let $e_\ell(k) = \zeta^{\ell k}$, $\ell = 0, \ldots, N-1$.

Lemma 1.1

The family $\{e_0, \ldots, e_{N-1}\}$ is orthogonal: $(e_m, e_\ell) = N$ if $m = \ell$ and 0 if $m \neq \ell$. Hence $\{e_\ell/\sqrt{N}\}$ is an orthonormal basis of V.

Proof. $(e_m, e_\ell) = \sum_{k=0}^{N-1} \zeta^{(m-\ell)k}$, which is N if $m = \ell$; otherwise $q = \zeta^{m-\ell} \neq 1$, $q^N = 1$ and the sum is $\frac{1-q^N}{1-q} = 0$. N orthogonal nonzero vectors form a basis. ∎

The n-th **Fourier coefficient** of F is $a_n = \frac{1}{N}(F, e_n) = \frac{1}{N}\sum_{k=0}^{N-1} F(k)\zeta^{-kn}$.

Theorem 1.2

If F is a function on $\mathbb{Z}(N)$, then $F(k) = \sum_{n=0}^{N-1} a_n \zeta^{nk}$ (inversion) and $\sum_{n=0}^{N-1} |a_n|^2 = \frac{1}{N}\sum_{k=0}^{N-1} |F(k)|^2$ (Parseval).

Proof. In the orthonormal basis $e_n^* = e_n/\sqrt{N}$: $F = \sum (F, e_n^*)e_n^* = \sum a_n e_n$, and Pythagoras gives $\|F\|^2 = \sum N|a_n|^2$. ∎

(If $F(k) = f(k/N)$ for 1-periodic f, a_n is a Riemann sum for $\hat{f}(n)$.)

7.1.3 The fast Fourier transform

Naively all a_n cost about N^2 operations. Let $\#(M)$ be the least number of operations needed for all Fourier coefficients on $\mathbb{Z}(M)$, with $\omega_M = e^{-2\pi i/M}$ and $a_k^M(F) = \frac{1}{M}\sum_r F(r)\omega_M^{kr}$.

Theorem 1.3

For $N = 2^n$, the Fourier coefficients of a function on $\mathbb{Z}(N)$ can be computed with at most $4 \cdot 2^n n = 4N \log_2 N = O(N \log N)$ operations, given ω_N.

Lemma 1.4

Given ω_{2M}, $\#(2M) \le 2\#(M) + 8M$.

Proof. $\omega_{2M}, \dots, \omega_{2M}^{2M}$ cost $\le 2M$ operations and give $\omega_M = \omega_{2M}^2$. With $F_0(r) = F(2r)$, $F_1(r) = F(2r+1)$ on $\mathbb{Z}(M)$, splitting into even and odd indices,

$$a_k^{2M}(F) = \frac{1}{2M}\sum_{r=0}^{2M-1} F(r)\omega_{2M}^{kr} = \frac{1}{2}\Big(\frac{1}{M}\sum_{\ell} F_0(\ell)\omega_M^{k\ell} + \frac{1}{M}\sum_{m} F_1(m)\omega_M^{km}\omega_{2M}^{k}\Big)$$
$$= \frac{1}{2}\big(a_k^M(F_0) + a_k^M(F_1)\omega_{2M}^k\big)$$

(a_k^M M-periodic in k). $a^M(F_0)$, $a^M(F_1)$ cost $2\#(M)$, then 3 operations per coefficient: $\#(2M) \le 2M + 2\#(M) + 6M$. ∎

Proof of Theorem 1.3. Induction on n. $N = 2$: $a_0 = \frac{1}{2}(F(0)+F(1))$, $a_1 = \frac{1}{2}(F(0)-F(1))$, ≤ 8 operations. If $\#(2^{n-1}) \le 4\cdot 2^{n-1}(n-1)$, Lemma 1.4 gives $\#(2^n) \le 2\cdot 4\cdot 2^{n-1}(n-1) + 8\cdot 2^{n-1} = 4\cdot 2^n n$. ∎

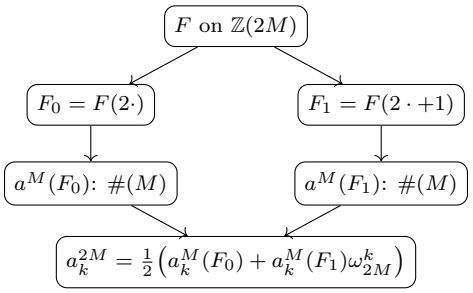


Figure. *One step of the fast Fourier transform (Lemma 1.4).*

7.2 Fourier analysis on finite abelian groups

Motivation. On a finite abelian group, characters replace exponentials; they are orthonormal and, by simultaneous diagonalization of translations, a basis.

7.2.1 Abelian groups

An **abelian group** $(G, \cdot)$ is associative and commutative with identity 1 and inverses: e.g. $\mathbb{Z}(N)$, $\mathbb{R}/\mathbb{Z}$, $\mathbb{R}$, $\mathbb{R}^+ = (0,\infty)$ under multiplication, and $\mathbb{Z}^*(q)$, the residues coprime to q under multiplication. A **homomorphism** satisfies $f(ab) = f(a)f(b)$; an **isomorphism** is a bijective one. $G_1 \times G_2$ multiplies componentwise ($\mathbb{Z}^*(8) \cong \mathbb{Z}(2) \times \mathbb{Z}(2)$, $\mathbb{Z}^*(5) \cong \mathbb{Z}(4)$). Every finite abelian group is a product of cyclic groups; we will not need this.

7.2.2 Characters

A **character** of G is a homomorphism $e : G \to S^1 = \{|z| = 1\}$; the **trivial character** is $e \equiv 1$, and $\hat{G}$ is the set of characters. On $\mathbb{Z}(N)$ these are exactly $e_\ell(k) = \zeta^{\ell k}$ ($e(1)$ is an N-th root of unity).

Lemma 2.1

$\hat{G}$ is an abelian group under $(e_1 e_2)(a) = e_1(a)e_2(a)$.

Proof. Products of characters are characters, the trivial one is the identity, $e^{-1} = \bar{e}$ ($|e| = 1$), and the laws hold pointwise. ∎

Lemma 2.2

If G is finite and $e : G \to \mathbb{C} - \{0\}$ satisfies $e(ab) = e(a)e(b)$, then e is a character.

Proof. If $|e(a)| > 1$ then $|e(a^n)| = |e(a)|^n \to \infty$, and if $|e(a)| < 1$ then $|e(a^n)| \to 0$; both impossible as G is finite. So $|e(a)| = 1$. ∎

7.2.3 The orthogonality relations

On functions on finite G use $(f, g) = \frac{1}{|G|} \sum_{a \in G} f(a)\overline{g(a)}$.

Theorem 2.3

The characters of G form an orthonormal family in V.

Lemma 2.4

If e is a nontrivial character, then $\sum_{a \in G} e(a) = 0$.

Proof. Take $e(b) \neq 1$. As $a \mapsto ba$ permutes G, $e(b) \sum_a e(a) = \sum_a e(ba) = \sum_a e(a)$, so $\sum_a e(a) = 0$. ∎

Proof of Theorem 2.3. $(e, e) = \frac{1}{|G|} \sum |e(a)|^2 = 1$; for $e \neq e'$, $e\overline{e'}$ is a nontrivial character, so $(e, e') = 0$ by Lemma 2.4. ∎

Hence $|\hat{G}| \leq |G|$.

7.2.4 Characters as a total family

Theorem 2.5

The characters of a finite abelian group G form a basis of the vector space V of functions on G; in particular $|\hat{G}| = |G|$.

Lemma 2.6

A commuting family $\{T_1, \ldots, T_k\}$ of unitary transformations of a finite-dimensional inner product space V ($T_i T_j = T_j T_i$) is simultaneously diagonalizable: V has a basis of common eigenvectors.

Proof. Induction on k; $k = 1$ is the spectral theorem for unitary operators. Write $V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_s}$ (eigenspaces of T_k). Each T_j, $j < k$, preserves V_{λ_i}: $T_k T_j v = T_j T_k v = \lambda_i T_j v$. By induction each V_{λ_i} has a basis of common eigenvectors of the restrictions; take the union. ∎

Proof of Theorem 2.5. Let $(T_a f)(x) = f(ax)$: commuting (G abelian) and unitary ($x \mapsto ax$ permutes G). Lemma 2.6 gives a basis of common eigenvectors v, $T_a v = \lambda_a v$. Then $v(1) \neq 0$, else $v(a) = (T_a v)(1) = \lambda_a v(1) = 0$ for all a. So $w = v/v(1)$ satisfies $w(x) = \lambda_x$ ($v(x) = (T_x v)(1)$), $w(ab) = \lambda_{ab} = \lambda_a \lambda_b$ (from $T_{ab} = T_a T_b$), and $w \neq 0$ (T_x invertible); by Lemma 2.2 w is a character. So characters span V, and with Theorem 2.3 form a basis. ∎

7.2.5 Fourier inversion and Plancherel formula

For f on G and $e \in \hat{G}$: the **Fourier coefficient** $\hat{f}(e) = (f, e) = \frac{1}{|G|}\sum_a f(a)\overline{e(a)}$ and the **Fourier series** $f \sim \sum_{e\in\hat{G}} \hat{f}(e)e$.

Theorem 2.7

The characters of a finite abelian group G form an orthonormal basis of V with $(f, g) = \frac{1}{|G|}\sum_a f(a)\overline{g(a)}$. For every f on G, $f = \sum_{e\in\hat{G}} \hat{f}(e)e$ (inversion) and $\|f\|^2 = \sum_{e\in\hat{G}} |\hat{f}(e)|^2$ (Plancherel).

Proof. Theorems 2.3 and 2.5, then expansion in an orthonormal basis and Pythagoras. ■

For $G = \mathbb{Z}(N)$ this is Theorem 1.2 (up to normalization).

7.3 Selected exercises

Motivation. Book exercises linking the finite theory to the circle, and finite Fourier series as convolution with a delta kernel.

Exercise 7.2 (Decay of discrete Fourier coefficients)

Let f be a C^1 function on the circle and, for $N \geq 1$, $a_N(n) = \frac{1}{N}\sum_{k=1}^{N} f(e^{2\pi ik/N})e^{-2\pi ikn/N}$. Prove that there is c, independent of N and n, with $|a_N(n)| \leq c/|n|$ whenever $0 < |n| \leq N/2$.

Solution. Let $g(x) = f(e^{2\pi ix})$ (1-periodic, C^1) and $M = \sup|g'|$.

Step 1. For integer ℓ, reindexing $k \mapsto k - \ell$ (N-periodic summand),

$$\frac{1}{N}\sum_{k=1}^{N} g\Big(\frac{k+\ell}{N}\Big)e^{-2\pi ikn/N} = e^{2\pi i\ell n/N}a_N(n).$$

Subtracting from the definition,

$$a_N(n)(1 - e^{2\pi i\ell n/N}) = \frac{1}{N}\sum_{k=1}^{N}\Big[g\Big(\frac{k}{N}\Big) - g\Big(\frac{k+\ell}{N}\Big)\Big]e^{-2\pi ikn/N}.$$

Step 2. By the mean value theorem the right side is at most $M\ell/N$.

Step 3. Let $\ell = \lfloor N/(2|n|)\rfloor \geq 1$. Then $\ell|n|/N \leq \frac{1}{2}$, and $\ell|n|/N \geq \max(|n|/N, \frac{1}{2} - |n|/N) \geq \frac{1}{4}$. So $s = \ell n/N$ has $\frac{1}{4} \leq |s| \leq \frac{1}{2}$ and

$$|1 - e^{2\pi is}| = 2|\sin(\pi s)| \geq 2\sin(\pi/4) = \sqrt{2}.$$

Step 4. So $|a_N(n)| \leq \frac{M\ell}{\sqrt{2}N} \leq \frac{M}{2\sqrt{2}\,|n|}$: take $c = M/(2\sqrt{2})$. ■

Exercise 7.8 (Sampling a trigonometric polynomial)

Let $P(x) = \sum_{n=1}^{N} a_n e^{2\pi inx}$.
(a) Show that $\int_0^1 |P(x)|^2 dx = \frac{1}{N}\sum_{j=1}^{N} |P(j/N)|^2$.

(b) Prove that $P(x) = \sum_{j=1}^{N} P(j/N)K(x - j/N)$, where

$$K(x) = \frac{1}{N}\sum_{n=1}^{N} e^{2\pi i n x} = \frac{e^{2\pi i x}}{N}\,\frac{1 - e^{2\pi i N x}}{1 - e^{2\pi i x}} \qquad (x \notin \mathbb{Z}),$$

and that $K(0) = 1$, $K(j/N) = 0$ for $j \not\equiv 0 \pmod N$.

Solution. Let $\zeta = e^{2\pi i/N}$, so $P(j/N) = \sum_n a_n \zeta^{nj}$.

Step 1. For $n, m \in \{1, \dots, N\}$, $\frac{1}{N}\sum_{j=1}^{N} \zeta^{(n-m)j} = \mathbb{1}\{n = m\}$ by Lemma 1.1 ($j = N$ may replace $j = 0$, and $n - m \in (-N, N)$ is $\equiv 0$ only if $n = m$).

Step 2. So

$$\frac{1}{N}\sum_{j=1}^{N} P(j/N)\zeta^{-mj} = \sum_{n=1}^{N} a_n \frac{1}{N}\sum_{j=1}^{N} \zeta^{(n-m)j} = a_m, \qquad 1 \le m \le N.$$

(a) $\int_0^1 e^{2\pi i(n-m)x}dx = \mathbb{1}\{n = m\}$ gives $\int_0^1 |P|^2 = \sum_n |a_n|^2$, and by Step 1

$$\frac{1}{N}\sum_{j=1}^{N} |P(j/N)|^2 = \sum_{n,m} a_n \bar{a}_m \frac{1}{N}\sum_{j=1}^{N} \zeta^{(n-m)j} = \sum_n |a_n|^2.$$

(Parseval on the circle and, by Step 2, on $\mathbb{Z}(N)$, Theorem 1.2.)

(b) Insert Step 2 into P:

$$P(x) = \sum_{n=1}^{N} \Big(\frac{1}{N}\sum_{j=1}^{N} P(j/N)e^{-2\pi i n j/N}\Big)e^{2\pi i n x} = \sum_{j=1}^{N} P(j/N)\,\frac{1}{N}\sum_{n=1}^{N} e^{2\pi i n(x - j/N)},$$

i.e. $\sum_j P(j/N)K(x - j/N)$. For $x \notin \mathbb{Z}$, $q = e^{2\pi i x} \neq 1$ and $\sum_{n=1}^{N} q^n = q\frac{1-q^N}{1-q}$. $K(0) = 1$, and for $j \not\equiv 0$, $q = \zeta^j \neq 1$, $q^N = 1$, so $K(j/N) = \frac{q}{N}\frac{1-q^N}{1-q} = 0$. So P is determined by its N samples. ∎

Exercise 7.13 (Convolution and the delta kernel on G)

Let G be a finite abelian group with unit 1_G and $(f * g)(a) = \frac{1}{|G|}\sum_{b\in G} f(b)g(ab^{-1})$.

(a) Show that $\widehat{f * g}(e) = \hat{f}(e)\hat{g}(e)$ for all $e \in \hat{G}$.

(b) Show that $\sum_{e\in\hat{G}} e(c) = 0$ whenever $c \neq 1_G$.

(c) Let $D(c) = \sum_{e\in\hat{G}} e(c)$. Show that the Fourier series $Sf = \sum_e \hat{f}(e)e$ equals $f * D$, and deduce $Sf = f$.

Solution.

(a) With $\overline{e(bc)} = \overline{e(b)}\,\overline{e(c)}$ and $c = ab^{-1}$ running over G,

$$\widehat{f * g}(e) = \frac{1}{|G|^2}\sum_{b\in G} f(b)\sum_{a\in G} g(ab^{-1})\overline{e(a)} = \frac{1}{|G|^2}\sum_{b\in G} f(b)\sum_{c\in G} g(c)\overline{e(bc)}$$
$$= \Big(\frac{1}{|G|}\sum_b f(b)\overline{e(b)}\Big)\Big(\frac{1}{|G|}\sum_c g(c)\overline{e(c)}\Big) = \hat{f}(e)\hat{g}(e).$$

(b)

Step 1. Some $e_0 \in \hat{G}$ has $e_0(c) \neq 1$: otherwise all characters, hence all functions (Theorem 2.5), agree at c and 1_G, false for the indicator of $\{1_G\}$.

Step 2. $e \mapsto e_0 e$ is a bijection of $\hat{G}$ (Lemma 2.1), so $\sum_e e(c) = e_0(c)\sum_e e(c)$, forcing $\sum_e e(c) = 0$.

(c)

Step 1. By (b) and $|\hat{G}| = |G|$ (Theorem 2.5), $D(1_G) = |G|$, $D(c) = 0$ for $c \neq 1_G$.

Step 2. With $\overline{e(b)} = e(b^{-1})$,

$$Sf(a) = \sum_{e \in \hat{G}} \frac{1}{|G|} \sum_{b \in G} f(b)\overline{e(b)}e(a) = \frac{1}{|G|} \sum_{b \in G} f(b) \sum_{e \in \hat{G}} e(ab^{-1}) = (f * D)(a).$$

Step 3. By Step 1 only $b = a$ contributes: $(f * D)(a) = f(a)$. So $Sf = f$, a second proof of inversion (Theorem 2.7). ■

Chapter 8. Dirichlet's Theorem

8.1 A little elementary number theory

Motivation. The arithmetic behind Dirichlet's theorem: unique factorization, and Euler's proof via $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ that $\sum 1/p$ diverges.

8.1.1 The fundamental theorem of arithmetic

$a \mid b$ means $b = ac$ for some $c \in \mathbb{Z}$. A prime is an integer $p > 1$ with only $1, p$ as positive divisors. $\gcd(a,b)$ is the largest positive common divisor; a, b are relatively prime if $\gcd(a,b) = 1$.

Theorem 1.1 (Euclid's algorithm)

For any integers a and b with $b > 0$ there exist unique integers q and r with $0 \le r < b$ such that $a = qb + r$.

Proof.

Step Existence. $\{a - qb : q \in \mathbb{Z}\} \cap \{0, 1, 2, \dots\}$ is nonempty, with least element $r = a - qb \ge 0$; $r \ge b$ would make $r - b \ge 0$ smaller. So $0 \le r < b$.

Step Uniqueness. $qb + r = q'b + r'$ gives $b \mid r - r'$ with $|r - r'| < b$, so $r = r'$, $q = q'$. ■

Theorem 1.2

If $\gcd(a,b) = d$, then there exist integers x and y such that $ax + by = d$.

Proof. Let $s = ax + by$ be the least positive integer of this form. Common divisors of a, b divide s, so $d \le s$. With $a = qs + r$, $0 \le r < s$, $r = a(1 - qx) + b(-qy)$ has the same form, so $r = 0$ and $s \mid a$; likewise $s \mid b$, so $s \le d$. ■

Corollary 1.3

Positive integers a and b are relatively prime if and only if $ax + by = 1$ for some integers x, y.

Proof. One direction is Theorem 1.2; conversely, $ax + by = 1$ means common divisors divide 1. ■

Corollary 1.4

If a and c are relatively prime and $c \mid ab$, then $c \mid b$. In particular, if a prime p does not divide a and $p \mid ab$, then $p \mid b$.

Proof. $1 = ax + cy$ gives $b = abx + cby$, divisible by c. For prime $p \nmid a$, $\gcd(a,p) = 1$. ■

Corollary 1.5

If a prime p divides $a_1 \cdots a_r$, then $p \mid a_i$ for some i.

Proof. If $p \nmid a_1$ then $p \mid a_2 \cdots a_r$ by Corollary 1.4; induct on r. ■

Theorem 1.6

Every integer greater than 1 factors uniquely into a product of primes (up to order).

Proof.

Step Existence. Otherwise take the least $n > 1$ without factorization; $n = ab$ with $1 < a, b < n$, both of which factor: a contradiction.

Step Uniqueness. If $p_1 \cdots p_r = q_1 \cdots q_s$, Corollary 1.5 gives $p_1 = q_j$ for some j; cancel and induct. ■

8.1.2 The infinitude of primes

Theorem 1.7

There are infinitely many primes.

Proof. If $p_1, \ldots, p_n$ were all the primes, a prime factor p_i of $N = p_1 \cdots p_n + 1$ would divide $N - p_1 \cdots p_n = 1$. ■

Euler's argument proves $\sum 1/p = \infty$ and is the model for Dirichlet's proof. For $s > 1$ let $\zeta(s) = \sum_{n \geq 1} n^{-s}$; comparing with $\int x^{-s} dx$,

$$\frac{1}{s-1} \leq \zeta(s) \leq 1 + \frac{1}{s-1}, \qquad \text{so } \zeta(s) \to \infty \text{ as } s \to 1^+. \tag{8.1}$$

(as $\int_n^{n+1} x^{-s} dx \leq n^{-s} \leq \int_{n-1}^n x^{-s} dx$). An infinite product $\prod A_n$ **converges** if $\prod_{n \leq N} A_n$ has a limit.

Lemma 1.8

The exponential and logarithm functions satisfy:
(i) $e^{\log x} = x$ for $x > 0$.
(ii) $\log(1 + x) = x + E(x)$ with $|E(x)| \leq x^2$ if $|x| < 1/2$.
(iii) If $\log(1 + x) = y$ and $|x| < 1/2$, then $|y| \leq 2|x|$.

Proof. (i) Definition of log.

(ii) For $|x| < 1$, $\log(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$, so $|E(x)| \leq \frac{x^2}{2(1-|x|)} \leq x^2$ for $|x| < 1/2$.

(iii) By (ii), $|y| \leq |x| + x^2 \leq |x| + \frac{1}{2}|x| \leq 2|x|$. ■

Proposition 1.9

If $A_n = 1 + a_n$ and $\sum |a_n|$ converges, then $\prod_n A_n$ converges, and the product vanishes if and only if some factor A_n vanishes. Also, if $a_n \neq 1$ for all n, then $\prod_n 1/(1 - a_n)$ converges.

Proof. For real a_n (complex: Proposition 3.2). The finitely many n with $|a_n| \geq 1/2$ give a fixed finite factor (zero iff some $A_n = 0$). For the rest, $\prod_{n \leq N} A_n = e^{B_N}$, $B_N = \sum_{n \leq N} \log(1 + a_n)$, with $|\log(1 + a_n)| \leq 2|a_n|$, so $B_N \to B$ and $\prod A_n \to e^B \neq 0$. Finally $1/(1 - a_n) = 1 + a_n/(1 - a_n)$ with $|a_n/(1 - a_n)| \leq 2|a_n|$ once $|a_n| < 1/2$. ■

Theorem 1.10

For every $s > 1$, $\zeta(s) = \displaystyle\prod_p \frac{1}{1 - p^{-s}}$, the product being over all primes.

Proof. The product converges (Proposition 1.9, $\sum_p p^{-s} < \infty$). For the primes $p_1 < \cdots < p_N$ up to M, expanding geometric series,

$$\prod_{j\le N} \frac{1}{1-p_j^{-s}} = \prod_{j\le N} \left(1 + p_j^{-s} + p_j^{-2s} + \cdots\right) = \sum_{n\in\mathcal{N}_M} n^{-s},$$

where by Theorem 1.6 $\mathcal{N}_M$ is the set of integers with all prime factors $\le M$, each once (absolutely convergent series may be multiplied and rearranged). As $\{1,\dots,M\} \subset \mathcal{N}_M$,

$$\sum_{n\le M} n^{-s} \le \prod_{p\le M} \frac{1}{1-p^{-s}} \le \zeta(s).$$

Let $M \to \infty$. ■

Proposition 1.11

The series $\sum_p 1/p$ over all primes diverges.

Proof. Take logarithms in Theorem 1.10 (the partial products tend to a positive limit); by Lemma 1.8(ii),

$$\log\zeta(s) = -\sum_p \log(1-p^{-s}) = \sum_p \left(p^{-s} + O(p^{-2s})\right) = \sum_p p^{-s} + O(1),$$

as $\sum_p p^{-2s} \le \sum n^{-2}$. As $s \to 1^+$, $\zeta(s) \to \infty$ by (8.1), so $\sum_p p^{-s} \to \infty$, and $p^{-s} \le p^{-1}$ gives $\sum_p 1/p = \infty$. ■

8.2 Dirichlet's theorem

Motivation. The indicator of the class ℓ expands in characters of $\mathbb{Z}^*(q)$, reducing the theorem to the boundedness of $\sum \chi(p)p^{-s}$ for $\chi \ne \chi_0$, i.e. to L-functions at $s = 1$.

Theorem 2.1 (Dirichlet)

If q and ℓ are relatively prime positive integers, then there are infinitely many primes of the form $\ell + kq$ with $k \in \mathbb{Z}$.

As Euler, we show $\sum_{p\equiv\ell\,(q)} p^{-s} \to \infty$ as $s \to 1^+$ (completed in Section 8.3).

8.2.1 Fourier analysis, Dirichlet characters, and reduction of the theorem

$\mathbb{Z}^*(q)$, the residues mod q prime to q, is a multiplicative group (Corollary 1.3) of order $\varphi(q)$. For the indicator δ_ℓ of ℓ, Theorem 2.7 gives

$$\delta_\ell(m) = \sum_{e\in\widehat{\mathbb{Z}^*(q)}} \hat\delta_\ell(e)e(m), \qquad \hat\delta_\ell(e) = \frac{1}{\varphi(q)} \sum_{m\in\mathbb{Z}^*(q)} \delta_\ell(m)\overline{e(m)} = \frac{1}{\varphi(q)}\overline{e(\ell)}.$$

A character e of $\mathbb{Z}^*(q)$ gives the **Dirichlet character** $\chi = \chi_e$ on $\mathbb{Z}$: $\chi(m) = e(m \bmod q)$ if $\gcd(m,q) = 1$, else 0; χ_0 comes from $e \equiv 1$. Extend δ_ℓ to $\mathbb{Z}$ by $\delta_\ell(m) = \mathbb{1}\{m \equiv \ell \pmod q\}$.

Lemma 2.2

The Dirichlet characters are multiplicative: $\chi(mn) = \chi(m)\chi(n)$ for all $m, n \in \mathbb{Z}$. Moreover

$$\delta_\ell(m) = \frac{1}{\varphi(q)} \sum_{\chi} \overline{\chi(\ell)}\chi(m),$$

summing over all Dirichlet characters mod q.

Proof. For m, n prime to q, $\chi(mn) = e(m)e(n)$; otherwise mn is not prime to q and both sides vanish. The formula holds for m prime to q by the expansion; for $\gcd(m, q) > 1$ both sides are 0 ($m \not\equiv \ell$, $\chi(m) = 0$). ∎

Hence, for $s > 1$,

$$\sum_{p \equiv \ell\,(q)} \frac{1}{p^s} = \sum_{p} \frac{\delta_\ell(p)}{p^s} = \frac{1}{\varphi(q)} \sum_{p \nmid q} \frac{1}{p^s} + \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \overline{\chi(\ell)} \sum_{p} \frac{\chi(p)}{p^s}. \tag{8.2}$$

The first term tends to ∞ as $s \to 1^+$ (Proposition 1.11; finitely many $p \mid q$), so Dirichlet's theorem follows from:

Theorem 2.3

If χ is a nontrivial Dirichlet character, then $\sum_p \chi(p)p^{-s}$ remains bounded as $s \to 1^+$.

The proof occupies Section 8.3.

8.2.2 Dirichlet L-functions

For a Dirichlet character χ and $s > 1$ define

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

Theorem 2.4 (Dirichlet product formula)

If $s > 1$, then

$$\sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_{p} \frac{1}{1 - \chi(p)p^{-s}}.$$

Proof. As for Theorem 1.10, with $|\chi(p)p^{-s}| \le p^{-s}$ (convergence by Proposition 3.2) and multiplicativity (Lemma 2.2): $\prod_{p \le M}(1 + \chi(p)p^{-s} + \chi(p)^2p^{-2s} + \cdots) = \sum_{n \in \mathcal{N}_M} \chi(n)n^{-s}$, which differs from $\sum_n \chi(n)n^{-s}$ by at most $\sum_{n>M} n^{-s} \to 0$. ∎

Formally, with $\log(1 + x) = x + O(x^2)$,

$$\log L(s, \chi) = -\sum_{p} \log\left(1 - \chi(p)p^{-s}\right) = \sum_{p} \frac{\chi(p)}{p^s} + O(1).$$

So if $L(1, \chi)$ is finite and nonzero, $\sum_p \chi(p)p^{-s}$ stays bounded. Rigour needs:

(a) a complex logarithm, as χ may be complex;

(b) continuity of $L(s, \chi)$ down to $s = 1$; and (c) the key fact $L(1, \chi) \neq 0$.

8.3 Proof of the theorem

Motivation. A power-series logarithm, continuation of $L(s,\chi)$ to $s>0$, and $L(1,\chi)\neq 0$ (positivity for complex χ, hyperbola sums for real χ) make the argument rigorous.

8.3.1 Logarithms

For $|z|<1$ define

$$\log_1\Big(\frac{1}{1-z}\Big)=\sum_{k=1}^{\infty}\frac{z^k}{k}.$$

Proposition 3.1

The function $\log_1$ satisfies:
(i) If $|z|<1$, then $e^{\log_1(\frac{1}{1-z})}=\frac{1}{1-z}$.
(ii) If $|z|<1$, then $\log_1\left(\frac{1}{1-z}\right)=z+E_1(z)$, where $|E_1(z)|\le|z|^2$ if $|z|<1/2$.
(iii) If $|z|<1/2$, then $\left|\log_1\left(\frac{1}{1-z}\right)\right|\le 2|z|$.

Proof. (i) With $z=re^{i\theta}$, show $(1-re^{i\theta})\,e^{\sum_k (re^{i\theta})^k/k}=1$. Its r-derivative is

$$\Big[-e^{i\theta}+(1-re^{i\theta})\,e^{i\theta}\sum_{k\ge 1}(re^{i\theta})^{k-1}\Big]e^{\sum_k (re^{i\theta})^k/k}=\Big[-e^{i\theta}+(1-re^{i\theta})\frac{e^{i\theta}}{1-re^{i\theta}}\Big]e^{\cdots}=0.$$

so it is constant in r, equal to 1 at $r=0$.

(ii) $|E_1(z)|=\left|\sum_{k\ge 2}z^k/k\right|\le\frac{|z|^2}{2}\sum_{j\ge 0}|z|^j=\frac{|z|^2}{2(1-|z|)}\le|z|^2$ for $|z|<1/2$.

(iii) By (ii), $|\log_1(\frac{1}{1-z})|\le|z|+|z|^2\le 2|z|$. ∎

Proposition 3.2

If $\sum|a_n|$ converges and $a_n\neq 1$ for all n, then $\prod_{n=1}^{\infty}\frac{1}{1-a_n}$ converges and is nonzero.

Proof. As in Proposition 1.9: $|a_n|<1/2$ for $n\ge n_0$, and by Proposition 3.1(i),

$$\prod_{n_0\le n\le N}\frac{1}{1-a_n}=\exp\Big(\sum_{n_0\le n\le N}\log_1\Big(\frac{1}{1-a_n}\Big)\Big).$$

The exponent converges absolutely by (iii), so the products tend to $e^B\neq 0$; the finitely many factors $n<n_0$ are finite and nonzero. ∎

8.3.2 L-functions

Proposition 3.3

Let χ_0 be the trivial Dirichlet character ($\chi_0(n)=1$ if $\gcd(n,q)=1$, 0 otherwise) and $q=p_1^{a_1}\cdots p_N^{a_N}$. Then

$$L(s,\chi_0)=(1-p_1^{-s})(1-p_2^{-s})\cdots(1-p_N^{-s})\,\zeta(s).$$

Therefore $L(s,\chi_0)\to\infty$ as $s\to 1^+$.

Proof. By Theorems 2.4 and 1.10, $L(s,\chi_0)=\prod_{p\nmid q}(1-p^{-s})^{-1}$ is the Euler product of ζ without the factors for $p_1,\dots,p_N$; then use (8.1). ∎

Proposition 3.4

If χ is a nontrivial Dirichlet character, then $\sum_{n\ge1}\chi(n)/n^s$ converges for $s>0$; we denote its sum by $L(s,\chi)$. Moreover:
(i) $L(s,\chi)$ is continuously differentiable for $0<s<\infty$.
(ii) There are constants $c,c'>0$ with $L(s,\chi)=1+O(e^{-cs})$ and $L'(s,\chi)=O(e^{-c's})$ as $s\to\infty$.

Lemma 3.5

If χ is a nontrivial Dirichlet character, then $\left|\sum_{n=1}^{k}\chi(n)\right|\le q$ for all k.

Proof. $S=\sum_{n=1}^{q}\chi(n)=0$: for $a\in\mathbb{Z}^*(q)$, $\chi(a)S=\sum_n\chi(an)=S$ ($n\mapsto an$ permutes residues), and $\chi(a)\neq1$ for some a. By q-periodicity, with $k=aq+b$, $0\le b<q$, $\sum_{n=1}^{k}\chi(n)$ has at most q remaining terms of modulus ≤1. ∎

Proof of Proposition 3.4. Let $s_k=\sum_{n=1}^{k}\chi(n)$, $s_0=0$, $|s_k|\le q$. Summation by parts:

$$\sum_{n=1}^{N}\frac{\chi(n)}{n^s}=\sum_{n=1}^{N}\frac{s_n-s_{n-1}}{n^s}=\sum_{n=1}^{N-1}s_n\Big[\frac{1}{n^s}-\frac{1}{(n+1)^s}\Big]+\frac{s_N}{N^s}.$$

By the mean value theorem $|n^{-s}-(n+1)^{-s}|\le s\,n^{-s-1}$. For $s\ge\delta>0$ the boundary term tends to 0 and the sum converges absolutely and uniformly on compacts of $[\delta,\infty)$, as $|s_n|\,s\,n^{-s-1}\le qs\,n^{-\delta-1}$. So $L(s,\chi)=\sum_n s_n[n^{-s}-(n+1)^{-s}]$ is continuous for $s>0$.

(i) The termwise derivative $\sum_n s_n[-\log n\,n^{-s}+\log(n+1)(n+1)^{-s}]$ converges uniformly on compacts too, since $g(x)=x^{-s}\log x$ has $|g'(x)|\le C_\delta x^{-1-\delta/2}$ for $s\ge\delta$. So $L(\cdot,\chi)\in C^1(0,\infty)$.

(ii) $\chi(1)=1$, so $|L(s,\chi)-1|\le\sum_{n\ge2}n^{-s}\le 2^{-s}(1+\frac{2}{s-1})=O(e^{-cs})$ for $s\ge2$, any $c<\log2$; likewise $|L'(s,\chi)|\le\sum_{n\ge2}(\log n)n^{-s}=O(e^{-c's})$. ∎

For nontrivial χ and $s>1$ ($L(s,\chi)\neq0$ by Theorem 2.4 and Proposition 3.2), as $L(s,\chi)\to1$ for $s\to\infty$, define

$$\log_2 L(s,\chi)=-\int_s^\infty\frac{L'(t,\chi)}{L(t,\chi)}\,dt,$$

convergent as $L'/L=O(e^{-c't})$ ($\log_2$ is a name, not base 2). If $L(1,\chi)\neq0$ the integral extends continuously to $s=1$, so $\log_2 L(s,\chi)$ stays bounded as $s\to1^+$.

Proposition 3.6

If $s>1$, then $e^{\log_2 L(s,\chi)}=L(s,\chi)$. Moreover

$$\log_2 L(s,\chi)=\sum_p\log_1\Big(\frac{1}{1-\chi(p)/p^s}\Big).$$

Proof.

Step First identity. $\frac{d}{ds}(e^{-\log_2 L}L)=e^{-\log_2 L}(-\frac{L'}{L}L+L')=0$, and as $s\to\infty$, $\log_2 L\to0$, $L\to1$; so the constant is 1.

Step Second identity. By Proposition 3.1(i) and Theorem 2.4,

$$e^{\sum_p\log_1(\frac{1}{1-\chi(p)p^{-s}})}=\prod_p\frac{1}{1-\chi(p)p^{-s}}=L(s,\chi).$$

So the two sides differ by $2\pi i M(s)$, $M(s) \in \mathbb{Z}$. Both are continuous for $s > 1$ (uniform convergence on $[\delta, \infty)$, $\delta > 1$, by Proposition 3.1(iii)), so M is constant, and both tend to 0 as $s \to \infty$: $M \equiv 0$. ■

By Proposition 3.1(ii),

$$\log_2 L(s,\chi) = \sum_p \log_1 \Big(\frac{1}{1-\chi(p)/p^s}\Big) = \sum_p \frac{\chi(p)}{p^s} + O\Big(\sum_p \frac{1}{p^{2s}}\Big) = \sum_p \frac{\chi(p)}{p^s} + O(1).$$

So $L(1,\chi) \neq 0$ makes $\sum_p \chi(p)p^{-s}$ bounded as $s \to 1^+$, and Theorem 2.3 (hence Dirichlet's theorem) reduces to:

8.3.3 Non-vanishing of the L-function

Theorem 3.7

If $\chi \neq \chi_0$, then $L(1,\chi) \neq 0$.

A Dirichlet character is **real** if it takes only real values $(0, \pm 1)$, i.e. $\chi = \bar{\chi}$, and **complex** otherwise.

Case I: complex Dirichlet characters

Lemma 3.8

If $s > 1$, then $\prod_\chi L(s,\chi) \geq 1$, the product being over all Dirichlet characters mod q. In particular the product is real-valued.

Proof. By Proposition 3.6 (and its analogue for χ_0 from the Euler product), $L(s,\chi) = \exp\big(\sum_p \log_1 \frac{1}{1-\chi(p)p^{-s}}\big)$, so

$$\prod_\chi L(s,\chi) = \exp\Big(\sum_\chi \sum_p \sum_{k\geq 1} \frac{1}{k}\frac{\chi(p^k)}{p^{ks}}\Big) = \exp\Big(\sum_p \sum_{k\geq 1} \frac{1}{kp^{ks}} \sum_\chi \chi(p^k)\Big),$$

using $\chi(p)^k = \chi(p^k)$ and absolute convergence. By Lemma 2.2 with $\ell = 1$, $\sum_\chi \chi(p^k) = \varphi(q)\delta_1(p^k) \geq 0$, so the exponent is ≥ 0. ■

Lemma 3.9

The following hold:
(i) If $L(1,\chi) = 0$, then $L(1,\bar{\chi}) = 0$.
(ii) If χ is nontrivial and $L(1,\chi) = 0$, then $|L(s,\chi)| \leq C|s-1|$ for $1 \leq s \leq 2$.
(iii) For the trivial character, $|L(s,\chi_0)| \leq \frac{C}{|s-1|}$ for $1 < s \leq 2$.

Proof. (i) $L(1,\bar{\chi}) = \overline{L(1,\chi)}$ (real n^{-s}).

(ii) Mean value theorem, $L(\cdot,\chi) \in C^1(0,\infty)$ (Proposition 3.4): $|L(s,\chi) - L(1,\chi)| \leq \sup_{[1,2]} |L'| \cdot |s-1|$.

(iii) By Proposition 3.3 and (8.1), $|L(s,\chi_0)| \leq \zeta(s) \leq \frac{2}{s-1}$. ■

Proof of Theorem 3.7 for complex χ. If $L(1,\chi) = 0$, then $L(1,\bar{\chi}) = 0$ with $\bar{\chi} \neq \chi$, so in $\prod_{\chi'} L(s,\chi')$ two factors are $O(|s-1|)$, the χ_0 factor is $O(1/|s-1|)$, and the rest are bounded (continuous on $[1,2]$) (Lemma 3.9). So the product tends to 0, contradicting Lemma 3.8. ■

Case II: real Dirichlet characters

For real χ only one factor could vanish, and the pole of $L(s,\chi_0)$ can cancel it; instead we use Dirichlet's *summation along hyperbolas*, first on the divisor problem.

Sums vs. integrals.

Proposition 3.10

If N is a positive integer, then:
(i) $\displaystyle\sum_{1\le n\le N}\frac{1}{n}=\int_1^N\frac{dx}{x}+O(1)=\log N+O(1)$.
(ii) More precisely, there is a real number γ, **Euler's constant**, with $\displaystyle\sum_{1\le n\le N}\frac{1}{n}=\log N+\gamma+O(1/N)$.

Proof. Prove (ii). Let $\gamma_n=\frac{1}{n}-\int_n^{n+1}\frac{dx}{x}$; as $1/x$ decreases, $0\le\gamma_n\le\frac{1}{n}-\frac{1}{n+1}\le\frac{1}{n^2}$, so $\sum\gamma_n=\gamma$ converges with $\sum_{n>N}\gamma_n\le\frac{1}{N}$. Then

$$\sum_{n=1}^{N}\frac{1}{n}-\int_1^N\frac{dx}{x}=\sum_{n=1}^{N}\gamma_n+\int_N^{N+1}\frac{dx}{x}=\gamma-\sum_{n>N}\gamma_n+\int_N^{N+1}\frac{dx}{x},$$

with the last two terms $O(1/N)$. ∎

Proposition 3.11

If N is a positive integer, then

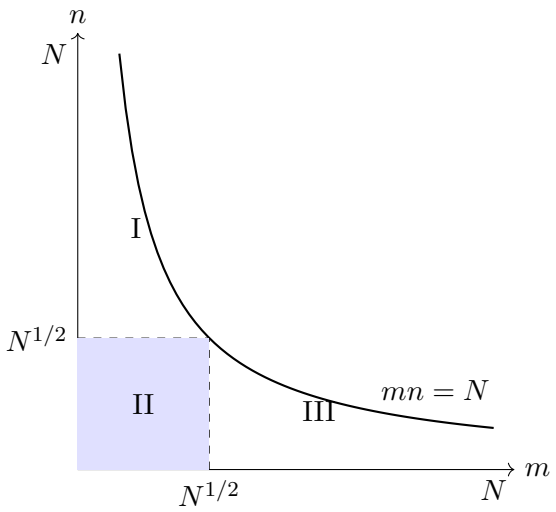
$$\sum_{1\le n\le N}\frac{1}{n^{1/2}}=\int_1^N\frac{dx}{x^{1/2}}+c'+O(N^{-1/2})=2N^{1/2}+c+O(N^{-1/2}).$$

Proof. [Supplement] Let $\gamma_n=n^{-1/2}-\int_n^{n+1}x^{-1/2}dx$; by the mean value theorem $0\le\gamma_n\le\frac{1}{2}n^{-3/2}$, so $\sum\gamma_n=c'$ with $\sum_{n>N}\gamma_n=O(N^{-1/2})$. As before,

$$\sum_{n\le N}n^{-1/2}-\int_1^N x^{-1/2}dx=c'-\sum_{n>N}\gamma_n+\int_N^{N+1}x^{-1/2}dx=c'+O(N^{-1/2}),$$

and $\int_1^N x^{-1/2}dx=2N^{1/2}-2$; take $c=c'-2$. ∎

Hyperbolic sums. For F on pairs of positive integers, $S_N=\sum\sum_{mn\le N}F(m,n)$ can be computed
(a) along hyperbolas: $S_N=\sum_{1\le k\le N}\left(\sum_{nm=k}F(m,n)\right)$;
(b) vertically: $S_N=\sum_{1\le m\le N}\left(\sum_{1\le n\le N/m}F(m,n)\right)$;
(c) horizontally: $S_N=\sum_{1\le n\le N}\left(\sum_{1\le m\le N/n}F(m,n)\right)$.



Split $\{mn \le N\}$ into $\mathrm{I} = \{1 \le m < N^{1/2},\ N^{1/2} < n \le N/m\}$, $\mathrm{II} = \{1 \le m \le N^{1/2},\ 1 \le n \le N^{1/2}\}$, and $\mathrm{III} = \{N^{1/2} < m \le N/n,\ 1 \le n < N^{1/2}\}$, with corresponding sums $S_{\mathrm{I}}, S_{\mathrm{II}}, S_{\mathrm{III}}$.

Let $d(k) = \sum_{nm=k} 1$, the number of divisors of k.

Theorem 3.12 (Dirichlet's divisor theorem)

For positive integers N,

$$\frac{1}{N}\sum_{k=1}^{N} d(k) = \log N + O(1).$$

More precisely, $\frac{1}{N}\sum_{k=1}^{N} d(k) = \log N + (2\gamma - 1) + O(N^{-1/2})$.

Proof. Let $S_N = \sum_{k \le N} d(k)$, the hyperbolic sum of $F \equiv 1$.

Step First estimate. Vertically, $S_N = \sum_{m \le N}[N/m] = N\sum_{m \le N}\frac{1}{m} + O(N)$, so $S_N/N = \log N + O(1)$ by Proposition 3.10(i).

Step Refined estimate. By symmetry $S_{\mathrm{I}} = S_{\mathrm{III}}$, so $S_N = 2(S_{\mathrm{I}} + S_{\mathrm{II}}) - S_{\mathrm{II}}$, and by Proposition 3.10(ii)

$$S_{\mathrm{I}} + S_{\mathrm{II}} = \sum_{m \le N^{1/2}} [N/m] = N \sum_{m \le N^{1/2}} \frac{1}{m} + O(N^{1/2}) = N \log N^{1/2} + N\gamma + O(N^{1/2}).$$

With $S_{\mathrm{II}} = [N^{1/2}]^2 = N + O(N^{1/2})$, $S_N = N \log N + (2\gamma - 1)N + O(N^{1/2})$. ∎

Non-vanishing for real characters. For a nontrivial real χ let $F(m,n) = \dfrac{\chi(n)}{(nm)^{1/2}}$ and $S_N = \sum\sum_{mn \le N} F(m,n)$.

Proposition 3.13

The following hold:
(i) $S_N \ge c \log N$ for some constant $c > 0$.
(ii) $S_N = 2N^{1/2}L(1,\chi) + O(1)$.

This gives Theorem 3.7 for real χ: $L(1,\chi) = 0$ would make $S_N = O(1)$, contradicting (i).

Along hyperbolas, $\sum_{nm=k} \frac{\chi(n)}{(nm)^{1/2}} = \frac{1}{k^{1/2}} \sum_{n \mid k} \chi(n)$, so (i) follows from:

Lemma 3.14

For every $k \geq 1$, $\sum_{n|k} \chi(n) \geq 0$, and $\sum_{n|k} \chi(n) \geq 1$ if $k = \ell^2$ for some $\ell \in \mathbb{Z}$.

Proof.

Step Prime powers. For $k = p^a$, $\sum_{n|k} \chi(n) = 1 + \chi(p) + \cdots + \chi(p)^a$ with $\chi(p) \in \{0, \pm 1\}$: $a+1$ if $\chi(p) = 1$; 1 or 0 if $\chi(p) = -1$ and a is even or odd; 1 if $\chi(p) = 0$. Always ≥ 0, and ≥ 1 for even a.

Step General k. For $k = p_1^{a_1} \cdots p_N^{a_N}$, by multiplicativity

$$\sum_{n|k} \chi(n) = \prod_{j=1}^{N} (1 + \chi(p_j) + \cdots + \chi(p_j^{a_j})).$$

Each factor is ≥ 0, and ≥ 1 for squares (all a_j even). ■

Proof of Proposition 3.13(i). By the lemma, $S_N \geq \sum_{\ell \leq N^{1/2}} \frac{1}{(\ell^2)^{1/2}} = \sum_{\ell \leq N^{1/2}} \frac{1}{\ell} \geq c \log N$ (Proposition 3.10(i)). ■

Lemma 3.15

For all integers $0 < a < b$:

(i) $\displaystyle\sum_{n=a}^{b} \frac{\chi(n)}{n^{1/2}} = O(a^{-1/2})$;

(ii) $\displaystyle\sum_{n=a}^{b} \frac{\chi(n)}{n} = O(a^{-1})$.

Proof. Summation by parts as in Proposition 3.4, $|s_n| \leq q$ (Lemma 3.5):

$$\sum_{n=a}^{b} \frac{\chi(n)}{n^{1/2}} = \sum_{n=a}^{b-1} s_n [n^{-1/2} - (n+1)^{-1/2}] + \frac{s_b}{b^{1/2}} - \frac{s_{a-1}}{a^{1/2}},$$

$$\Big|\sum_{n=a}^{b} \frac{\chi(n)}{n^{1/2}}\Big| \leq q \sum_{n=a}^{b-1} [n^{-1/2} - (n+1)^{-1/2}] + \frac{2q}{a^{1/2}} \leq \frac{3q}{a^{1/2}},$$

by telescoping; (ii) likewise with n^{-1}, bound $3q/a$. ■

Proof of Proposition 3.13(ii). Write $S_N = S_{\mathrm{I}} + (S_{\mathrm{II}} + S_{\mathrm{III}})$.

Step Region I, summed vertically.

$$S_{\mathrm{I}} = \sum_{m < N^{1/2}} \frac{1}{m^{1/2}} \Big(\sum_{N^{1/2} < n \leq N/m} \frac{\chi(n)}{n^{1/2}} \Big).$$

The inner sum is $O(N^{-1/4})$ (Lemma 3.15(i)) and $\sum_{m<N^{1/2}} m^{-1/2} = O(N^{1/4})$ (Proposition 3.11), so $S_{\mathrm{I}} = O(1)$.

Step Regions II and III, summed horizontally.

$$S_{\mathrm{II}} + S_{\mathrm{III}} = \sum_{n \leq N^{1/2}} \frac{\chi(n)}{n^{1/2}} \Big(\sum_{m \leq N/n} \frac{1}{m^{1/2}} \Big) = \sum_{n \leq N^{1/2}} \frac{\chi(n)}{n^{1/2}} \Big(2\frac{N^{1/2}}{n^{1/2}} + c + O\Big(\frac{n^{1/2}}{N^{1/2}}\Big) \Big)$$

by Proposition 3.11, i.e.

$$2N^{1/2} \sum_{n \leq N^{1/2}} \frac{\chi(n)}{n} + c \sum_{n \leq N^{1/2}} \frac{\chi(n)}{n^{1/2}} + O\Big(\frac{1}{N^{1/2}} \sum_{n \leq N^{1/2}} 1 \Big).$$

The third is $O(1)$, the second $O(1)$ by Lemma 3.15(i), and Lemma 3.15(ii) gives $\sum_{n\le N^{1/2}} \frac{\chi(n)}{n} = L(1,\chi) + O(N^{-1/2})$. So $S_N = 2N^{1/2}L(1,\chi) + O(1)$. ■

This proves Theorem 3.7, hence Theorem 2.3, and by (8.2) $\sum_{p\equiv \ell\,(q)} p^{-s} \to \infty$ as $s \to 1^+$: infinitely many primes $p \equiv \ell \pmod q$ (Theorem 2.1).

Appendix. Integration

A.1 Definition of the Riemann integral

Motivation. The Riemann integral, approximation by continuous functions, and Lebesgue's criterion (integrable iff the discontinuities have measure zero).

Let f be bounded and real on $[a,b]$. A **partition** P is $a = x_0 < x_1 < \cdots < x_N = b$, with $I_j = [x_{j-1}, x_j]$, $|I_j| = x_j - x_{j-1}$; the upper and lower sums are

$$\mathcal{U}(P,f) = \sum_{j=1}^{N} \Big[\sup_{x\in I_j} f(x)\Big]|I_j|, \qquad \mathcal{L}(P,f) = \sum_{j=1}^{N} \Big[\inf_{x\in I_j} f(x)\Big]|I_j|.$$

$P' \supset P$ is a **refinement** of P, and then

$$\mathcal{L}(P,f) \le \mathcal{L}(P',f) \le \mathcal{U}(P',f) \le \mathcal{U}(P,f).$$

(splitting I_j does not increase the sups). With the common refinement $P_1 \cup P_2$, every lower sum is $\le$ every upper sum, so $\mathcal{L} = \sup_P \mathcal{L}(P,f) \le \mathcal{U} = \inf_P \mathcal{U}(P,f)$.

Definition

f is **(Riemann) integrable** if for every $\epsilon > 0$ there is a partition P with $\mathcal{U}(P,f) - \mathcal{L}(P,f) < \epsilon$. Then $\mathcal{U} = \mathcal{L}$, and this common value is $\int_a^b f(x)\,dx$. For complex $f = u + iv$, f is integrable if u and v are, and $\int f = \int u + i\int v$.

Equivalently, with Riemann sums $\mathcal{R}(P,f) = \sum f(x_j^*)|I_j|$, f is integrable iff $\mathcal{R}(P,f)$ converges as the mesh $\max|I_j| \to 0$. Continuous f are integrable: for small mesh $\sup_{I_j} f - \inf_{I_j} f < \epsilon$, so $\mathcal{U} - \mathcal{L} \le \epsilon(b-a)$.

A.1.1 Basic properties

Proposition 1.1

If f and g are integrable on $[a,b]$, then:
(i) $f+g$ is integrable, and $\int_a^b (f+g) = \int_a^b f + \int_a^b g$.
(ii) If $c \in \mathbb{C}$, then cf is integrable and $\int_a^b cf = c\int_a^b f$.
(iii) If f, g are real-valued and $f \le g$, then $\int_a^b f \le \int_a^b g$.
(iv) If $c \in [a,b]$, then f is integrable on $[a,c]$ and $[c,b]$, and $\int_a^b f = \int_a^c f + \int_c^b f$.

Proof. (i) Let f, g be real. $\mathcal{U}(P, f+g) \le \mathcal{U}(P,f) + \mathcal{U}(P,g)$ and $\mathcal{L}(P, f+g) \ge \mathcal{L}(P,f) + \mathcal{L}(P,g)$. With P_1, P_2 giving $\mathcal{U} - \mathcal{L} < \epsilon$ for f, g and a common refinement P_0, $\mathcal{U}(P_0, f+g) - \mathcal{L}(P_0, f+g) < 2\epsilon$, and the integral I satisfies

$$I \le \mathcal{U}(P_0, f+g) \le \mathcal{U}(P_0,f) + \mathcal{U}(P_0,g) \le \int f + \int g + 2\epsilon,$$

and likewise $I \ge \int f + \int g - 2\epsilon$.

(ii) For $c \ge 0$, $\mathcal{U}(P, cf) = c\,\mathcal{U}(P,f)$, $\mathcal{L}(P,cf) = c\,\mathcal{L}(P,f)$; $\mathcal{U}(P,-f) = -\mathcal{L}(P,f)$, $\mathcal{L}(P,-f) = -\mathcal{U}(P,f)$; complex c by (i).

(iii) $g - f \ge 0$ gives $\mathcal{L}(P, g-f) \ge 0$, so $\int(g-f) \ge 0$; apply (i) and (ii).

(iv) Take partitions containing c; then $\mathcal{U}(P,f) = \mathcal{U}(P',f) + \mathcal{U}(P'',f)$ for the induced partitions of $[a,c]$, $[c,b]$, likewise $\mathcal{L}$, so $\mathcal{U} - \mathcal{L} < \epsilon$ on each piece and the integrals add. ∎

Consequently $|f|$ is integrable with $\left|\int f\right| \le \int |f|$ (Lemma 1.2 with $\varphi(x) = |x|$; for complex f, $\varphi(u,v)$ on the bounded range, and $\int f = e^{i\theta}|\int f|$ gives $|\int f| = \int \mathrm{Re}(e^{-i\theta} f) \le \int |f|$), and $fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$ is integrable.

Lemma 1.2

If f is real-valued and integrable on $[a,b]$, and φ is a real-valued continuous function on $\mathbb{R}$, then $\varphi \circ f$ is integrable on $[a,b]$.

Proof. Let $|f| \le M$; by uniform continuity on $[-M,M]$ choose $\delta < \epsilon$ with $|\varphi(y_1) - \varphi(y_2)| < \epsilon$ for $|y_1 - y_2| < \delta$, and P with $\mathcal{U}(P,f) - \mathcal{L}(P,f) < \delta^2$. Let Λ be the j with $\sup_{I_j} f - \inf_{I_j} f < \delta$ (there $\varphi \circ f$ oscillates $\le \epsilon$) and Λ' the rest. Then

$$\delta \sum_{j \in \Lambda'} |I_j| \le \sum_{j \in \Lambda'} \Big(\sup_{I_j} f - \inf_{I_j} f \Big) |I_j| < \delta^2,$$

so $\sum_{\Lambda'} |I_j| < \delta$, and with $B = \sup_{[-M,M]} |\varphi|$,

$$\mathcal{U}(P, \varphi \circ f) - \mathcal{L}(P, \varphi \circ f) \le \epsilon(b-a) + 2B\delta < \epsilon(b - a + 2B).$$

■

Proposition 1.3

A bounded monotonic function f on $[a,b]$ is integrable.

Proof. Take $a = 0$, $b = 1$, f increasing, and $x_j = j/N$:

$$\mathcal{U}(P_N, f) - \mathcal{L}(P_N, f) = \sum_{j=1}^{N} \frac{1}{N} [f(x_j) - f(x_{j-1})] = \frac{f(1) - f(0)}{N} \to 0.$$

■

Proposition 1.4

Let f be a bounded function on $[a,b]$. If $c \in (a,b)$, and for all small $\delta > 0$ the function f is integrable on $[a, c-\delta]$ and $[c+\delta, b]$, then f is integrable on $[a,b]$.

Proof. Let $|f| \le M$. Choose $4\delta M \le \epsilon/3$ and partitions P_1 of $[a, c-\delta]$, P_2 of $[c+\delta, b]$ with $\mathcal{U} - \mathcal{L} < \epsilon/3$; $P = P_1 \cup P_2$ has $[c-\delta, c+\delta]$ as a subinterval, so

$$\mathcal{U}(P,f) - \mathcal{L}(P,f) \le (\mathcal{U}(P_1,f) - \mathcal{L}(P_1,f)) + 2M \cdot 2\delta + (\mathcal{U}(P_2,f) - \mathcal{L}(P_2,f)) < \epsilon.$$

■

By induction, bounded functions continuous except at finitely many points are integrable.

A function on the circle (2π-periodic on $\mathbb{R}$) is **integrable** if integrable on $[-\pi, \pi]$.

Lemma 1.5

Suppose f is integrable on the circle and $|f| \le B$. Then there is a sequence $\{f_k\}$ of continuous functions on the circle with

$$\sup_{x \in [-\pi,\pi]} |f_k(x)| \le B \ \text{ for all } k, \qquad \int_{-\pi}^{\pi} |f(x) - f_k(x)|\, dx \to 0 \ \text{ as } k \to \infty.$$

Proof. Take f real (complex: treat parts separately, then truncate radially to B, a 1-Lipschitz operation).

Step Step function. Choose $-\pi = x_0 < \cdots < x_N = \pi$ with $\mathcal{U} - \mathcal{L} < \epsilon$ and $f^* = \sup_{[x_{j-1},x_j]} f$ on $[x_{j-1}, x_j)$. Then $|f^*| \le B$ and

$$\int_{-\pi}^{\pi} |f^* - f| = \int_{-\pi}^{\pi} (f^* - f) \le \mathcal{U} - \mathcal{L} < \epsilon.$$

Step Smoothing the jumps. For small δ, let $\tilde{f} = f^*$ away from the x_j, linear on $[x_j - \delta, x_j + \delta]$ $(1 \le j \le N-1)$ matching $f^*(x_j \pm \delta)$, and linear on $[-\pi, -\pi + \delta]$ and $[\pi - \delta, \pi]$ with $\tilde{f}(\pm\pi) = 0$. Then $\tilde{f}$ is continuous, 2π-periodic, $|\tilde{f}| \le B$, and differs from f^* on N intervals of length $\le 2\delta$: $\int |f^* - \tilde{f}| \le 4BN\delta < \epsilon$.

So $\int |f - \tilde{f}| < 2\epsilon$; take $\epsilon = 1/k$. ■

A.1.2 Sets of measure zero and discontinuities of integrable functions

$E \subset \mathbb{R}$ has **measure** 0 if for every $\epsilon > 0$ countably many open intervals I_k cover E with $\sum |I_k| < \epsilon$. Countable sets have measure 0 (lengths $\epsilon 2^{-k}$); a compact set of measure 0 is covered by finitely many I_k.

Lemma 1.6

The union of countably many sets of measure 0 has measure 0.

Proof. Cover E_i by $\{I_{k,i}\}_k$ with $\sum_k |I_{k,i}| < \epsilon/2^i$; together they cover $\bigcup_i E_i$ with total length $< \epsilon$. ■

Theorem 1.7 (Lebesgue)

A bounded function f on $[a, b]$ is integrable if and only if its set of discontinuities has measure 0.

Let $J = [a, b]$, $I(c, r) = (c - r, c + r)$, $\operatorname{osc}(f, c, r) = \sup_{x,y \in J \cap I(c,r)} |f(x) - f(y)|$ and $\operatorname{osc}(f, c) = \lim_{r \to 0} \operatorname{osc}(f, c, r)$; f is continuous at c iff $\operatorname{osc}(f, c) = 0$. With $A_\epsilon = \{c \in J : \operatorname{osc}(f, c) \ge \epsilon\}$, the discontinuity set is $D = \bigcup_n A_{1/n}$.

Lemma 1.8

For $\epsilon > 0$, the set A_ϵ is closed, hence compact.

Proof. If $c_n \in A_\epsilon$, $c_n \to c \notin A_\epsilon$, write $\operatorname{osc}(f, c) = \epsilon - \delta$, $\delta > 0$, choose r with $\operatorname{osc}(f, c, r) < \epsilon - \delta/2$ and $|c_n - c| < r/2$. As $I(c_n, r/2) \subset I(c, r)$, $\operatorname{osc}(f, c_n) < \epsilon$: a contradiction. ■

Proof of Theorem 1.7. Let $|f| \le M$.

Step Measure zero ⇒ integrable. $A_\epsilon \subset D$ is compact of measure 0, so finitely many open $I_1, \ldots, I_N$ of total length $< \epsilon$ cover it. Each point z of the compact rest of J has an open interval F_z on which f oscillates $\le \epsilon$; take a finite subcover $I_{N+1}, \ldots, I_{N'}$. The endpoints give a partition P whose subintervals lie in the closure of some I_k, $k \le N$ (contributing $\le 2M$ times length), or in some I_k, $k > N$ (oscillation $\le \epsilon$). So

$$\mathcal{U}(P, f) - \mathcal{L}(P, f) \le 2M \sum_{j=1}^{N} |I_j| + \epsilon(b - a) \le (2M + b - a)\epsilon.$$

Step Integrable ⇒ measure zero. By Lemma 1.6 show each $A_{1/n}$ has measure 0. Take P with $\mathcal{U}(P,f)-\mathcal{L}(P,f)<\epsilon/n$. If $A_{1/n}$ meets $I_j=(x_{j-1},x_j)$, then $\sup_{I_j} f-\inf_{I_j} f\geq 1/n$, so

$$\frac{1}{n}\sum_{j:\, I_j\cap A_{1/n}\neq\emptyset}|I_j|\leq \mathcal{U}(P,f)-\mathcal{L}(P,f)<\frac{\epsilon}{n}.$$

Enlarging these intervals slightly and adding tiny intervals around the x_j covers $A_{1/n}$ with total length $\leq 2\epsilon$. ■

(So fg is integrable: its discontinuities lie in those of f and g.)

A.2 Multiple integrals

Motivation. Integrals over rectangles in $\mathbb{R}^d$: Fubini, change of variables with $|\det Dg|$, and spherical coordinates.

A.2.1 The Riemann integral in $\mathbb{R}^d$

A closed rectangle is $R=[a_1,b_1]\times\cdots\times[a_d,b_d]$; partitions P_j of $[a_j,b_j]$ give a partition P of R into $S=S_1\times\cdots\times S_d$, $|S|=|S_1|\cdots|S_d|$. Upper and lower sums

$$\mathcal{U}(P,f)=\sum_S\Big[\sup_S f\Big]|S|,\qquad \mathcal{L}(P,f)=\sum_S\Big[\inf_S f\Big]|S|,$$

refinements, integrability and $\int_R f$ are as in one dimension; continuous f are integrable. For f continuous on a closed ball B (or a set with boundary of measure zero), $g=f\mathbb{1}_B$ is integrable on any rectangle $R\supset B$ (discontinuities on ∂B), and $\int_B f=\int_R g$ is independent of R.

A.2.2 Repeated integrals

Theorem 2.1 (Fubini, continuous case)

Let f be continuous on a closed rectangle $R=R_1\times R_2\subset\mathbb{R}^d$, $R_i\subset\mathbb{R}^{d_i}$, $d=d_1+d_2$. Writing $x=(x_1,x_2)$, the function $F(x_1)=\int_{R_2} f(x_1,x_2)\,dx_2$ is continuous on R_1, and

$$\int_R f(x)\,dx=\int_{R_1}\Big(\int_{R_2} f(x_1,x_2)\,dx_2\Big)dx_1.$$

Proof.

Step Continuity. $|F(x_1)-F(x_1')|\leq|R_2|\sup_{x_2}|f(x_1,x_2)-f(x_1',x_2)|$, small for x_1' near x_1 by uniform continuity.

Step Identity. For partitions P_1,P_2 of R_1,R_2 with subrectangles S,T,

$$\sup_{S\times T} f\geq\sup_{x_1\in S}\Big(\sup_{x_2\in T} f(x_1,x_2)\Big),\qquad \inf_{S\times T} f\leq\inf_{x_1\in S}\Big(\inf_{x_2\in T} f(x_1,x_2)\Big).$$

For each x_1,

$$\sum_T\Big[\inf_{x_2\in T} f(x_1,x_2)\Big]|T|\leq F(x_1)\leq\sum_T\Big[\sup_{x_2\in T} f(x_1,x_2)\Big]|T|.$$

Taking $\sup_{x_1\in S}$ and using the first inequality,

$$\mathcal{U}(P_1,F)\leq\sum_S\sum_T\Big[\sup_{S\times T} f\Big]|T||S|=\mathcal{U}(P,f),$$

and likewise $\mathcal{L}(P,f) \le \mathcal{L}(P_1,F)$. As $\mathcal{U}(P,f) - \mathcal{L}(P,f)$ can be made small, $\int_{R_1} F = \int_R f$. ■

By induction $\int_R f = \int_{a_1}^{b_1} \cdots \int_{a_d}^{b_d} f\, dx_d \cdots dx_1$, in any order.

A.2.3 The change of variables formula

$g : A \to B$ between compact sets is a C^1 **diffeomorphism** if it is a bijection extending to a C^1 map near A, with inverse extending to a C^1 map near B, and $\det Dg \neq 0$ on A.

Theorem 2.2 (change of variables)

Suppose A and B are compact subsets of $\mathbb{R}^d$ and $g : A \to B$ is a C^1 diffeomorphism. If f is continuous on B, then

$$\int_{g(A)} f(x)\, dx = \int_A f(g(y))\, |\det(Dg)(y)|\, dy.$$

Proof. [Supplement] Assume (implicitly) that A, B have boundaries of measure zero, and write $\mathrm{vol}(E) = \int_E 1$.

Step Linear maps. For invertible linear L and a rectangle R, $\mathrm{vol}(L(R)) = |\det L|\, \mathrm{vol}(R)$. Both sides are multiplicative in L ($\mathrm{vol}(L(E)) = c_L \mathrm{vol}(E)$, approximating E by rectangles), so check elementary matrices: permutations and scalings map rectangles to rectangles with the right volume, and for a shear $x_1 \mapsto x_1 + t x_2$ Fubini (Theorem 2.1) shows each x_1-slice is translated, so volume is preserved ($\det = 1$).

Step Local estimate. Fix $\epsilon \in (0,1)$. By uniform continuity of Dg and $(Dg)^{-1}$ on A there is $\eta > 0$ such that for every cube $Q \subset A$ of side $< \eta$ and centre a,

$$g(a) + Dg(a)\big((1-\epsilon)(Q-a)\big) \subset g(Q) \subset g(a) + Dg(a)\big((1+\epsilon)(Q-a)\big). \tag{A.1}$$

Indeed $h = Dg(a)^{-1}(g - g(a))$ satisfies $|h(y) - (y-a)| \le \epsilon' |y-a|$ (mean value inequality, ϵ' uniform in a), so h maps $Q-a$ into $(1+\epsilon)(Q-a)$, and, being a homeomorphism moving ∂Q by less than the gap, covers $(1-\epsilon)(Q-a)$. By the linear case and (A.1),

$$(1-\epsilon)^d |\det Dg(a)|\, |Q| \le \mathrm{vol}(g(Q)) \le (1+\epsilon)^d |\det Dg(a)|\, |Q|.$$

Step Summation. Take $f \ge 0$ (general f by linearity). Partition a rectangle containing A into cubes Q_k of side $< \eta$; let $\mathcal{K}$ be those inside A. The others meet ∂A, and they and their images (g Lipschitz) have volume $\to 0$ as $\eta \to 0$. The $g(Q_k)$ overlap only on null boundaries, so

$$\int_{g(A)} f = \sum_{k \in \mathcal{K}} \int_{g(Q_k)} f + o(1) = \sum_{k \in \mathcal{K}} f(g(a_k))\, \mathrm{vol}(g(Q_k)) + o(1),$$

with a_k the centre of Q_k (uniform continuity of f). By the local estimate this is within a factor $(1 \pm \epsilon)^d$ of the Riemann sum $\sum_k f(g(a_k)) |\det Dg(a_k)| |Q_k|$ for $\int_A f(g(y)) |\det Dg(y)| dy$. Let $\eta \to 0$, then $\epsilon \to 0$. ■

$|\det Dg|$ is the new volume element: $|L(R)| = |\det L|\, |R|$ for linear L.

A.2.4 Spherical coordinates

Spherical coordinates in $\mathbb{R}^d$ are $x = g(r, \theta_1, \dots, \theta_{d-1})$ with

$$\begin{aligned} x_1 &= r\sin\theta_1\sin\theta_2\cdots\sin\theta_{d-2}\cos\theta_{d-1},\\ x_2 &= r\sin\theta_1\sin\theta_2\cdots\sin\theta_{d-2}\sin\theta_{d-1},\\ &\vdots\\ x_{d-1} &= r\sin\theta_1\sin\theta_2, \qquad x_d = r\cos\theta_1, \end{aligned}$$

with $0 \le \theta_i \le \pi$ $(i \le d-2)$, $0 \le \theta_{d-1} \le 2\pi$, and Jacobian $r^{d-1}\sin^{d-2}\theta_1 \sin^{d-3}\theta_2 \cdots \sin\theta_{d-2}$. Every $x \neq 0$ is uniquely $r\gamma$, $r > 0$, $\gamma \in S^{d-1}$. Define

$$\int_{S^{d-1}} f(\gamma)\, d\sigma(\gamma) = \int_0^\pi \cdots \int_0^\pi \int_0^{2\pi} f(g(1,\theta))\sin^{d-2}\theta_1 \cdots \sin\theta_{d-2}\, d\theta_{d-1}\cdots d\theta_1.$$

Change of variables gives, for the ball $B(0,N)$,

$$\int_{B(0,N)} f(x)\, dx = \int_{S^{d-1}} \int_0^N f(r\gamma)\, r^{d-1}\, dr\, d\sigma(\gamma). \tag{A.2}$$

(g fails to be a diffeomorphism on a null set: apply Theorem 2.2 on $\{\delta \le r,\ \delta \le \theta_i \le \pi - \delta\}$ and let $\delta \to 0$.) The unit sphere has area $\omega_d = \int_{S^{d-1}} d\sigma$.

For $A(R_1, R_2) = \{R_1 \le |x| \le R_2\}$ and $\lambda \in \mathbb{R}$,

$$\int_{A(R_1,R_2)} |x|^\lambda\, dx = \omega_d \int_{R_1}^{R_2} r^{\lambda+d-1} dr = \begin{cases} \dfrac{\omega_d}{\lambda+d}\left(R_2^{\lambda+d} - R_1^{\lambda+d}\right), & \lambda \neq -d,\\ \omega_d \log(R_2/R_1), & \lambda = -d. \end{cases}$$

So $|x|^\lambda$ is integrable near 0 iff $\lambda > -d$, and near ∞ iff $\lambda < -d$.

A.3 Improper integrals. Integration over $\mathbb{R}^d$

Motivation. Over all of $\mathbb{R}^d$, integrals of continuous functions of moderate decrease are limits over growing cubes or balls; Fubini and polar coordinates extend.

A.3.1 Integration of functions of moderate decrease

Let $Q_N = \{|x_j| \le N/2\}$; for continuous f, $\int_{\mathbb{R}^d} f = \lim_N \int_{Q_N} f$ if it exists. f is of **moderate decrease** if

$$|f(x)| \le \frac{A}{1+|x|^{d+1}} \quad \text{for some } A > 0.$$

(For $d = 1$ as in Chapter 5; e.g. $\mathcal{P}_y(x) = \frac{1}{\pi}\frac{y}{x^2+y^2}$.)

Then the limit exists: with $I_N = \int_{Q_N} f$ and $M > N$, $Q_M - Q_N \subset A(aN, bM)$, $a = 1/2$, $b = \sqrt{d}/2$ (as $\{|x| < N/2\} \subset Q_N \subset \{|x| \le \sqrt{d}N/2\}$), so by the annulus formula with $\lambda = -d-1$

$$|I_M - I_N| \le A\int_{aN \le |x| \le bM} |x|^{-d-1} dx = A\omega_d\Big(\frac{1}{aN} - \frac{1}{bM}\Big) \le \frac{C}{N}.$$

So $\{I_N\}$ is Cauchy; over balls the limit is the same, as $|\int_{Q_N} f - \int_{B_{N/2}} f| \le C/N$.

The integral is linear and invariant under translations and rotations, and $\int f(\delta x)\, dx = \delta^{-d}\int f$: let $N \to \infty$ in the statements on cubes or balls (Theorem 2.2, $|\det| = 1$ or δ^d), translates of Q_N differing from Q_N where $|f|$ contributes $O(1/N)$.

A.3.2 Repeated integrals

Theorem 3.1

Suppose f is continuous on $\mathbb{R}^2$ and of moderate decrease. Then $F(x_1) = \int_{\mathbb{R}} f(x_1, x_2)\, dx_2$ is of moderate decrease on $\mathbb{R}$, and

$$\int_{\mathbb{R}^2} f(x)\, dx = \int_{\mathbb{R}} \Big(\int_{\mathbb{R}} f(x_1, x_2)\, dx_2 \Big) dx_1.$$

Proof.

Step Decrease of F. With $|f| \le A/(1+|x|^3)$, split at $|x_2| = |x_1|$: on $|x_2| \le |x_1|$ the integrand is $\le A/(1+|x_1|^3)$, giving $\le 2A|x_1|/(1+|x_1|^3) \le A'/(1+|x_1|^2)$; on $|x_2| \ge |x_1|$ it is $\le A/(1+|x_2|^3)$, giving $\le A'''/(1+|x_1|^2)$. F is continuous: integrals over $|x_2| \le N$ are (Theorem 2.1) and converge uniformly, the tails being $\le \int_{|x_2|>N} A/(1+|x_2|^3)$.

Step Identity. Choose N with $\big|\int_{\mathbb{R}^2} f - \int_{Q_N} f\big| < \epsilon$, $\int_{|x_1|>N/2} |F| < \epsilon$ and $\int_{|x_1|\le N/2} \int_{|x_2|>N/2} |f| < \epsilon$ (the inner integral is $\le \min\big(\frac{C}{1+|x_1|^2}, \frac{C}{N}\big)$). By Theorem 2.1,

$$\int_{Q_N} f = \int_{|x_1|\le N/2} \int_{|x_2|\le N/2} f\, dx_2\, dx_1,$$

within ϵ of $\int_{|x_1|\le N/2} F$, itself within ϵ of $\int_{\mathbb{R}} F$. So $\big|\int_{\mathbb{R}^2} f - \int_{\mathbb{R}} F\big| < 3\epsilon$. ∎

Likewise in $\mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$.

A.3.3 Spherical coordinates

For f of moderate decrease and fixed $\gamma \in S^{d-1}$,

$$|f(r\gamma) r^{d-1}| \le \frac{A r^{d-1}}{1 + r^{d+1}} \le \frac{B}{1+r^2},$$

so $r \mapsto f(r\gamma) r^{d-1}$ is of moderate decrease, and $N \to \infty$ in (A.2) (uniformly in γ) gives

$$\int_{\mathbb{R}^d} f(x)\, dx = \int_{S^{d-1}} \int_0^\infty f(r\gamma)\, r^{d-1}\, dr\, d\sigma(\gamma).$$

With rotation invariance of $\int_{\mathbb{R}^d}$, this gives rotation invariance of $d\sigma$:

$$\int_{S^{d-1}} f(R(\gamma))\, d\sigma(\gamma) = \int_{S^{d-1}} f(\gamma)\, d\sigma(\gamma) \tag{A.3}$$

for continuous f on S^{d-1} (apply the formula to $F(x) = f(x/|x|)\varphi(|x|)$, $\varphi \ge 0$ continuous with compact support in $(0,\infty)$ and $\int_0^\infty \varphi(r) r^{d-1} dr = 1$).